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Jet Physics of MG3 J163554+3629 and Its Impact on Supermassive Black Hole Evolution in the Early Universe

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"And you run, and you run
To catch up with the sun, but it's sinking
And racing around
To come up behind you again
The sun is the same in a relative way
But you're older
Shorter of breath
And one day closer to death"

— Pink Floyd, *Time* (1973)

Abstract

The origin and growth of supermassive black holes (SMBHs) in the early universe, particularly in the first billion years after the Big Bang, remains one of the most intriguing challenges in astrophysics. Detecting active galactic nuclei (AGN) in this era is hindered by their faint emissions due to the vast distances involved. However, blazars, a unique subclass of AGN with relativistic jets pointed directly towards Earth, provide an exceptional opportunity to study SMBH evolution at high redshifts. In this thesis we model the broadband spectral energy distribution (SED) of MG3 J163554+3629[1] at $z = 3.65$, one of the highest-redshift blazars detected by the *Fermi* Large Area Telescope. The model includes recent *NuSTAR* hard X-ray data and is built with a one-zone leptonic model in `JetSeT`, for accretion efficiencies $\eta = 0.057$–0.3, corresponding to black hole spins from $a = 0$ to nearly maximal. We find a black hole mass $M_{\rm BH} \simeq 1.8 \times 10^9\, M_\odot$ and an Eddington ratio $\lambda \simeq 0.08$ for an inner disk radius of $8\, R_{\rm S}$. The *NuSTAR* band is produced by a mix of synchrotron self-Compton and external Compton emission, and the γ-rays by Compton scattering of torus photons. The jet is particle-dominated and carries more power than the disk radiates. If the black hole had always accreted at its present Eddington ratio, the seed at $z = 30$ would need to be $\gtrsim 10^7\, M_\odot$, above the most massive direct-collapse seeds. The black hole must therefore have grown through an earlier phase of faster accretion, with a radiative efficiency well below the value for a maximally spinning black hole.

[1] Also known as NVSS J163547+362930 in some earlier work; in this thesis we adopt the canonical designation MG3 J163554+3629 throughout.

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Chapter 1

Introduction

Supermassive black holes (SMBHs) are now understood to reside at the centres of almost all massive galaxies and to play a fundamental role in their formation and evolution. Observations across cosmic time reveal that SMBHs with masses exceeding $10^9\,M_\odot$ were already in place within the first billion years after the Big Bang (Fan et al., 2006; Mortlock et al., 2011; Wu et al., 2015), posing significant challenges to current models of black hole growth. The existence of such massive black holes at early cosmic epochs is inferred from the detection of luminous active galactic nuclei and blazars at high redshift, whose extreme luminosities and relativistically beamed jet emission require rapidly accreting, massive central engines. Explaining how these objects formed and evolved so rapidly therefore demands a detailed understanding of both accretion processes and relativistic jet activity in the early Universe.

Active Galactic Nuclei (AGN) provide a direct observational window into SMBH growth through the release of gravitational energy as matter accretes onto the black hole. Among AGN, blazars represent an extreme and particularly valuable subclass, characterised by relativistic jets closely aligned with the observer's line of sight (Urry & Padovani, 1995). This geometric configuration results in strong Doppler boosting, making blazars observable out to very high redshifts and across the entire electromagnetic spectrum, from radio wavelengths to γ-rays. As a consequence, high-redshift blazars act as powerful beacons that trace the most massive and actively accreting black holes at early cosmic epochs.

The broadband emission of blazars is dominated by non-thermal radiation produced within the jet, typically described by a two-component spectral energy distribution (SED). The low-energy component is attributed to synchrotron emission from relativistic electrons, while the high-energy component is commonly interpreted as inverse Compton scattering, either through synchrotron self-Compton (SSC) or external Compton (EC) processes (Blumenthal & Gould, 1970). Modelling the SED therefore provides direct constraints on the physical conditions within the jet, including particle energy distributions, magnetic fields, bulk motion, and the properties of external radiation fields associated

with the accretion disk, broad-line region, or dusty torus.

High-redshift blazars offer additional insight beyond their low-redshift counterparts. Their extreme luminosities, combined with the effects of cosmic evolution, allow investigations of jet–accretion coupling, black hole spin, and accretion efficiency under conditions that are likely different from those observed in the local Universe (Ghisellini et al., 2010a; Sbarrato et al., 2014). In particular, the relative contributions of external photon fields to the high-energy emission can provide indirect constraints on the structure and evolution of the central engine at early times.

1.1 Aim of the Thesis

In this thesis, we focus on the high-redshift flat-spectrum radio quasar (FSRQ) MG3 J163554+3629 ($z \simeq 3.65$), a rare and powerful source that provides a unique opportunity to investigate relativistic jet physics and supermassive black hole (SMBH) growth in the early Universe.

Building upon previous research (Sánchez Zaballa et al., 2025), we analyse and use recent *NuSTAR* observations of the source. We combine them with multi-wavelength data spanning from radio to γ-rays, and we perform detailed SED modelling with the `JetSeT` framework (Tramacere et al., 2009, 2011; Tramacere, 2020). In particular, we aim to constrain the physical conditions within the jet, such as the electron energy distribution, the magnetic field strength and the bulk relativistic motion. We also explore the role of the accretion efficiency and of the black hole spin within the context of a standard Shakura–Sunyaev accretion disk (Shakura & Sunyaev, 1973).

A key advancement of this work is the inclusion of hard X-ray observations from *NuSTAR* (Harrison et al., 2013), which provide critical constraints on the high-energy component of the SED. These data allow a more robust characterisation of the inverse Compton emission and constrain the relative contributions of the synchrotron self-Compton and external Compton components, strengthening the physical interpretation of the jet–disk system.

Overall, this thesis aims to improve our understanding of the physical connection between accretion and jet production in high-redshift blazars, and to assess the role of such sources as probes of SMBH formation and rapid growth in the early Universe.

Chapter 2

Active Galactic Nuclei: From Local to High Redshift

2.1 Active Galactic Nuclei

Active Galactic Nuclei (AGN) are compact and extremely luminous regions located at the centres of galaxies, in which the radiative output can rival or exceed the integrated stellar emission of the host. The AGN power source is the accretion of matter onto a central supermassive black hole (SMBH): gravitational potential energy is converted into radiation through viscous dissipation in an accretion disk (Shakura & Sunyaev, 1973), and in many systems a fraction of the available energy is also channelled into collimated, relativistic jets.

AGN exhibit a broad range of observational manifestations, including continuum emission from radio to γ-rays, strong variability on timescales from minutes to years, and prominent emission-line features. This diversity reflects both intrinsic physical parameters (e.g. black hole mass, accretion rate, jet power) and geometric effects tied to the orientation of the nucleus relative to the line of sight.

A widely adopted interpretation is provided by the AGN unified model (Antonucci, 1993), in which many of the apparent differences between classes arise primarily from orientation-dependent obscuration by a dusty molecular torus surrounding the accretion disk and the broad-line region. As schematically illustrated in Figure 2.1, when the system is viewed face-on the nuclear continuum and broad emission lines can be directly observed (Type 1 objects), whereas at larger inclinations the torus obscures the inner regions and the spectrum becomes dominated by narrow emission lines and reprocessed radiation (Type 2 objects). Within this framework, Seyfert galaxies represent relatively low-luminosity AGN (commonly hosted by spirals) classified into Type 1 and Type 2 according to the visibility of the broad-line region. Radio-loud AGN additionally host powerful jets and extended radio structures, and are often subdivided into Fanaroff–Riley

Type I and Type II sources based on their large-scale radio morphology and luminosity, as also indicated in Figure 2.1.

Quasars correspond to the most luminous AGN population and are detectable out to the highest redshifts, implying the existence of massive, rapidly accreting SMBHs at early cosmic epochs. When the relativistic jet is closely aligned with the observer's line of sight (top of Figure 2.1), Doppler boosting strongly amplifies the observed non-thermal emission and variability, giving rise to blazars (Urry & Padovani, 1995). Owing to this extreme orientation effect, blazars form a distinct observational class and provide particularly powerful probes of jet physics, accretion processes, and SMBH growth across cosmic time.

Studying AGN at high redshift nevertheless remains challenging, even with the most advanced telescopes. A striking example is the population of compact, red sources discovered by the *James Webb Space Telescope* (JWST) at $z \gtrsim 4$, the so-called "little red dots" (LRDs). Their broad Balmer emission lines suggest the presence of accreting black holes (Matthee et al., 2024; Greene et al., 2024). However, much of their light can also be explained by stellar emission (Pérez-González et al., 2024), and they are remarkably weak in X-rays (Yue et al., 2024), so that whether or not they are AGN is still a matter of debate. Blazars do not suffer from this ambiguity: their beamed jet emission makes them bright at all frequencies, and its non-thermal origin is clear even at high redshift.

2.2 Blazars

Blazars constitute an extreme subclass of radio-loud active galactic nuclei in which a relativistic jet is oriented at a small angle with respect to the observer's line of sight (Urry & Padovani, 1995). This particular geometry leads to strong relativistic beaming effects, resulting in amplified luminosities, rapid variability, and a non-thermal continuum that dominates the emission across the entire electromagnetic spectrum. Because of these effects, blazars are among the most luminous persistent sources in the Universe and can be detected out to very high redshifts.

The broadband SED of blazars is typically characterised by two broad components. The low-energy component, extending from radio to optical or X-ray frequencies, is attributed to synchrotron radiation emitted by relativistic electrons within the jet. The high-energy component, spanning from X-rays to γ-rays, is generally interpreted as inverse Compton scattering (Blumenthal & Gould, 1970), where the same population of electrons upscatters low-energy photons. These seed photons may originate from within the jet itself, in the case of synchrotron self-Compton (SSC), or from external radiation fields such as the accretion disk, broad-line region, or dusty torus, in the case of external Compton (EC) processes.

Blazars are commonly divided into two main observational classes: *BL Lacertae ob-*

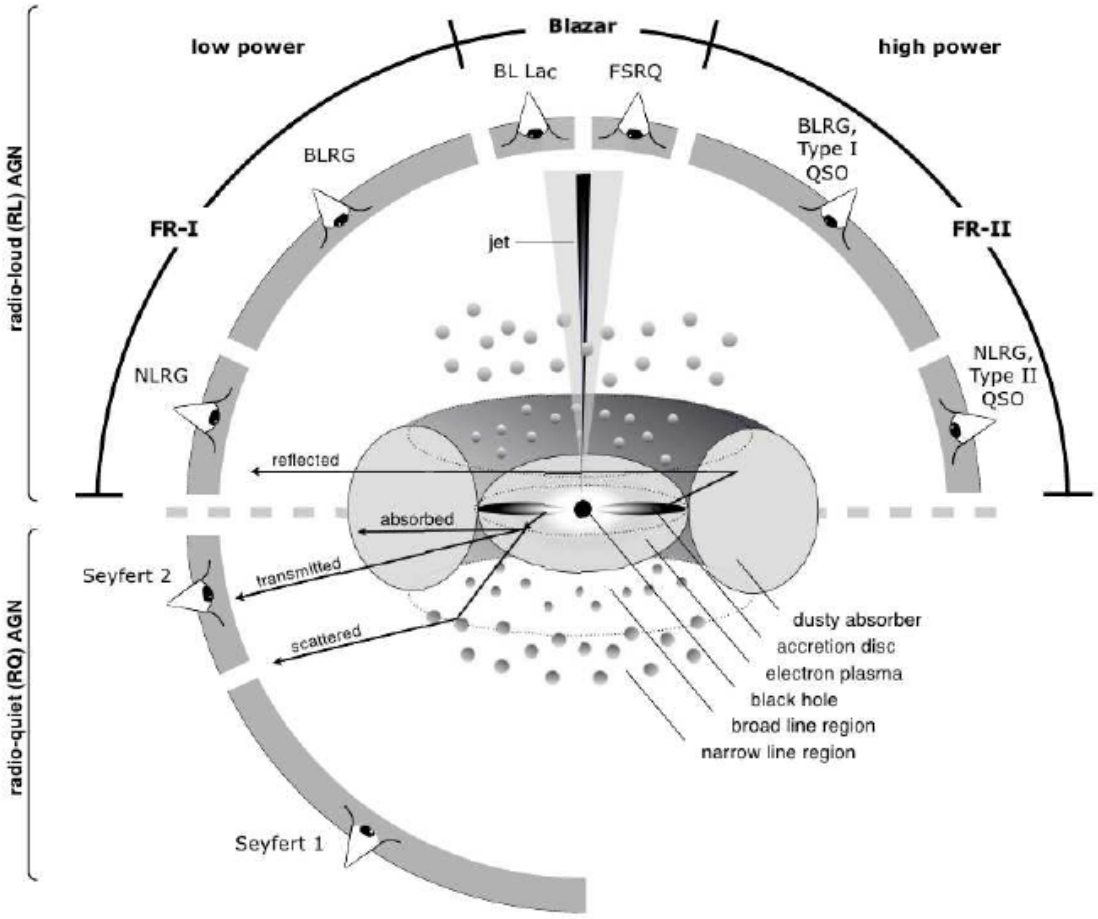


Figure 2.1: Schematic representation of the AGN unified scheme, illustrating how different observed AGN classes arise from the same central engine when viewed at different orientations, and how radio-loud sources with jets (including blazars for small viewing angles) fit within this framework. Reproduced from Beckmann & Shrader (2012).

jects (BL Lacs) and *flat-spectrum radio quasars* (FSRQs). BL Lacs exhibit weak or absent emission lines and are typically associated with lower accretion rates, while FSRQs display prominent broad emission lines and are linked to radiatively efficient accretion disks and strong external photon fields. This distinction is reflected in both their spectral properties and their dominant high-energy emission mechanisms. MG3 J163554+3629 belongs to the FSRQ class.

Due to their strong Doppler boosting, blazars provide direct access to the physical conditions within relativistic jets, including particle acceleration, magnetic field structure, and jet–accretion coupling. As a result, they serve as key laboratories for studying jet physics and the interplay between accretion and outflow in AGN. Their extreme luminosities further make them uniquely suited for probing the most massive and rapidly accreting black holes at cosmological distances.

2.3 High-Redshift Blazars

High-redshift blazars represent the most extreme and rare population of active galactic nuclei, hosting powerful relativistic jets in systems observed within the first few billion years of cosmic history. Their detection at redshifts $z \gtrsim 3$ implies the presence of supermassive black holes with masses exceeding $10^9\,M_\odot$ at early epochs, placing strong constraints on models of black hole seed formation, accretion efficiency, and growth timescales (Fan et al., 2006; Mortlock et al., 2011).

Due to their relativistic beaming, high-redshift blazars are preferentially selected in flux-limited surveys, making them among the few AGN populations observable at such large distances. However, this selection bias also implies that the observed sources represent only the tip of a much larger parent population of misaligned radio-loud AGN. Consequently, the study of high-redshift blazars provides indirect information on the abundance and evolution of massive black holes in the early Universe.

From a physical perspective, high-redshift blazars offer insight into jet and accretion physics under conditions distinct from those observed locally. Their SEDs often show a pronounced high-energy component, frequently dominated by external Compton processes involving photons from the accretion disk, broad-line region, or dusty torus. This behaviour is commonly associated with high accretion rates and radiatively efficient disks, suggesting that powerful jets and efficient accretion can coexist at early cosmic times (Ghisellini et al., 2010a; Sánchez Zaballa et al., 2025).

Observations in the X-ray and γ-ray bands are particularly critical for constraining the emission mechanisms of high-redshift blazars. Hard X-ray measurements can probe the transition region between the synchrotron and inverse Compton components of the SED (Bottacini et al., 2010), providing essential constraints on the particle energy distribution and the relative contributions of synchrotron self-Compton and external Compton

emission. As a result, broadband modelling that includes hard X-ray data is fundamental for a robust physical interpretation of these sources.

In this context, high-redshift blazars serve as powerful laboratories for studying the interplay between accretion and jet production in the early Universe and for testing theoretical models of rapid SMBH growth. Detailed multi-wavelength observations and physically motivated SED modelling are therefore essential to fully exploit their potential as probes of cosmic evolution.

2.4 Structure of Active Galactic Nuclei

While Active Galactic Nuclei display a wide range of observational properties, their phenomenology can be understood in terms of a common underlying physical structure. In this section, we describe the main components of an AGN central engine, focusing on those that are most relevant for high-energy emission and SED modelling.

2.4.1 Supermassive Black Hole and Accretion Disk

At the centre of every AGN lies a supermassive black hole (SMBH) with masses typically ranging from 10^6 to $10^{10}\,M_\odot$. The primary source of AGN luminosity is the accretion of matter onto the SMBH through a rotating accretion disk. As matter spirals inward, gravitational potential energy is converted into radiation via viscous dissipation (Shakura & Sunyaev, 1973), producing thermal emission that peaks in the optical–ultraviolet band for radiatively efficient disks.

A fundamental quantity describing the accretion process is the accretion luminosity, which can be written as

$$L_{\rm acc} = \eta\,\dot{M}\,c^2, \tag{2.1}$$

where $\dot{M}$ is the mass accretion rate, c is the speed of light, and η is the radiative efficiency, which depends on the black hole spin and on the geometry of the accretion flow.

The maximum luminosity achievable through spherical accretion is set by the Eddington limit, defined as

$$L_{\rm Edd} = \frac{4\pi G M_{\rm BH} m_p c}{\sigma_T} \simeq 1.26\times10^{38}\left(\frac{M_{\rm BH}}{M_\odot}\right)\ {\rm erg\ s^{-1}}, \tag{2.2}$$

where G is the gravitational constant, m_p is the proton mass, and σ_T is the Thomson cross section. Note that throughout this thesis we adopt the value 1.26×10^{38} erg s^{-1} for the Eddington luminosity constant, consistent with $\sigma_T = 6.652\times10^{-25}$ cm^2.

It is convenient to express the accretion regime in terms of the dimensionless accretion

rate

$$\dot{m} = \frac{\dot{M}}{\dot{M}_{\rm Edd}}, \tag{2.3}$$

where $\dot{M}_{\rm Edd} = L_{\rm Edd}/(\eta c^2)$. High values of $\dot{m}$ are associated with radiatively efficient thin disks, while lower accretion rates may lead to radiatively inefficient flows.

2.4.2 Broad Line and Narrow Line Regions

Surrounding the accretion disk are regions of ionized gas responsible for the emission-line features observed in AGN optical spectra (Peterson, 1993). The Broad Line Region (BLR) consists of dense clouds located at sub-parsec scales, whose high orbital velocities produce Doppler-broadened emission lines. At larger distances, between the dusty torus and ∼1000 pc, the Narrow Line Region (NLR) is composed of lower-density gas, giving rise to narrower emission lines.

The BLR plays an important role in high-energy emission models, as its radiation field can act as a source of external seed photons for inverse Compton scattering in the jet, particularly in powerful radio-loud AGN and blazars.

2.4.3 Dusty Torus

Within ∼10 pc, the central engine is surrounded by a dusty, molecular structure commonly referred to as the torus (Antonucci, 1993). This component absorbs a fraction of the optical and ultraviolet radiation emitted by the accretion disk and re-emits it thermally in the infrared band. The dusty torus is a key element of AGN unification schemes and strongly influences the observed properties of the source as a function of viewing angle.

In the context of blazar emission, infrared photons from the torus provide an additional external radiation field that can be upscattered by relativistic electrons in the jet via external Compton processes, contributing significantly to the high-energy component of the SED.

2.4.4 Relativistic Jets

A fraction of AGN launch collimated relativistic jets perpendicular to the accretion disk. These jets transport energy and momentum away from the central regions and are responsible for non-thermal emission extending from radio wavelengths to γ-ray energies. The jet power is thought to be linked to the accretion process and possibly to the extraction of rotational energy from the spinning black hole (Blandford & Znajek, 1977).

When the jet is oriented at a small angle with respect to the observer's line of sight, relativistic beaming effects strongly enhance the observed emission. In this case, the

observed luminosity $L_{\rm obs}$ is related to the intrinsic luminosity $L_{\rm int}$ through

$$L_{\rm obs} \simeq \delta^p L_{\rm int}, \tag{2.4}$$

where δ is the Doppler factor and p depends on the emission geometry and radiation mechanism (Urry & Padovani, 1995). This configuration gives rise to blazars, which dominate the extragalactic γ-ray sky and are key targets for broadband SED modelling.

2.5 Blazar SED and Its Interpretation

The SED of a blazar provides a global view of its emission across the electromagnetic spectrum and represents a key diagnostic for probing the physical conditions within relativistic jets. When expressed in the νF_ν representation, blazar SEDs typically display a characteristic double-peaked structure, reflecting the interplay between different non-thermal radiation mechanisms operating in the jet.

The low-energy component, extending from radio frequencies to the optical or X-ray band, is commonly attributed to synchrotron radiation emitted by relativistic electrons spiralling in the jet magnetic field. The location and shape of this synchrotron peak encode information about the particle energy distribution, magnetic field strength, and efficiency of particle acceleration processes.

The high-energy component, spanning from X-rays to γ-rays, is generally interpreted within a leptonic framework as inverse Compton emission (Blumenthal & Gould, 1970) produced by the same population of relativistic electrons. In this scenario, seed photons can originate internally from the synchrotron radiation itself, leading to synchrotron self-Compton (SSC) emission, or externally from radiation fields associated with the accretion disk, broad-line region (BLR), or dusty torus (DT), giving rise to external Compton (EC) emission. The relative contributions of SSC and EC processes depend on the accretion environment, jet location, and external photon energy densities. For completeness, we note that the high-energy component can also be explained within the competing lepto-hadronic framework, in which relativistic protons contribute to the γ-ray emission. This framework, however, faces uncertainties related to the extreme energies required to accelerate the particles (Bottacini et al., 2016, 2018), and in this work we adopt a purely leptonic model.

Figure 2.2 illustrates how the superposition of these leptonic radiative components reproduces the observed broadband emission. Such a decomposition provides a physically intuitive interpretation of the SED and forms the basis for quantitative modelling.

In high-redshift blazars, the interpretation of the SED is further influenced by cosmological redshifting and by the enhanced importance of external radiation fields, often resulting in a dominant inverse Compton component at high energies. A detailed discus-

sion of the individual radiative mechanisms contributing to the leptonic SED is presented in the following chapter, where the physical processes underlying the modelling approach adopted in this work are described.

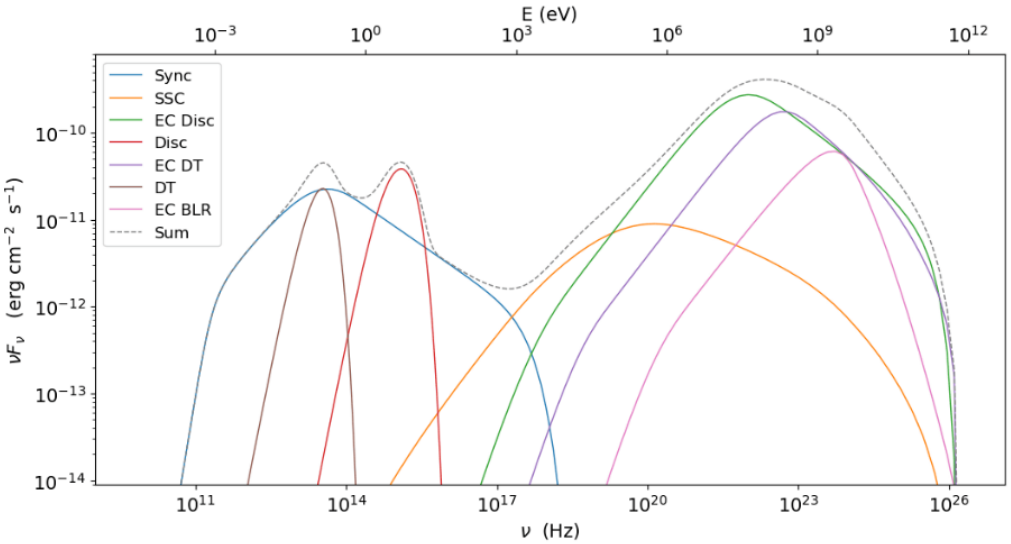


Figure 2.2: Illustrative blazar SED decomposed into its individual radiative components within a one-zone leptonic model. The low-energy peak is produced by synchrotron emission (Sync), while the high-energy emission arises from inverse Compton processes: synchrotron self-Compton (SSC) and external Compton scattering of photons from the accretion disk (EC Disc), the broad-line region (EC BLR) and the dusty torus (EC DT). The thermal emission of the accretion disk (Disc) and of the dusty torus (DT) is also shown, together with the total emission (Sum).

Chapter 3

Jet Radiative Processes and MG3 J163554+3629

The broadband SED of MG3 J163554+3629 at $z = 3.65$ encodes the physical conditions within a relativistic jet, the thermal state of a standard accretion disk, and the geometry of the nuclear radiation environment. Reproducing this SED self-consistently requires a framework that accounts for synchrotron emission and its associated self-absorption, inverse Compton scattering in both the Thomson and Klein–Nishina regimes, the external photon fields of the broad-line region (BLR) and dusty torus (DT), and the multi-colour thermal continuum of the Shakura–Sunyaev disk. All of these components are implemented in the `JetSeT` framework (Tramacere et al., 2009, 2011; Tramacere, 2020; Tramacere et al., 2022), which forms the computational basis of the modelling performed in this thesis.

A recent new observational addition to the SED of MG3 J163554+3629 is provided by the *NuSTAR* (Harrison et al., 2013) high-energy broad-band observation (3–79 keV). It provides direct spectral constraints at the energies where the synchrotron self-Compton (SSC) and external Compton (EC) components overlap. Its photon index Γ_X therefore helps to disentangle the two components, which softer X-ray data alone cannot do.

Sánchez Zaballa et al. (2025) also explored the dependence of the inferred physical parameters on the accretion efficiency η, varying it across six values (0.057, 0.083, 0.100, 0.150, 0.200, 0.300) corresponding to a range of black hole spin parameters from non-rotating (Schwarzschild) to rapidly rotating (Kerr) (Bardeen et al., 1972; Novikov & Thorne, 1973). In this work we repeat this exploration, now with the additional constraint provided by the *NuSTAR* data and with the additional value $\eta = 0.070$, to assess how the inclusion of hard X-ray observations modifies the inferred jet and disk parameters at each spin value.

The remainder of this chapter is organised as follows. Section 3.1 introduces the one-zone leptonic framework and its assumptions. Section 3.2 presents synchrotron the-

ory, including the electron energy distributions used in this work, the spectral peak, self-absorption, and radiative cooling. Section 3.3 covers inverse Compton scattering, distinguishing the SSC and EC contributions and discussing the Klein–Nishina suppression relevant for high-redshift FSRQs. Section 3.4 defines the jet parameters and the total jet power budget. Section 3.5 describes the Shakura–Sunyaev disk model and its observational role. Section 3.6 presents the relationship between black hole spin, ISCO radius, and accretion efficiency, and motivates the parameter grid adopted here. Section 3.7 discusses the specific diagnostic power of the *NuSTAR* band for this source.

3.1 One-Zone Leptonic Model

In the one-zone leptonic model, all non-thermal jet emission is attributed to a single, homogeneous, spherical emission region (hereafter the "blob") of radius R, permeated by a randomly oriented magnetic field of strength B and populated by a relativistic electron distribution $n(\gamma)$. The blob moves along the jet axis with bulk Lorentz factor Γ at a small angle θ to the observer's line of sight. The corresponding Doppler factor is

$$\delta = \frac{1}{\Gamma\,(1 - \beta\cos\theta)}, \qquad \beta = \sqrt{1 - \Gamma^{-2}}, \tag{3.1}$$

and for $\theta \approx 1/\Gamma$ one obtains $\delta \approx \Gamma$. Relativistic beaming transforms observed quantities from the comoving frame (primed) to the observer frame via

$$\nu_{\rm obs} = \frac{\delta}{1+z}\,\nu', \qquad F_{\nu_{\rm obs}} = \delta^{3+\alpha} F'_{\nu'}, \tag{3.2}$$

where α is the local spectral index (Urry & Padovani, 1995). The strong δ-dependence of the observed flux is precisely what renders blazars visible at cosmological distances, while misaligned sources of the same intrinsic power would be much fainter.

The one-zone model is a deliberate simplification: the complex, stratified structure of a real jet is replaced by a single emission zone whose averaged properties reproduce the time-averaged SED. This approximation is well-motivated for FSRQs in quiescent or mildly variable states, where the emission is dominated by a compact, active region (Ghisellini & Tavecchio, 2009). Its principal limitation is the inability to account for emission arising from spatially separated regions of the jet, such as the low-frequency radio emission, which is therefore not included in the fits.

The framework adopted follows Sánchez Zaballa et al. (2025), implemented in `JetSeT` (Tramacere et al., 2009, 2011; Tramacere, 2020). It is important to note that two distinct electron energy distributions (EEDs) are used in this work for different purposes. For the quantitative SED fitting of MG3 J163554+3629, we adopt a broken power-law EED (Section 3.2.2), which captures the two distinct power-law regimes observed in the

broadband data. The log-parabolic EED (Section 3.2.1) is used exclusively as a tool for the parameter exploration of Section 4.3, because its smooth, continuously curved shape allows a systematic study of parameter degeneracies without discrete breaks; it is not applied to the real observational data. The key parameters of the broken power-law model and the multi-wavelength observables that constrain them are summarised in Table 3.1.

3.2 Synchrotron Radiation

The low-energy component of the blazar SED, extending from radio frequencies to the optical or soft X-ray band depending on the source type, arises from synchrotron emission produced by relativistic electrons spiralling in the jet magnetic field. For MG3 J163554+3629, a low-synchrotron-peaked (LSP) FSRQ, the synchrotron peak lies in the infrared.

3.2.1 Log-Parabolic Electron Energy Distribution

The shape of the EED reflects the integrated history of particle acceleration and cooling processes within the blob. In the parameter exploration (Section 4.3), we adopt a log-parabolic EED of the form (Massaro et al., 2004; Tramacere et al., 2011)

$$n(\gamma) = N_0 \left(\frac{\gamma}{\gamma_0}\right)^{-(s + r \log_{10}(\gamma/\gamma_0))}, \qquad \gamma_{\min} \leq \gamma \leq \gamma_{\max}, \tag{3.3}$$

where:

- γ_0 is the reference Lorentz factor, around which the curvature of the distribution is defined;
- s is the local spectral index at γ_0, controlling the low-energy slope;
- $r > 0$ is the curvature parameter, with smaller r corresponding to a broader, flatter distribution and larger r to a narrower, more peaked one;
- N_0 is the normalisation in units of cm^{-3}.

The log-parabolic form has no direct physical origin and does not arise naturally from first principles: it is a phenomenological description, introduced to reproduce the curved spectra observed in blazars (Massaro et al., 2004). In this work it is used as a convenient tool. The curvature parameter r provides a simple way of controlling the width of the distribution, and the smooth, continuously curved shape allows a systematic variation of each physical parameter in isolation, without the discrete spectral break introduced by a broken power-law.

3.2.2 Broken Power-Law Electron Energy Distribution

For the quantitative SED modelling of MG3 J163554+3629, following Sánchez Zaballa et al. (2025), we adopt a broken power-law EED of the form:

$$n(\gamma) = N_0\, f(\gamma), \qquad f(\gamma) = \begin{cases} \gamma^{-p}, & \gamma_{\rm min} \le \gamma < \gamma_{\rm break}, \\ \gamma^{-p_1}, & \gamma_{\rm break} \le \gamma \le \gamma_{\rm max}, \end{cases} \tag{3.4}$$

where p and p_1 are the low- and high-energy spectral indices respectively, and $\gamma_{\rm break}$ is the break Lorentz factor. The number density of emitting electrons is $N = \int_{\gamma_{\rm min}}^{\gamma_{\rm max}} n(\gamma)\, d\gamma$. This parametrisation is described by five parameters $(N_0, p, p_1, \gamma_{\rm break}, \gamma_{\rm min})$, of which N_0, p_1 and $\gamma_{\rm break}$ are left free in the fits, and provides a physically direct description of the electron population in a source where the break is associated with radiative cooling or a change in the acceleration efficiency. Below the peaks of the inverse Compton components, which is where the *NuSTAR* band lies for this source, the X-ray photon index is set by the low-energy index through $\Gamma_X \simeq (p+1)/2$, while p_1 shapes the spectrum above the peaks and is constrained mainly by the *Fermi*-LAT data (Section 3.7). The break Lorentz factor $\gamma_{\rm break}$ determines the synchrotron peak frequency via $\nu_{\rm sync}^{\rm peak} \propto \gamma_{\rm break}^2 B\delta/(1+z)$.

3.2.3 Single-Electron Synchrotron Power and Characteristic Frequency

A relativistic electron of Lorentz factor γ and velocity βc moving at pitch angle $\alpha_{\rm p}$ to a magnetic field B radiates synchrotron power (Rybicki & Lightman, 1979)

$$P_{\rm sync}(\gamma) = 2\,\sigma_T c\,\gamma^2\,\beta^2\,U_B\,\sin^2\!\alpha_{\rm p}, \qquad U_B = \frac{B^2}{8\pi}, \tag{3.5}$$

where $\sigma_T = 6.65\times10^{-25}$ cm^2 is the Thomson cross section and U_B is the magnetic energy density. For ultra-relativistic electrons $\beta \approx 1$ and this reduces to $2\sigma_T c\gamma^2 U_B \sin^2\alpha_{\rm p}$. Averaged over an isotropic pitch-angle distribution, $\langle\sin^2\alpha_{\rm p}\rangle = 2/3$, so the pitch-angle-averaged power is $(4/3)\sigma_T c\gamma^2\beta^2 U_B$. The spectrum of a single electron peaks at the critical frequency

$$\nu_c = \frac{3eB}{4\pi m_e c}\,\gamma^2 \sin\alpha_{\rm p}, \tag{3.6}$$

with the single-electron power spectrum peaking at $\simeq 0.29\,\nu_c$ and falling exponentially above it (Longair, 2011).

3.2.4 Synchrotron Spectrum from the Broken Power-Law EED

For a power-law EED with index p, the pitch-angle-averaged synchrotron emissivity in the optically thin regime is a power law $j_\nu \propto \nu^{-(p-1)/2}$ (Blumenthal & Gould, 1970).

For the broken power-law EED (Equation 3.4), the synchrotron spectrum consists of two power-law segments joined smoothly at the frequency corresponding to $\gamma_{\rm break}$. In the νF_ν representation the synchrotron component peaks at an observer-frame frequency following the scaling

$$\nu_{\rm sync}^{\rm peak} \propto \nu_B \, \gamma_{\rm break}^2 \, \frac{\delta}{1+z}, \qquad \nu_B = \frac{eB}{2\pi m_e c}, \tag{3.7}$$

where $\nu_B = eB/(2\pi m_e c)$ is the non-relativistic electron gyrofrequency (Tramacere et al., 2011). Equation 3.7 establishes the fundamental degeneracy among $\gamma_{\rm break}$, B, and δ: observations of $\nu_{\rm sync}^{\rm peak}$ alone constrain only the product $\gamma_{\rm break}^2 B\delta$. Breaking this degeneracy requires independent constraints on at least one of these parameters, which the full broadband SED (including the SSC component and the *NuSTAR* hard X-ray data) provides.

3.2.5 Radiative Cooling and the Cooling Break

Synchrotron-emitting electrons lose energy at a rate (Rybicki & Lightman, 1979)

$$\left.\frac{d\gamma}{dt}\right|_{\rm sync} = -\frac{4\sigma_T U_B}{3 m_e c}\,\gamma^2 \equiv -b_{\rm sync}\,\gamma^2, \tag{3.8}$$

where $b_{\rm sync} = 4\sigma_T U_B/(3 m_e c)$. Electrons cool on a timescale

$$t_{\rm cool}(\gamma) = \frac{1}{b_{\rm sync}\,\gamma} = \frac{3 m_e c}{4\sigma_T U_B\,\gamma}. \tag{3.9}$$

The critical "cooling Lorentz factor" $\gamma_{\rm cool}$ at which $t_{\rm cool} = t_{\rm cross} = R/c$ (the blob light-crossing time) separates the fast-cooling regime ($\gamma > \gamma_{\rm cool}$) from the slow-cooling regime ($\gamma < \gamma_{\rm cool}$):

$$\gamma_{\rm cool} = \frac{3 m_e c^2}{4\sigma_T U_B\, R}. \tag{3.10}$$

In the fast-cooling regime, the synchrotron spectrum steepens by $\Delta\alpha = 1/2$ above the cooling-break frequency $\nu_{\rm cool} \propto \gamma_{\rm cool}^2 B\delta/(1+z)$ (Sari et al., 1998). In the broken power-law EED used for fitting MG3 J163554+3629, the spectral break at $\gamma_{\rm break}$ and the steepening of the slope from p to p_1 naturally encode the cumulative effect of radiative cooling and any change in the acceleration efficiency.

3.2.6 Synchrotron Self-Absorption

At sufficiently low frequencies the synchrotron-emitting plasma becomes optically thick to its own radiation. The synchrotron self-absorption coefficient for a power-law electron distribution of index p scales as $\alpha_\nu^{\rm SSA} \propto \nu^{-(p+4)/2}$, and the self-absorption frequency $\nu_{\rm SSA}$

is defined by the condition $\tau(\nu_{\rm SSA}) = \alpha_\nu^{\rm SSA} R = 1$ (Rybicki & Lightman, 1979; Longair, 2011). Below $\nu_{\rm SSA}$ the source is optically thick and the emerging spectrum scales as $F_\nu \propto \nu^{5/2}$, rather than as the optically thin power law $F_\nu \propto \nu^{-(p-1)/2}$. `JetSeT` handles the radiative transfer through the blob numerically across the self-absorbed regime (Tramacere et al., 2011). For a compact emission region such as the one modelled here, the self-absorption frequency falls well above the radio band, and the low-frequency radio emission of MG3 J163554+3629 is produced by more extended regions of the jet. In this work the radio points below 10^{10} Hz are therefore not fitted, and $\nu_{\rm SSA}$ and $\gamma_{\rm min}$ are not constrained by the fit (Section 4.4.1).

3.3 Inverse Compton Scattering

The high-energy SED component of MG3 J163554+3629, covering the X-ray to γ-ray band, is produced by inverse Compton (IC) upscattering of low-energy seed photons by the relativistic jet electrons. The kinematic and energetic properties of IC scattering depend critically on the energy of the seed photons in the electron rest frame, determining whether the interaction proceeds in the classical Thomson regime or the quantum Klein–Nishina regime.

3.3.1 Thomson and Klein-Nishina Regimes

The scattering cross section is the key quantity governing IC emission (Blumenthal & Gould, 1970). Defining the dimensionless seed photon energy in the electron rest frame as

$$x = \frac{\gamma\, h\nu_{\rm seed}}{m_e c^2}, \tag{3.11}$$

the scattering regime is determined by the magnitude of x:

- **Thomson regime** ($x \ll 1$): the cross section is constant at σ_T, and the scattered photon energy is $\epsilon_{\rm out} \approx \frac{4}{3}\gamma^2 \epsilon_{\rm seed}$. The IC power for an isotropic photon field of energy density $U_{\rm ph}$ is
$$P_{\rm IC}^{\rm Th} = \frac{4}{3}\,\sigma_T c\,\gamma^2\,U_{\rm ph}, \tag{3.12}$$
yielding the Compton dominance ratio $P_{\rm IC}/P_{\rm sync} = U_{\rm ph}/U_B$ (Blumenthal & Gould, 1970).

- **Klein-Nishina regime** ($x \gg 1$): the cross section falls as
$$\sigma_{\rm KN}(x) \approx \frac{3\sigma_T}{8x}\left(\ln(2x) + \frac{1}{2}\right), \tag{3.13}$$

significantly suppressing the IC power relative to the Thomson prediction (Blumenthal & Gould, 1970). The scattered photon carries a large fraction of the electron energy, $\epsilon_{\rm out} \approx \gamma m_e c^2$, independent of the seed photon energy.

The transition between regimes occurs around $x \sim 1$, i.e., when the seed photon energy in the electron rest frame satisfies $\gamma h\nu_{\rm seed} \sim m_e c^2$. For external photon fields the seed energy must be evaluated in the blob frame, where it is boosted by a factor $\sim \Gamma$. For MG3 J163554+3629, torus photons ($h\nu \approx 0.25$ eV for $T_{\rm DT} \approx 1000$ K) reach a few eV in the blob frame and enter the KN regime only for $\gamma \gtrsim 10^4$–10^5, well above the break Lorentz factor ($\gamma_{\rm break} \approx 10^3$): the EC(DT) emission around the γ-ray peak is therefore produced in the Thomson regime. BLR photons ($h\nu \approx 10$ eV) are about 40 times more energetic, and their scattering is affected by KN suppression at correspondingly lower electron energies. The SSC seed photons are synchrotron photons emitted in the far infrared, whose energy in the blob frame is only $\sim 10^{-3}$ eV, so that SSC scattering occurs in the Thomson regime. `JetSeT` implements the full Compton redistribution function $\mathcal{R}(x, \epsilon_{\rm seed}, \epsilon_{\rm out})$ across the Thomson-to-KN transition, following the formalism of Blumenthal & Gould (1970), as detailed in Tramacere et al. (2011).

3.3.2 Synchrotron Self-Compton

In the SSC process, the same electron population that produces the synchrotron radiation upscatters the resulting synchrotron photons to X-ray and γ-ray energies. In the Thomson limit, the SSC peak frequency scales with the synchrotron peak (Equation 3.7) as

$$\nu_{\rm SSC}^{\rm peak} \propto \gamma_{\rm break}^2 \, \nu_{\rm sync}^{\rm peak}. \tag{3.14}$$

The ratio of the two peaks, $\nu_{\rm SSC}^{\rm peak}/\nu_{\rm sync}^{\rm peak} \propto \gamma_{\rm break}^2$, provides a direct measurement of $\gamma_{\rm break}$ independently of B and δ, breaking part of the degeneracy identified in Section 3.2.4.

The SSC luminosity in the Thomson limit is

$$L_{\rm SSC} = \frac{4}{3}\,\sigma_T c\, U'_{\rm sync} \int \gamma^2\, n(\gamma)\, d\gamma, \tag{3.15}$$

where $U'_{\rm sync} = L_{\rm sync}/(4\pi R^2 c)$ is the comoving synchrotron photon energy density (Ghisellini & Tavecchio, 2009). Combining with the synchrotron power gives the Compton dominance $A_C = L_{\rm SSC}/L_{\rm sync} = U'_{\rm sync}/U_B$.

3.3.3 External Compton Emission from the Dusty Torus

In powerful FSRQs such as MG3 J163554+3629, the dominant γ-ray emission mechanism is external Compton (EC) scattering of photons from radiation fields outside the blob: the direct accretion disk continuum, the BLR line emission, or the DT infrared emission.

The relative importance of each field depends on the location of the blob along the jet and on the geometry of the nuclear environment.

The comoving energy density of an external isotropic radiation field of luminosity $L_{\rm ext}$ and characteristic radius $R_{\rm ext}$ is (Ghisellini & Tavecchio, 2009)

$$U'_{\rm ext} = \frac{\Gamma^2 \, L_{\rm ext}}{4\pi R_{\rm ext}^2 \, c}, \tag{3.16}$$

where the factor Γ^2 accounts for the Doppler compression of the external radiation field in the blob frame. For a power-law electron distribution of index p_1 (the high-energy slope), the EC luminosity at small viewing angles scales as (Georganopoulos et al., 2001)

$$L_{\rm EC} \propto \delta^{4+2\alpha} \, U'_{\rm ext} \, \nu^{-\alpha}, \qquad \alpha = \frac{p_1 - 1}{2}, \tag{3.17}$$

so that $L_{\rm EC} \propto \delta^{3+p_1}$. This strong Doppler dependence, combined with the large Γ^2 amplification in Equation 3.16, makes EC dominant over SSC in powerful FSRQs where the external radiation energy density $U'_{\rm ext}$ exceeds $U'_{\rm sync}$.

The seed photon field that dominates the γ-ray emission depends on the location of the emission region. Upscattering of BLR photons ($h\nu_{\rm BLR} \approx 10$ eV) by electrons with $\gamma \approx \gamma_{\rm break} \approx 10^3$ produces γ-rays of energy $\approx \Gamma\delta\gamma^2 h\nu_{\rm BLR}/(1+z) \sim 1$ GeV in the observer frame, in the *Fermi*-LAT band. If the emission region were located inside the BLR, the γ-rays would also be absorbed through γ-γ pair production on the BLR photons, producing a cutoff above a few GeV in the observer frame (Costamante et al., 2018). No such absorption signature is evident in the *Fermi*-LAT spectrum of MG3 J163554+3629, which suggests that the emission region is located outside the BLR. Whether the torus or the BLR then provides the dominant seed photons depends on the distance of the blob from the two regions (Section 4.6.4).

The DT is modelled as a greybody of temperature $T_{\rm DT}$ and total luminosity $\xi_{\rm DT} \, L_{\rm disc}$, where $\xi_{\rm DT}$ is the covering factor, set to 0.1 in this work. Its characteristic radius is

$$R_{\rm DT} \simeq 2.5 \times 10^{18} \left(\frac{L_{\rm disc}}{10^{45} \, {\rm erg \, s^{-1}}} \right)^{1/2} \ {\rm cm}, \tag{3.18}$$

following the scaling of Ghisellini & Tavecchio (2009); the `JetSeT` models of Chapter 4 adopt a normalisation larger by a factor of about 2.5 (Equation 4.4). For a torus temperature $T_{\rm DT} \approx 1000$ K the DT seed photons are near-to-mid-infrared photons with $h\nu_{\rm DT} \approx 0.1$–$0.3$ eV, and their EC upscattering to GeV energies occurs in the Thomson regime for the electron Lorentz factors relevant to MG3 J163554+3629. The disk luminosity $L_{\rm disc}$, measured from the optical-UV SED in Chapter 4, therefore directly sets both $R_{\rm DT}$ and $U'_{\rm ext}$, connecting the accretion disk properties to the high-energy emission without introducing additional free parameters.

3.4 Jet Parameters and Power Budget

3.4.1 Emission Region Geometry

The blob radius R is connected to observable variability through the light travel time argument: a source cannot vary coherently on a timescale $t_{\rm var}$ unless it is causally connected across a region of size

$$R \lesssim \frac{c\, t_{\rm var}\, \delta}{1+z}. \tag{3.19}$$

For sources where variability is not well-constrained, R is treated as a free parameter within physically plausible bounds (10^{14}–10^{17} cm for FSRQs). The distance of the blob from the central black hole, $d_{\rm blob}$, determines which external radiation fields the blob intercepts. As discussed in Section 3.3.3, the absence of γ-γ absorption in MG3 J163554+3629 suggests that the blob resides outside the BLR, $d_{\rm blob} \gtrsim R_{\rm BLR}$. In our fits the dissipation radius is $R_H \simeq 1.2 \times 10^{18}$ cm, outside the BLR and inside the torus (Section 4.5).

3.4.2 Jet Power Components

The total power transported along the jet is decomposed into contributions from relativistic electrons, the magnetic field, radiation, and cold (non-relativistic) protons (Ghisellini et al., 2014):

$$P_{\rm jet} = P_e + P_B + P_{\rm rad} + P_p. \tag{3.20}$$

Each component is expressed in the observer frame as

$$P_i = \pi R^2\, \Gamma^2\, \beta\, c\, U'_i, \tag{3.21}$$

where U'_i is the comoving energy density of component i. The individual contributions are:

$$U'_e = m_e c^2 \int_{\gamma_{\rm min}}^{\gamma_{\rm max}} \gamma\, n(\gamma)\, d\gamma, \tag{3.22}$$

$$U'_B = \frac{B^2}{8\pi}, \tag{3.23}$$

$$U'_{\rm rad} = \frac{L_{\rm sync} + L_{\rm IC}}{4\pi R^2 c\, \delta^4}, \tag{3.24}$$

$$U'_p = m_p c^2 \int_{\gamma_{\rm min}}^{\gamma_{\rm max}} n(\gamma)\, d\gamma, \tag{3.25}$$

where the last expression assumes one cold proton per relativistic electron (Celotti & Ghisellini, 2008).

A particularly informative ratio is the magnetisation parameter

$$\sigma_{\rm eq} = \frac{U'_B}{U'_e}, \tag{3.26}$$

which measures the departure from energy equipartition between the magnetic field and the radiating electrons. Powerful FSRQs typically show $\sigma_{\rm eq} \ll 1$, indicating particle dominance (Ghisellini et al., 2014). The total jet power $P_{\rm jet}$ can be compared with the disk accretion luminosity $L_{\rm disc}$ as a consistency check on the jet–accretion coupling, and is discussed in Chapter 4 for each value of η explored here.

3.5 Thermal Emission from the Accretion Disk

The optical-to-ultraviolet emission of MG3 J163554+3629, detected by *Swift*-UVOT and *Gaia*, is dominated by thermal radiation from the accretion disk. At $z = 3.65$, rest-frame UV photons are redshifted into the observed optical band, and the spectral shape provides a direct diagnostic of the disk temperature structure, the black hole mass, and the accretion rate.

3.5.1 Shakura-Sunyaev Disk: Structure and Flux

The disk is modelled as a geometrically thin ($H/R \ll 1$), optically thick, azimuthally symmetric, steady-state accretion flow (Shakura & Sunyaev, 1973). The accretion rate

$$\dot{M} = -2\pi R\,\Sigma(R)\,v_R = \text{const} \tag{3.27}$$

is constant at all radii. Applying mass and angular momentum conservation and imposing the zero-torque boundary condition at the inner edge $R_{\rm in} = R_{\rm ISCO}$, the radiated flux per unit disk area is (Shakura & Sunyaev, 1973)

$$F(R) = \frac{3GM_{\rm BH}\dot{M}}{8\pi R^3}\left(1 - \sqrt{\frac{R_{\rm ISCO}}{R}}\right). \tag{3.28}$$

Since the disk is optically thick, each annulus radiates as a local blackbody at effective temperature

$$T(R) = \left[\frac{F(R)}{\sigma_{\rm SB}}\right]^{1/4}, \tag{3.29}$$

where $\sigma_{\rm SB}$ is the Stefan-Boltzmann constant. The temperature profile reaches a maximum at

$$R_{\rm max} = \frac{49}{36}\,R_{\rm ISCO}, \tag{3.30}$$

Observable / SED feature	**JetSeT parameter(s)**	**Physical constraint / degeneracy breaking**
Synchrotron peak frequency $\nu_{\rm sync}^{\rm peak}$	$\gamma_{\rm break}$, B, δ	$\nu_{\rm sync}^{\rm peak} \propto \gamma_{\rm break}^2 B\,\delta/(1+z)$; constrains the product $\gamma_{\rm break}^2 B\delta$. Degeneracy partially broken by the SSC peak ratio (see row 4) or variability timescale.
Radio–IR spectral slope (optically thin)	p, $\gamma_{\rm min}$	Optically thin synchrotron slope $= (p-1)/2$; constrains the low-energy electron index p and, below $\nu_{\rm SSA}$, $\gamma_{\rm min}$. In this source the radio points are dominated by extended regions of the jet and are not fitted.
Synchrotron self-absorption turnover $\nu_{\rm SSA}$	B, N_0, R	Turnover frequency and $F_\nu \propto \nu^{5/2}$ optically thick slope jointly bound the magnetic field B, electron normalisation N_0, and blob size R; not constrained by the present data.
SSC peak frequency ratio $\nu_{\rm SSC}^{\rm peak}/\nu_{\rm sync}^{\rm peak}$	$\gamma_{\rm break}$	$\nu_{\rm SSC}^{\rm peak}/\nu_{\rm sync}^{\rm peak} \propto \gamma_{\rm break}^2$ (Thomson limit); gives $\gamma_{\rm break}$ independently of B and δ, breaking the synchrotron peak degeneracy.
Compton dominance $A_C = L_{\rm IC}/L_{\rm sync}$	$U'_{\rm ph}/U'_B$	$A_C = U'_{\rm ph}/U'_B$; for SSC, $U'_{\rm ph} = U'_{\rm sync}$, while for EC the external field is boosted as $U'_{\rm ext} = \Gamma^2 L_{\rm ext}/(4\pi R_{\rm ext}^2 c)$. The large A_C observed in MG3 J163554+3629 ($A_C \gg 1$) is consistent only with EC dominance.
Hard X-ray photon index Γ_X (*NuSTAR*, 3–79 keV)	**p (and the SSC/EC balance)**	**Below the inverse Compton peaks $\Gamma_X \simeq (p+1)/2$. The measured $\Gamma_X = 0.94 \pm 0.34$ (§ 4.1.2) is harder than a falling SSC tail; in the full model the band is produced by SSC close to its peak plus rising EC emission (§ 4.5.4).**
γ-ray spectral slope (*Fermi*-LAT)	p_1, $\gamma_{\rm break}$, δ, $T_{\rm DT}$	EC post-peak slope constrains p_1; EC peak frequency $\nu_{\rm EC}^{\rm peak} \propto \gamma_{\rm break}^2\,\Gamma\,\delta\,T_{\rm DT}/(1+z)$ jointly constrains $\gamma_{\rm break}$ and the dust torus temperature $T_{\rm DT}$.
Optical/UV thermal bump	$M_{\rm BH}$, $\dot{M}$, η	Multi-colour disk blackbody: $\nu_p L_{\nu_p} \propto \dot{M}$ and $\nu_p \propto \dot{M}^{1/4} M_{\rm BH}^{-1/2}$ fix $L_{\rm disc}$, hence $R_{\rm BLR}, R_{\rm DT}$ (Eq. 3.36), propagating η uncertainty to EC seed photons.

Table 3.1: Mapping between multi-wavelength observational diagnostics and `JetSeT` model parameters for the one-zone leptonic model with a broken power-law EED applied to MG3 J163554+3629. The **NuSTAR** row highlights the novel hard X-ray observational constraint introduced in this work.

giving

$$T_{\max} \propto \dot{M}^{1/4}\, M_{\rm BH}^{-1/2}. \tag{3.31}$$

For $M_{\rm BH} = 10^9\, M_\odot$ and $\dot{M} = \dot{M}_{\rm Edd}$ with $\eta = 0.1$, one obtains $T_{\max} \approx 7.5 \times 10^4$ K, placing the peak of the disk emission at $h\nu_p \approx 16$ eV in the rest-frame ultraviolet. At $z = 3.65$, this peak is shifted to ≈ 3.4 eV, within the *Swift*-UVOT and *Gaia* passbands.

3.5.2 Observed Spectrum: Multi-Colour Blackbody and Angular Correction

The specific luminosity of the disk is

$$L_\nu = \frac{16\pi^2 h\nu^3 \cos\theta_v}{c^2} \int_{R_{\rm ISCO}}^{R_{\rm out}} \frac{R\, dR}{\exp[h\nu/k_B T(R)] - 1}, \tag{3.32}$$

where $\cos\theta_v \approx 1$ for blazars (Cunningham, 1975; Calderone et al., 2013).

3.5.3 Asymptotic Spectral Behaviours

In the νL_ν representation, the SSD spectrum has three regimes: a Rayleigh-Jeans tail ($\nu L_\nu \propto \nu^3$) at low frequencies, the characteristic multi-colour power law $\nu L_\nu \propto \nu^{4/3}$ at intermediate frequencies, and a Wien exponential cutoff $\nu L_\nu \propto \nu^4 \exp(-h\nu/k_B T_{\max})$ at high frequencies (Shakura & Sunyaev, 1973; Frank et al., 2002; Campitiello et al., 2021).

3.5.4 Spectral Peak: Analytical Relations

The peak frequency and luminosity of the disk SED are (Calderone et al., 2013; Campitiello et al., 2021):

$$\frac{\nu_p}{[\rm Hz]} = A \left(\frac{\dot{M}}{M_\odot\, \rm yr^{-1}}\right)^{1/4} \left(\frac{M_{\rm BH}}{10^9\, M_\odot}\right)^{-1/2}, \quad \log_{10} A = 15.25, \tag{3.33}$$

$$\nu_p L_{\nu_p} = B \left(\frac{\dot{M}}{M_\odot\, \rm yr^{-1}}\right) \cos\theta_v, \quad \log_{10} B = 45.66. \tag{3.34}$$

Together, these form a two-equation system in $M_{\rm BH}$ and $\dot{M}$ once ν_p and $\nu_p L_{\nu_p}$ are measured, provided η is specified. The bolometric disk luminosity is

$$L_{\rm disc} \simeq \nu_p L_{\nu_p} / \cos\theta_v = \eta\, \dot{M}\, c^2, \tag{3.35}$$

where the constant B already incorporates $\eta_{\rm SS} = 1/12 \approx 0.083$. The BLR and DT radii follow as

$$R_{\rm BLR} \simeq 10^{17} \left(\frac{L_{\rm disc}}{10^{45}\,{\rm erg\,s^{-1}}} \right)^{1/2} \ {\rm cm}, \quad R_{\rm DT} \simeq 2.5 \times 10^{18} \left(\frac{L_{\rm disc}}{10^{45}\,{\rm erg\,s^{-1}}} \right)^{1/2} \ {\rm cm}, \quad (3.36)$$

following Kaspi et al. (2000) and Cleary et al. (2007) as parameterised by Ghisellini & Tavecchio (2009).

3.5.5 Novikov-Thorne Relativistic Disk Model

The Shakura-Sunyaev treatment neglects General Relativistic effects. The Novikov-Thorne relativistic flux (Novikov & Thorne, 1973; Page & Thorne, 1974) is

$$F(x) = \frac{3\dot{M}c^6}{8\pi G^2 M_{\rm BH}^2} \frac{1}{x^4\,(x^3 - 3x + 2a)} \left[x - x_0 - \frac{3a}{2} \ln \frac{x}{x_0} - \sum_{i=1}^{3} \frac{3(x_i - a)^2}{x_i (x_i - x_j)(x_i - x_k)} \ln \frac{x - x_i}{x_0 - x_i} \right], \quad (3.37)$$

where $x = (Rc^2/GM_{\rm BH})^{1/2}$, x_0 is its value at the ISCO, and x_1, x_2, x_3 are the roots of $x^3 - 3x + 2a = 0$. In the `JetSeT` implementation used here, the disk is described by a multi-colour blackbody whose inner radius is a free parameter in units of $R_{\rm S}$, not tied to the ISCO of a black hole of given spin (Section 4.6.5).

3.6 Black Hole Spin and Accretion Efficiency

3.6.1 The Kerr Metric and the ISCO

The ISCO radius in the Kerr metric is (Bardeen et al., 1972)

$$R_{\rm ISCO}(a) = \frac{GM_{\rm BH}}{c^2} \left(3 + Z_2 \mp \sqrt{(3 - Z_1)(3 + Z_1 + 2Z_2)} \right), \quad (3.38)$$

where

$$Z_1 = 1 + (1 - a^2)^{1/3} \left[(1 + a)^{1/3} + (1 - a)^{1/3} \right], \qquad Z_2 = \sqrt{3a^2 + Z_1^2}, \quad (3.39)$$

and the upper (lower) sign refers to prograde (retrograde) orbits. $R_{\rm ISCO}$ ranges from $6\,GM_{\rm BH}/c^2$ at $a = 0$ to $GM_{\rm BH}/c^2$ at $a = +1$.

3.6.2 Radiative Efficiency as a Function of Spin

The radiative efficiency equals the specific binding energy at the ISCO (Novikov & Thorne, 1973):

$$\eta(a) = 1 - \sqrt{1 - \frac{2}{3}\frac{GM_{\rm BH}/c^2}{R_{\rm ISCO}(a)}}, \tag{3.40}$$

giving $\eta \approx 0.057$ at $a = 0$, $\eta \approx 0.423$ at $a = +1$, and $\eta \approx 0.038$ at $a = -1$. The classical Shakura-Sunyaev value $\eta_{\rm SS} = 1/12 \approx 0.083$ overestimates the true Novikov-Thorne value at $a = 0$ by $\sim$46% because it neglects relativistic corrections to the effective potential (Campitiello et al., 2021).

3.6.3 Black Hole Growth and Seed Mass Constraints

The mass e-folding time for Eddington-limited accretion is the Salpeter timescale (Salpeter, 1964):

$$t_{\rm Salp} = \frac{\eta}{1-\eta}\,\frac{\sigma_T c}{4\pi G m_p\,\mu_e} \;\equiv\; \frac{\eta}{1-\eta}\,\tau_{\rm acc}, \tag{3.41}$$

where $\tau_{\rm acc} \equiv \sigma_T c/(4\pi G m_p \mu_e)$ is the rest-mass accretion timescale and $\mu_e = (1 - Y/2)^{-1}$ is the mean molecular weight per scattering electron. For a fully ionised, zero-metallicity primordial plasma with $Y \simeq 0.25$ (Cyburt et al., 2003), $\mu_e = 8/7 \simeq 1.143$, giving

$$\tau_{\rm acc} \simeq 0.45\,\mu_e^{-1}\ {\rm Gyr} \simeq 0.394\ {\rm Gyr}. \tag{3.42}$$

In this thesis we adopt the pure-hydrogen value, $\mu_e = 1$ and $\tau_{\rm acc} = 0.450$ Gyr, consistent with the Eddington luminosity of Equation 2.2; the effect of the helium content is discussed in Section 4.10. The Salpeter time is then $t_{\rm Salp} \simeq 0.45\,\eta/(1-\eta)$ Gyr, i.e. $\simeq 0.05$ Gyr for $\eta = 0.1$.

The required seed mass to reach $M_{\rm BH}$ at cosmic time $t_{\rm obs}$ is

$$M_{\rm seed} = M_{\rm BH}\,\exp\!\left(-\lambda\,\frac{1-\eta}{\eta}\,\frac{\Delta t}{\tau_{\rm acc}}\right), \tag{3.43}$$

with $\lambda = L_{\rm disc}/L_{\rm Edd}$. Adopting $z_{\rm form} = 30$ ($t_{\rm form} = 0.099$ Gyr) and the observed epoch $z_{\rm obs} = 3.65$ ($t_{\rm obs} = 1.715$ Gyr), the available growth time is $\Delta t \simeq 1.62$ Gyr, and the number of e-folding times available at the Eddington limit is

$$N_{\rm Salp} = \frac{\Delta t}{t_{\rm Salp}} = \frac{\Delta t}{\tau_{\rm acc}}\,\frac{1-\eta}{\eta} \simeq 3.6\,\frac{1-\eta}{\eta}. \tag{3.44}$$

Table 3.2 gives the values of $N_{\rm Salp}$ and of the seed mass required at $z = 30$ for growth at the observed Eddington ratio, for the seven values of η explored in this work, computed using the MCMC posterior medians of $M_{\rm BH}$ and λ obtained in Chapter 4. The $\eta = 0.083$

row corresponds to the baseline model.

Table 3.2: Number of Salpeter e-folding times available at the Eddington limit between $z = 30$ and $z = 3.65$, and seed black hole mass required at $z = 30$ to grow to the MCMC posterior median $M_{\rm BH}$ of this work if the black hole always accreted at its observed Eddington ratio, for each accretion efficiency η ($\tau_{\rm acc} = 0.450$ Gyr). The $\eta = 0.083$ row corresponds to the baseline model. Approximate prograde spin parameters a are given for reference.

η	a (approx.)	$t_{\rm Salp}$ [Myr]	$N_{\rm Salp}$	$M_{\rm seed}$ $[M_\odot]$ ($\lambda = \lambda_{\rm obs}$)
0.057	0.00	27	59.5	2×10^7
0.070	0.31	34	47.7	4×10^7
0.083	0.51	41	39.5	6×10^7
0.100	0.67	50	32.3	1.3×10^8
0.150	0.89	79	20.4	3×10^8
0.200	0.96	112	14.4	1.6×10^8
0.300	0.996	193	8.4	1×10^9

The values in Table 3.2 illustrate a tension between early black hole formation and the observed mass of MG3 J163554+3629. If the black hole had always accreted at its present Eddington ratio, $\lambda \simeq 0.08$, the seed required at $z = 30$ would be of 2×10^7–$10^9\,M_\odot$ for every efficiency considered, one to three orders of magnitude above the heaviest seeds expected from direct collapse ($\sim 10^4$–$10^6\,M_\odot$). At the Eddington limit, on the other hand, more than 30 e-folding times are available for $\eta \lesssim 0.1$. The tension therefore arises from the low present Eddington ratio, and it implies that the black hole accreted at a much higher rate in the past. The growth histories compatible with the observed mass are discussed in Section 4.10.

3.7 Hard X-ray Diagnostics: The Role of *NuSTAR*

3.7.1 Spectral Position and Diagnostic Power

NuSTAR (Harrison et al., 2013) is the first focusing hard X-ray telescope, operating in the 3–79 keV band. For MG3 J163554+3629 at $z = 3.65$, the observed 3–79 keV band corresponds to rest-frame energies of 14–370 keV, or equivalently photon frequencies of $\nu_{\rm rest} \simeq 3.4 \times 10^{18}$–$9.0 \times 10^{19}$ Hz.

This energy range occupies a uniquely diagnostic position in the blazar SED. For the parameters of MG3 J163554+3629 the SSC component peaks close to this band, while the external Compton components are still rising towards their peaks in the MeV range. The hard X-ray photon index Γ_X is therefore sensitive to:

1. the relative normalisations of the SSC and EC components (i.e. the Compton dominance and the external photon field density),

2. the low-energy electron index p, which sets the slope of both components below their peaks, and

3. the position of the SSC peak, which depends on $\gamma_{\rm break}$, B and δ.

3.7.2 Resolving the SSC/EC Degeneracy

Without hard X-ray data, the SED of MG3 J163554+3629 in the *XMM-Newton* band (0.5–10 keV observer frame; 2–47 keV rest-frame) is degenerate between an SSC-dominated and an EC-dominated scenario. Both can reproduce the soft X-ray flux and spectral slope, as well as the *Fermi*-LAT γ-ray data, by adjusting the magnetic field strength, Doppler factor, and electron distribution parameters.

The *NuSTAR* photon index $\Gamma_X = 0.94 \pm 0.34$ (Section 4.1.2) constrains this ambiguity. A hard photon index $\Gamma_X \lesssim 1.5$ is inconsistent with the falling tail of an SSC component alone, which steepens past the SSC peak, and requires either an SSC component close to its peak or a rising EC component. The full modelling of Chapter 4 shows that both contribute (Section 4.5.4).

3.7.3 Constraints on the Electron Distribution

The *NuSTAR* photon index constrains the low-energy slope p of the electron distribution, through $\Gamma_X \simeq (p+1)/2$ below the inverse Compton peaks. For the value $p = 1.59$ adopted in the fits this gives $\Gamma_X \simeq 1.3$, consistent within about 1σ with the measured $\Gamma_X = 0.94 \pm 0.34$. The high-energy index p_1, which shapes the spectrum above $\gamma_{\rm break}$, is instead constrained by the *Fermi*-LAT data. Together, the *NuSTAR* and *Fermi*-LAT data provide complementary constraints: *NuSTAR* fixes the spectral shape at the SSC/EC transition, while *Fermi*-LAT constrains the peak and high-energy shape of the EC component. This combination constrains the relative weight of the SSC and EC components and the electron distribution far better than *XMM-Newton* data alone.

The incorporation of *NuSTAR* data represents the principal new observational contribution of this thesis. MG3 J163554+3629 had not previously been observed with a focusing hard X-ray telescope; Sánchez Zaballa et al. (2025) relied on *XMM-Newton* (0.5–10 keV) and *Fermi*-LAT data for the high-energy SED constraints. The *NuSTAR* observation (ObsID 60901025002; exposure $\sim$ 32.8 ks; 27 October 2023) and its data reduction are described in Chapter 4.

Chapter 4

Data Analysis and Modelling

4.1 Data Acquisition

After studying the characteristics of blazars, especially our target source MG3 J163554+3629, and understanding the key radiative mechanisms that shape the central engine and its interaction with the environment, we now turn to constructing the source's SED. First, we gather publicly available data across the radio to γ-ray spectrum to build a raw SED. Using the `JetSeT` framework (Tramacere et al., 2009, 2011; Tramacere, 2020) with a one-zone leptonic model, we then create an initial model, focusing on a standard Shakura–Sunyaev disk with an accretion efficiency $\eta = 0.083$ ($= 1/12$). Finally, we explore various accretion efficiency values to refine and relate the key parameters for all models, considering the potential types of black holes formed at high redshifts.

4.1.1 Low-Energy Observations

The low-energy observations for the flat-spectrum radio quasar MG3 J163554+3629 were obtained using several key telescopes across multiple wavelengths. The NRAO Green Bank Telescope and the Very Large Array (VLA) provided valuable radio frequency data, aiding in the detection and localization of emissions across the radio-IR range. The Wide-field Infrared Survey Explorer (WISE) contributed infrared data, detecting the source in the W1 and W2 bands with significant measurements, while providing upper limits for the W3 and W4 bands. Gaia, a mission primarily focused on astrometry, also contributed high-precision photometry in the optical range, helping to confirm the presence of the source in the optical spectrum. Additionally, the Swift UV–Optical Telescope (UVOT) observed the source in the optical and ultraviolet bands, providing key data for modelling the blazar's emission, particularly in the UV–optical range.

4.1.2 High-Energy Observations

The high-energy observations for MG3 J163554+3629 were obtained from three major telescopes: XMM-Newton, the Fermi Large Area Telescope (LAT), and NuSTAR (Nuclear Spectroscopic Telescope Array). XMM-Newton, with its high sensitivity in the X-ray band, provided key data from the source, allowing detailed spectral analysis in the 0.5–10 keV range. The Fermi LAT, which observes the sky in the GeV energy range, contributed gamma-ray data, aiding in the characterization of the high-energy tail of the blazar's emission. In addition to these, NuSTAR data was incorporated into the study, providing important information in the hard X-ray band (3–79 keV) (Harrison et al., 2013).

Incorporating the NuSTAR data

The broadband SED of MG3 J163554+3629 discussed in previous work (Sánchez Zaballa et al., 2025) did not include hard X-ray measurements above the soft X-ray band. The inclusion of *NuSTAR* data therefore represents a substantial improvement, as it provides direct spectral constraints in the 3–79 keV range. This energy range is particularly important because it probes the high-energy tail of the X-ray continuum, close to the transition region between the synchrotron self-Compton (SSC) component and the emerging external Compton (EC) contribution. As a result, the inclusion of *NuSTAR* data can significantly modify the shape of the high-energy SED and the interpretation of the emission.

NuSTAR (Harrison et al., 2013) is the first satellite mission employing focusing optics in the hard X-ray band, operating up to ∼79 keV. This technological breakthrough allows hard X-rays to be focused onto the detectors rather than collimated, leading to a dramatic reduction in background and a corresponding increase in sensitivity and spectral accuracy compared to previous non-focusing instruments. This capability enables reliable imaging and spectral analysis of faint hard X-ray emission, making *NuSTAR* uniquely suited for constraining the high-energy emission processes in blazars.

NuSTAR data reduction and detector selection

The NuSTAR observation of MG3 J163554+3629 (ObsID 60901025002; exposure ∼ 32.8 ks; 27 Oct 2023) was processed using the `HEASOFT` software package and `NUSTARDAS` v1.9.6. The event files were reprocessed with `NUPIPELINE` using the most recent calibration files available in the NuSTAR CALDB.

NuSTAR consists of two focal plane modules, FPMA and FPMB, which observe the source simultaneously. Figure 4.1 shows the convolved spectra from both detectors. While one module exhibits a clean spectral shape, the other shows significant degradation consistent with enhanced background contamination, possibly of proton origin. Based on

this comparison, we exclude the lower-quality module from the spectral analysis and proceed using only the FPMB data, which provide the most reliable spectral information for this observation.

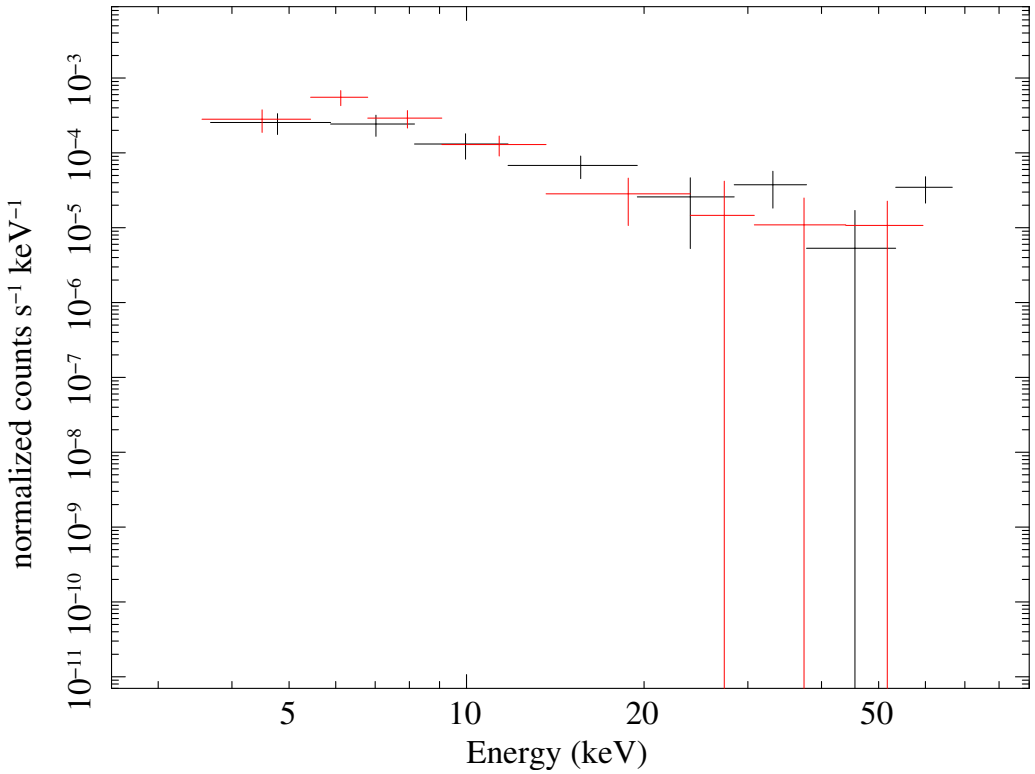


Figure 4.1: Convolved NuSTAR spectra from FPMA and FPMB for MG3 J163554+3629. FPMA is strongly affected by background contamination, while the FPMB spectrum shows higher quality and is therefore adopted for the spectral analysis.

Spectral modelling and statistical tests

The unfolded NuSTAR FPMB spectrum is shown in Figure 4.2. Spectral fitting was performed in XSPEC using an absorbed power-law model, `wabs*powerlaw`, with the Galactic absorption fixed at $n_{\mathrm{H}} = 1.8 \times 10^{-2} \times 10^{22}\,\mathrm{cm}^{-2}$.

Model 1: `wabs*powerlaw`. The best-fit parameters are:

- Photon index: $\Gamma = 0.94 \pm 0.34$,
- Normalization: $(5.73 \pm 4.68) \times 10^{-6}$,
- $\chi^2 = 8.51$ for 6 degrees of freedom,
- Null-hypothesis probability $P \simeq 0.20$.

This result indicates that a single absorbed power law provides an acceptable description of the NuSTAR spectrum over the full energy range.

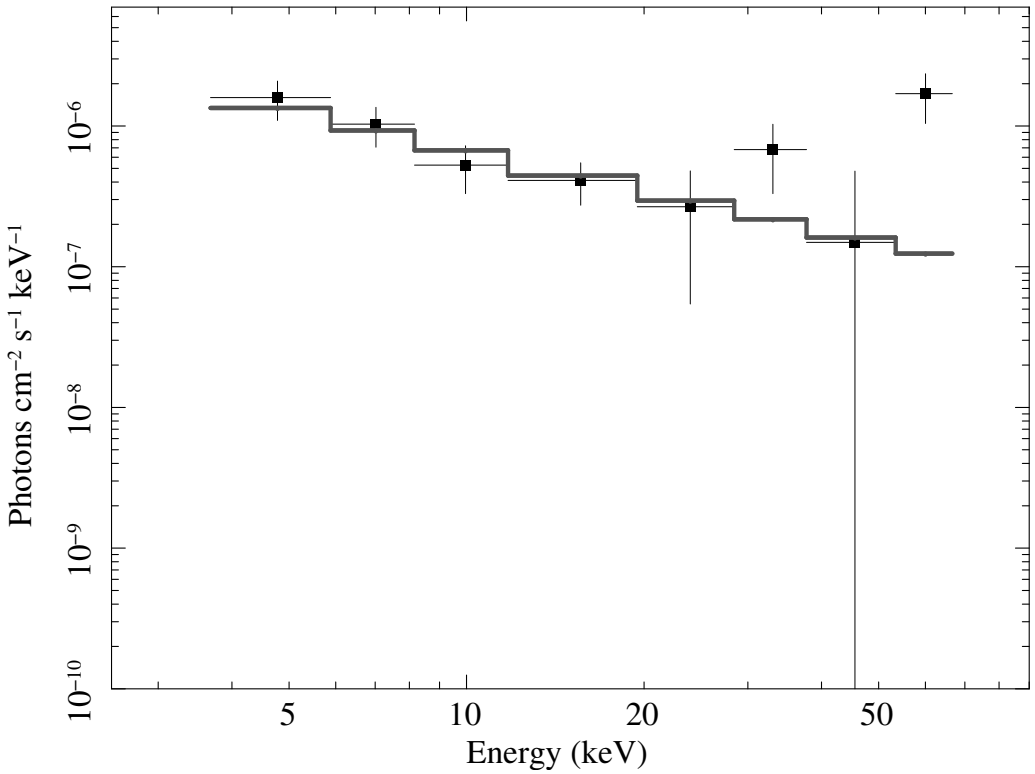


Figure 4.2: Unfolded NuSTAR FPMB spectrum of MG3 J163554+3629.

Model 2: `wabs*bknpower`**.** To test for possible spectral curvature at higher energies, we also fitted an absorbed broken power-law model. The resulting parameters are:

- $\Gamma_1 = 1.24 \pm 0.39$,
- $E_{\rm break} = 28.36 \pm 12.06$ keV,
- $\Gamma_2 = -2.50 \pm 2.41$,
- $\chi^2 = 4.20$ for 4 degrees of freedom,
- Null-hypothesis probability $P \simeq 0.38$.

Although the broken power-law model yields a lower value of χ^2, an F-test comparing the two nested models gives $F = 2.05$ with an associated probability $P = 0.24$. This result indicates that the reduction in χ^2 is not statistically significant when the increase in the number of free parameters is taken into account. We therefore adopt the simpler absorbed power-law model as the preferred description of the NuSTAR spectrum.

χ^2 profiles and parameter constraints

To further assess the robustness of the spectral fit, we computed χ^2 profiles for the photon index and normalization parameters and their joint confidence contours. Figures 4.3 and 4.4 show the variation of χ^2 as a function of each parameter individually, demonstrating that both parameters exhibit a well-defined minimum. The horizontal lines mark $\Delta\chi^2 = 2.71$, the 90% confidence level for one parameter, confirming that the best-fit values are statistically well constrained.

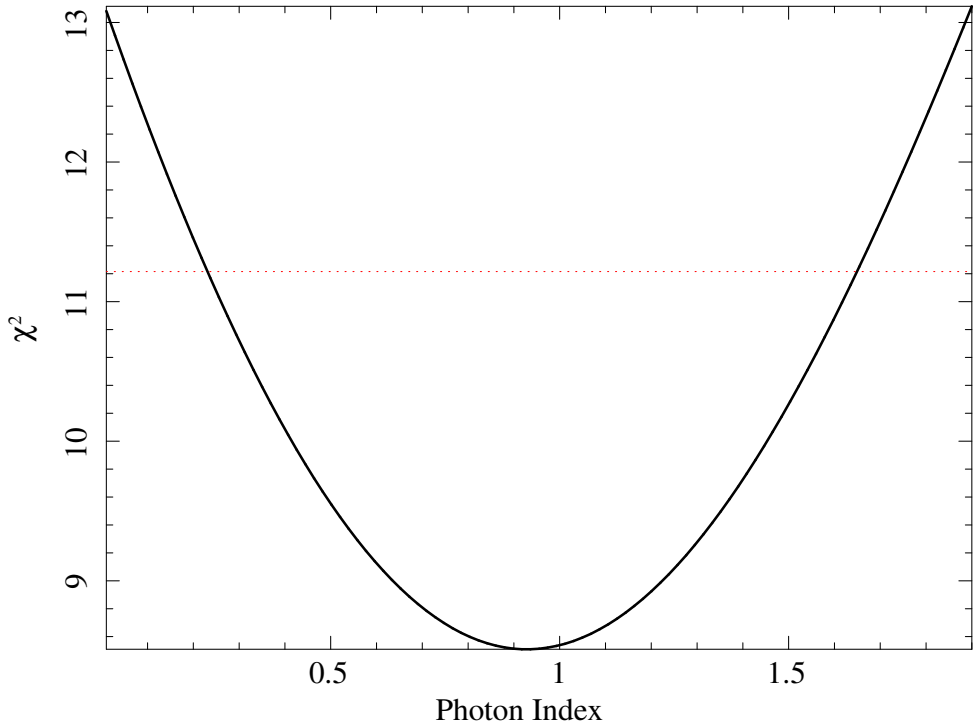


Figure 4.3: χ^2 profile for the photon index Γ, showing a well-defined minimum. The horizontal line marks the 90% confidence level ($\Delta\chi^2 = 2.71$).

4.2 JetSeT Overview

`JetSeT` (Tramacere et al., 2009, 2011; Tramacere, 2020) is a Python package designed for modelling the radiative emission of relativistic jets in active galactic nuclei (AGN) and gamma-ray bursts (GRBs). The framework implements a one-zone leptonic model where a homogeneous, spherical emission region (commonly referred to as the "blob") containing a relativistic electron distribution moves down the jet. This approach has been successfully applied to a wide range of blazar observations across the electromagnetic spectrum.

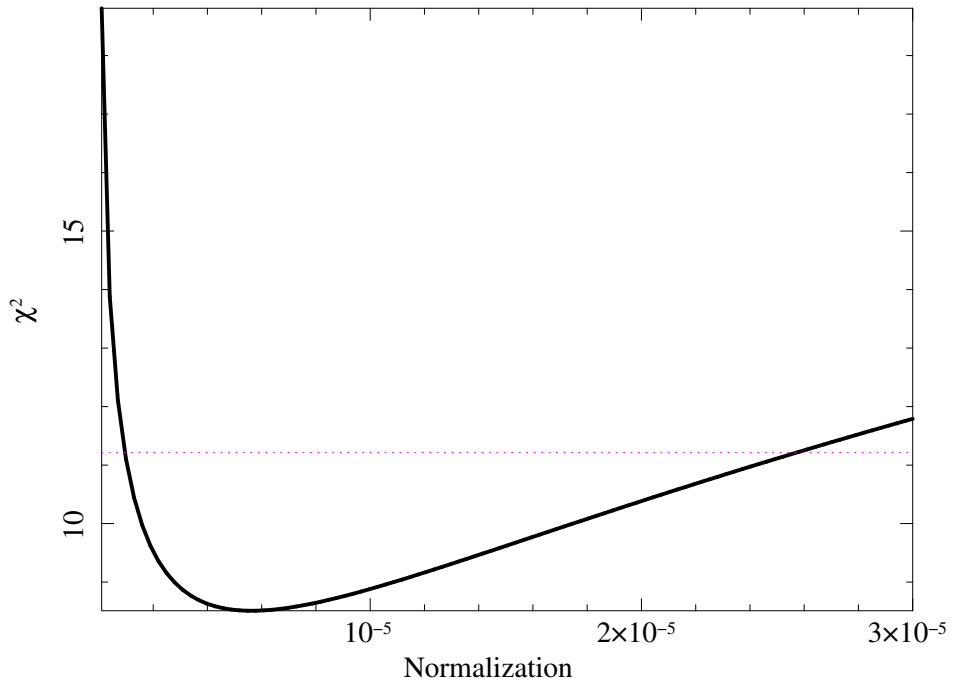


Figure 4.4: χ^2 profile for the power-law normalization, indicating a stable minimum. The horizontal line marks the 90% confidence level ($\Delta\chi^2 = 2.71$).

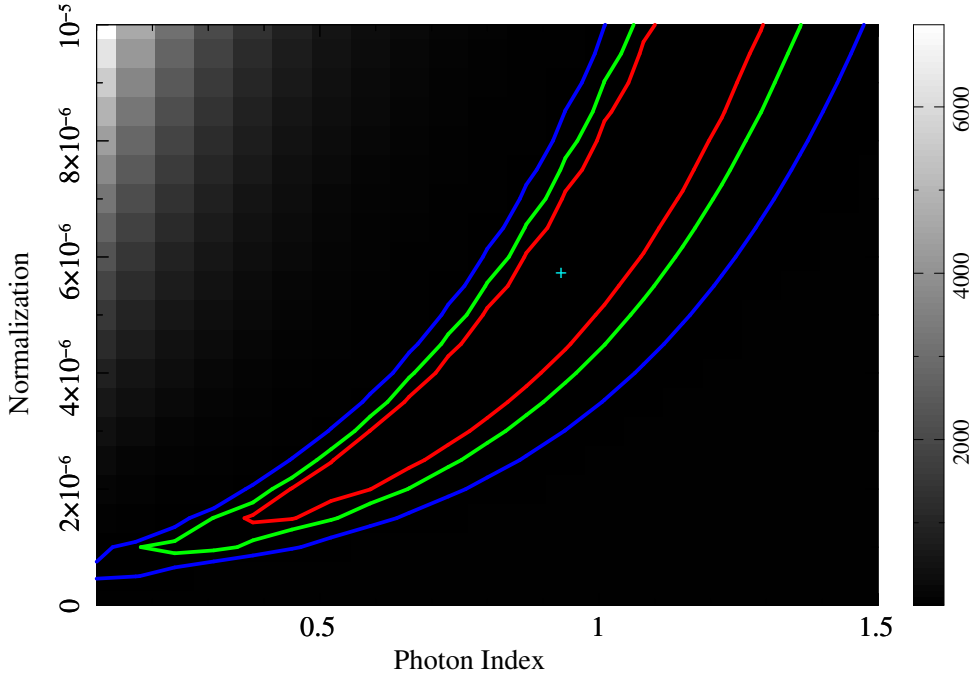


Figure 4.5: Joint confidence contours for photon index Γ and normalization, showing a strong correlation between the two parameters.

4.2.1 Key Features of JetSeT

The main features of the `JetSeT` framework relevant to this work include:

- **Self-consistent radiative calculation:** `JetSeT` computes synchrotron, synchrotron self-Compton (SSC), and external Compton (EC) emission from a single population of relativistic electrons.
- **Flexible electron distribution parametrization:** The code supports multiple electron energy distribution (EED) shapes, including power-law, broken power-law, and log-parabolic distributions (Massaro et al., 2004; Tramacere et al., 2011).
- **External photon fields:** `JetSeT` can include external radiation fields from the accretion disk (Shakura-Sunyaev model), broad-line region (BLR), and dusty torus (DT).
- **Relativistic effects:** The framework properly handles Doppler boosting, relativistic transformations, and cosmological redshifting (Urry & Padovani, 1995).
- **Fitting capabilities:** `JetSeT` includes MCMC (Markov Chain Monte Carlo) fitting tools for constraining model parameters using multi-wavelength observations.

4.2.2 The One-Zone Leptonic Model

In the one-zone leptonic model implemented in `JetSeT`, the emission region is characterised by the following physical parameters:

Table 4.1: Key parameters in the `JetSeT` one-zone leptonic model

Parameter	Description	Typical Range
R	Radius of the emission region (blob)	$10^{14} - 10^{17}$ cm
B	Magnetic field strength	$0.01 - 10$ G
δ	Doppler factor (relativistic beaming)	$5 - 50$
N	Electron density normalization	$10^2 - 10^5$ cm^{-3}
$\gamma_{\rm min}, \gamma_{\rm max}$	Minimum/maximum electron Lorentz factor	$10^2 - 10^8$
z	Redshift	Variable

4.2.3 The Log-Parabolic Electron Distribution

As described in Section 3.1, two electron energy distributions are used in this work for distinct purposes. The following description of the log-parabolic EED applies exclusively to the parameter exploration in Section 4.3. The actual SED fitting of MG3 J163554+3629 uses the broken power-law EED (Section 3.2.2).

For the parameter exploration, we adopt a log-parabolic EED of the form (see also Equation 3.3):

$$n(\gamma) = N_0 \left(\frac{\gamma}{\gamma_0} \right)^{-(s + r \log_{10}(\gamma/\gamma_0))} \tag{4.1}$$

where:

- γ is the electron Lorentz factor;
- γ_0 is the peak Lorentz factor (turn-over energy);
- s is the spectral index at γ_0 (low-energy slope);
- r is the curvature parameter (controls the width of the distribution);
- N_0 is the normalization constant.

This parametrization provides a flexible description of the electron population while requiring fewer parameters than a broken power-law, making it particularly suitable for exploring parameter sensitivities because its smooth, continuously curved shape allows systematic variation of each physical parameter in isolation without a discrete spectral break (Massaro et al., 2004, 2006).

4.2.4 Radiative Processes in JetSeT

`JetSeT` computes the following radiative components:

Synchrotron Emission

Relativistic electrons gyrating in the magnetic field emit synchrotron radiation (Rybicki & Lightman, 1979). The characteristic frequency is:

$$\nu_{\rm sync} \approx \frac{3}{4\pi} \frac{eB}{m_e c} \gamma^2 \tag{4.2}$$

`JetSeT` handles synchrotron self-absorption (SSA) at low frequencies, which becomes important for compact emission regions.

Synchrotron Self-Compton (SSC) Emission

The same population of relativistic electrons upscatters the synchrotron photons to higher energies via inverse Compton scattering (Blumenthal & Gould, 1970). In the Thomson regime, the characteristic SSC frequency is:

$$\nu_{\rm SSC} \approx \frac{4}{3} \gamma^2 \nu_{\rm sync} \tag{4.3}$$

JetSeT correctly handles the transition from the Thomson to the Klein-Nishina (KN) regime, where the Compton cross section decreases at high energies (Blumenthal & Gould, 1970).

External Compton (EC) Emission

When external photon fields (e.g., from the accretion disk, BLR, or DT) are present, JetSeT can compute EC emission from the upscattering of these seed photons (Ghisellini & Tavecchio, 2009).

4.2.5 The Inverse Compton Redistribution Function

A key feature of JetSeT is its accurate treatment of the inverse Compton redistribution function F_c (Blumenthal & Gould, 1970), which determines how electron energy maps to scattered photon energy. In the Thomson regime ($\Gamma \ll 1$), F_c approximates a δ-function, while in the Klein-Nishina regime ($\Gamma \gg 1$), it becomes broad and flat. This transition is critical for understanding the high-energy emission from blazars (Tramacere et al., 2011).

4.2.6 Validation and Parameter Studies

The JetSeT documentation provides extensive validation of the code against analytical predictions. Key theoretical relations that our parameter study will verify include:

1. The synchrotron spectral index $\alpha_{\rm sync}$ relates to the electron spectral index s as $\alpha_{\rm sync} = (s+1)/2$ in the optically thin regime (Blumenthal & Gould, 1970).

2. The SSC peak frequency relates to the synchrotron peak frequency and electron peak Lorentz factor as $\nu_{\rm SSC} \approx (4/3)\gamma_0^2 \nu_{\rm sync}$ in the Thomson regime (Tramacere et al., 2011).

3. The curvature of the SSC component is related to that of the electron distribution, and in the Thomson regime it is smaller than the curvature of the synchrotron component (Massaro et al., 2006).

4.3 Parameter Exploration with JetSeT

Before applying JetSeT to the multi-wavelength observations of MG3 J163554+3629, we perform a systematic exploration of the model parameter space using synthetic data. This exercise serves three purposes: (i) to validate our understanding of the radiative processes implemented in JetSeT, (ii) to identify degeneracies and sensitivities among key physical parameters, and (iii) to establish physically motivated priors and parameter ranges for the subsequent fitting of the real source.

4.3.1 Baseline Model Setup

Table 4.2 lists the baseline parameters adopted for this study, following the setup described in the `JetSeT` documentation.

Table 4.2: Baseline parameters for the `JetSeT` parameter study

Parameter	Symbol	Baseline Value
Magnetic field	B	0.2 G
Peak Lorentz factor	γ_0	5×10^3
Minimum Lorentz factor	$\gamma_{\min}$	10^2
Maximum Lorentz factor	$\gamma_{\max}$	10^8
Blob radius	R	10^{15} cm
Electron density	N	10^3 cm^{-3}
Curvature parameter	r	0.4
Redshift	z	3.65

Figure 4.6 shows the baseline electron energy distribution. The distribution peaks at $\gamma_0 = 5 \times 10^3$ and spans from $\gamma_{\min} = 10^2$ to $\gamma_{\max} = 10^8$, covering the electron energies responsible for synchrotron emission from radio to X-ray bands.

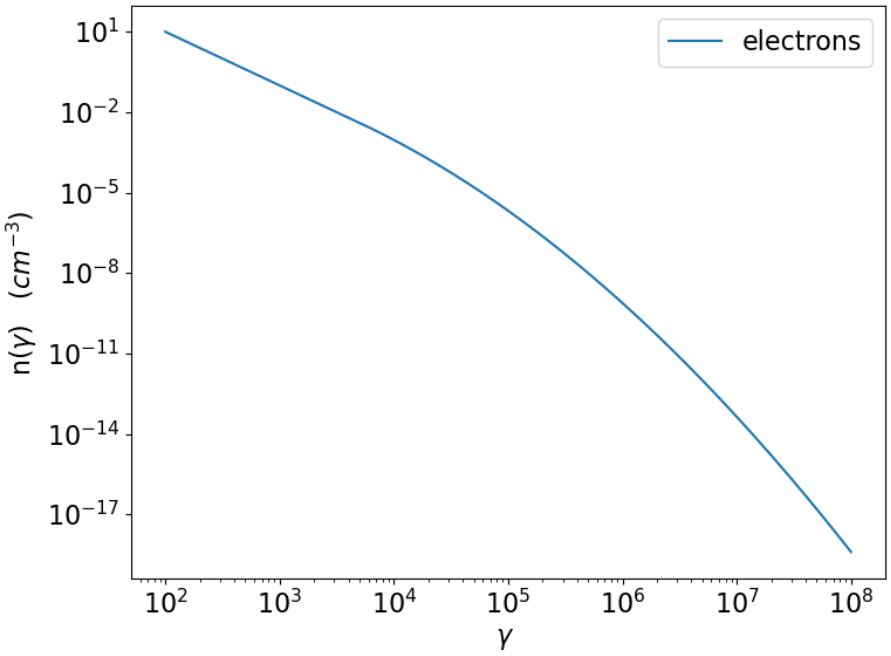


Figure 4.6: Baseline electron energy distribution for the `JetSeT` parameter study, using a log-parabolic model with $\gamma_0 = 5 \times 10^3$, $s = 2.0$, $r = 0.4$. The distribution spans $\gamma_{\min} = 10^2$ to $\gamma_{\max} = 10^8$.

4.3.2 Effect of the Minimum Electron Lorentz Factor ($\gamma_{\min}$)

Figure 4.7 shows the synchrotron-only SED evolution when $\gamma_{\min}$ is varied from ~ 1.3 to $\sim 3.2 \times 10^4$. All curves are normalized to produce $\nu F_\nu = 10^{-12}$ erg cm^{-2} s^{-1} at

$\nu = 10^{18}$ Hz (X-rays, $\simeq 4$ keV).

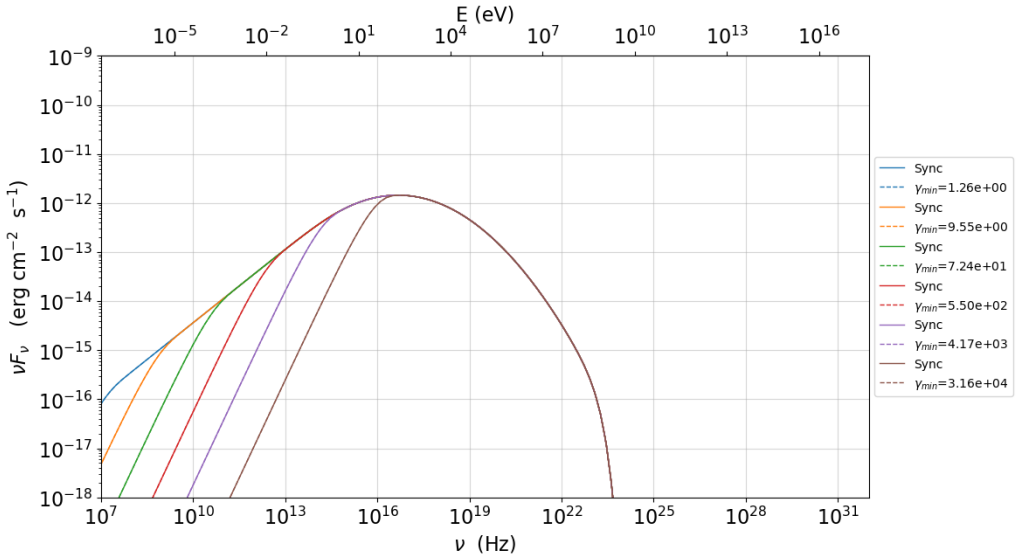


Figure 4.7: Synchrotron-only SED for different values of the minimum electron Lorentz factor $\gamma_{\rm min}$ (1.3 to 3.2 $\times$ 10^4). All models are normalized at $\nu = 10^{18}$ Hz ($\simeq$ 4 keV). Lower $\gamma_{\rm min}$ produces significantly brighter radio emission due to the increased population of low-energy electrons. Above the normalization frequency, all curves converge.

The low-frequency emission shows strong sensitivity to $\gamma_{\rm min}$, with lower values producing flatter radio spectra. This behaviour is expected from synchrotron theory: electrons with lower Lorentz factors radiate at lower frequencies ($\nu \propto \gamma^2 B$). The forced crossing at 10^{18} Hz reveals a degeneracy between $\gamma_{\rm min}$ and the electron normalization N_0. For MG3 J163554+3629, the radio data will constrain $\gamma_{\rm min}$, with typical values for FSRQs in the range $\gamma_{\rm min} \sim 10^2$–$10^3$.

As demonstrated in the `JetSeT` documentation (Tramacere et al., 2011), the synchrotron spectral index measured at low frequencies approaches the asymptotic value of 1/3 for very low $\gamma_{\rm min}$, consistent with the canonical synchrotron theory prediction for a power-law electron distribution (Blumenthal & Gould, 1970).

4.3.3 Effect of the Low-Energy Spectral Index (s)

Figure 4.8 shows the SED evolution as the spectral index s is varied from 1.5 to 2.5. The normalization is fixed at $\nu = 5 \times 10^{13}$ Hz (mid-IR) with $\nu F_\nu = 10^{-11}$ erg cm^{-2} s^{-1}.

The radio-to-IR slope steepens significantly with increasing s, consistent with synchrotron theory where the observed photon index α relates to the particle index as $\alpha = (s + 1)/2$ in the optically thin regime (Blumenthal & Gould, 1970). Crucially, this effect propagates to the high-energy component, as the same electron population is responsible for both synchrotron and inverse Compton emission. This parameter is well-constrained by broadband data and shows no strong degeneracies with other parameters.

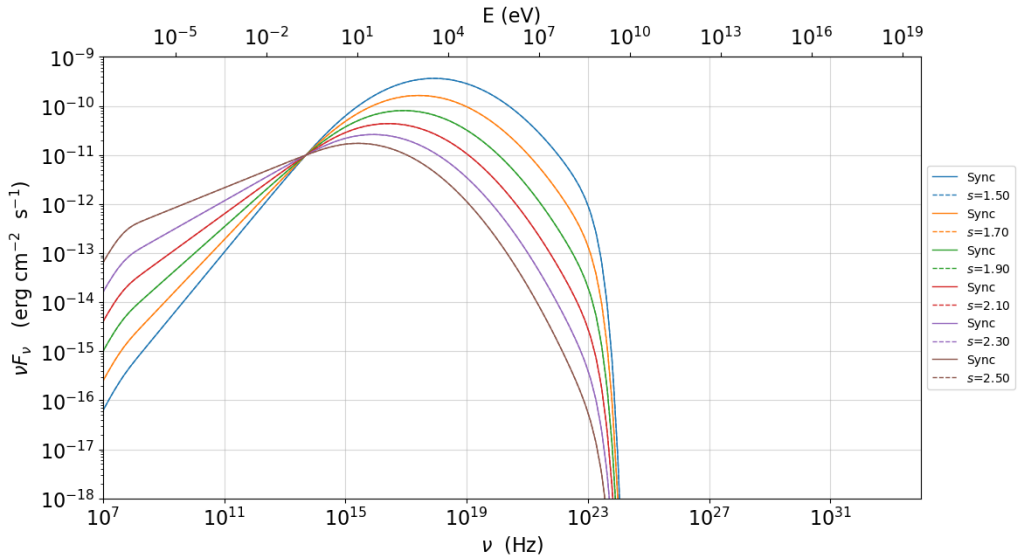


Figure 4.8: SED variation with the low-energy spectral index s (1.5 to 2.5). All models are normalized at $\nu = 5 \times 10^{13}$ Hz (mid-IR). The radio-to-IR spectral slope steepens with increasing s, consistent with $\alpha = (s+1)/2$ in the optically thin regime. The effect propagates to the high-energy component because the same electron population produces both synchrotron and SSC emission.

4.3.4 Effect of the Peak Lorentz Factor (γ_0)

The parameter γ_0 controls the energy at which the electron distribution peaks, and consequently determines the frequency of the synchrotron peak. Figure 4.9 shows the SED evolution as γ_0 is varied from 10^2 to 10^6.

This parameter is the *master control* for blazar classification (Fossati et al., 1998; Ghisellini et al., 1998):

- **Low γ_0 (10^2–10^3):** Synchrotron peak in the IR/optical band, characteristic of low-synchrotron-peaked (LSP) blazars and many FSRQs, including MG3 J163554+3629.
- **Medium γ_0 (10^4):** Synchrotron peak in the UV, typical of intermediate-synchrotron-peaked (ISP) sources.
- **High γ_0 (10^5–10^6):** Synchrotron peak in the X-ray band, characteristic of high-synchrotron-peaked (HSP) BL Lacs.

A critical effect is observed for $\gamma_0 \gtrsim 10^5$: the SSC component begins to suppress due to Klein-Nishina (KN) effects (Blumenthal & Gould, 1970). When the upscattered photon energy in the electron rest frame approaches or exceeds the electron rest mass energy ($h\nu' \gtrsim m_e c^2$), the Compton cross section decreases dramatically. This KN suppression has important implications for high-redshift blazars: if γ_0 is too high, the γ-ray emission may be dominated by external Compton rather than SSC processes. For MG3 J163554+3629,

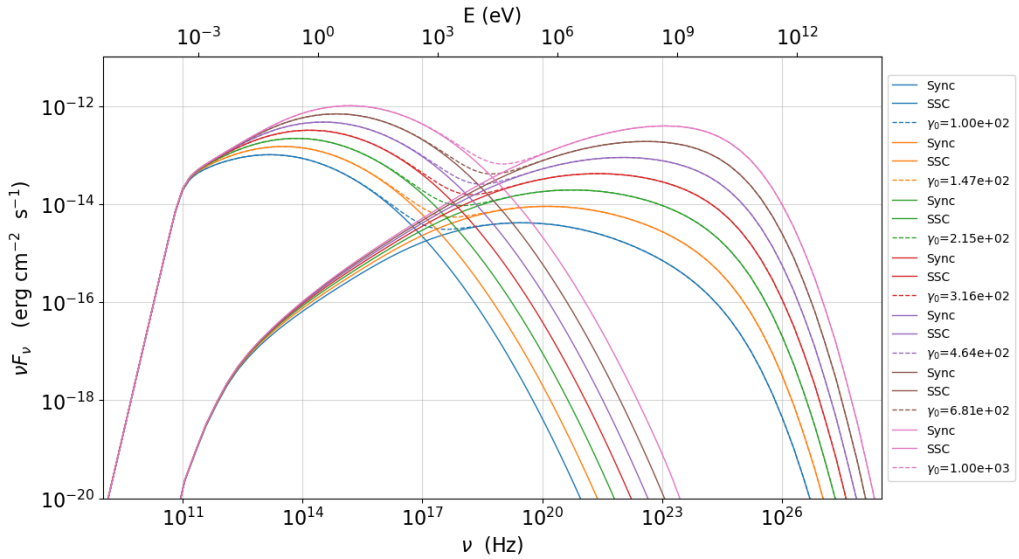


Figure 4.9: Variation of the peak Lorentz factor γ_0 (10^2 to 10^6). Low γ_0 (10^2–10^3) produces synchrotron peaks in the IR/optical; medium γ_0 (10^4) peaks in the UV; high γ_0 (10^5–10^6) peaks in X-rays. For $\gamma_0 \gtrsim 10^5$, Klein-Nishina effects suppress the SSC component.

the low value of $\gamma_0 \sim$ a few $\times 10^3$ places the source firmly in the LSP/FSRQ regime where EC dominates, consistent with the analysis in Section 3.3.3.

4.3.5 The Inverse Compton Redistribution Function

To understand the transition from Thomson to Klein-Nishina regimes, we examine the inverse Compton redistribution function F_c (Blumenthal & Gould, 1970), which determines how electron energy maps to scattered photon energy. Figure 4.10 shows F_c as a function of $\nu_{\rm out}/\nu_{\rm out,max}$ for different values of $\Gamma = 4\gamma h\nu_{\rm in}/(m_e c^2)$.

The behaviour can be classified into three regimes:

1. **Thomson regime ($\Gamma \ll 1$):** The Compton cross section is constant (σ_T). Each electron upscatters photons to a characteristic energy $\nu_{\rm out} \approx 4\gamma^2 \nu_{\rm in}$, with very little spread.

2. **Transition regime ($\Gamma \sim 0.1$–1):** Quantum effects become important, the cross section begins to decrease, and the distribution broadens.

3. **Klein-Nishina regime ($\Gamma \gg 1$):** The cross section scales as $\sigma \propto 1/\Gamma$ (decreasing with energy). The redistribution function becomes very broad, producing a flat νF_ν spectrum over a wide range of output frequencies.

For high-redshift blazars like MG3 J163554+3629, understanding this transition is crucial. If the γ-ray emission peaks at energies $\gtrsim 10^{25}$ Hz ($\sim$ 100 GeV in the source frame), KN effects may significantly modify the expected SSC spectrum.

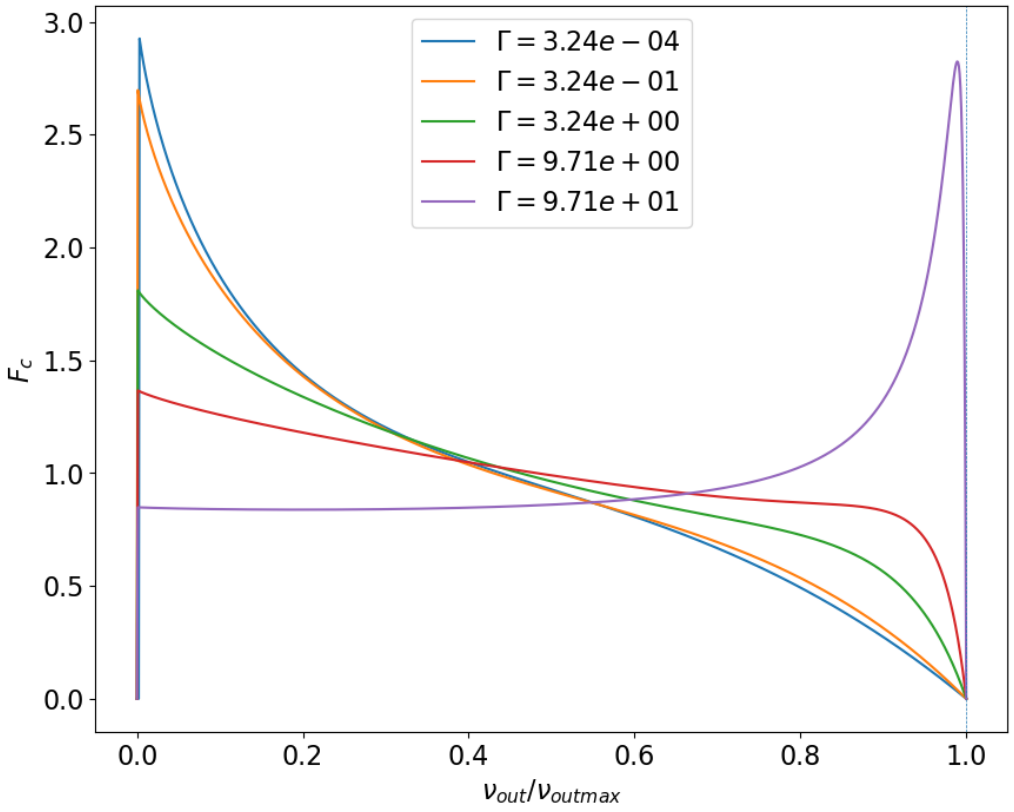


Figure 4.10: The inverse Compton redistribution function F_c for different values of $\Gamma = 4\gamma h\nu_{\rm in}/(m_e c^2)$. In the Thomson regime ($\Gamma \ll 1$), F_c approximates a δ-function, producing nearly mono-energetic scattered photons. As Γ increases into the Klein-Nishina regime ($\Gamma \gtrsim 1$), F_c broadens dramatically, producing a flat distribution of output frequencies. This transition fundamentally shapes the high-energy SED of blazars. This figure reproduces the classic results of Blumenthal & Gould (1970).

4.3.6 Effect of Curvature Parameter (r)

The curvature parameter r controls the width of the log-parabolic electron distribution (Massaro et al., 2004). Figure 4.11 shows the electron distributions for r varying from 0.5 (broad) to 2.0 (narrow), while Figure 4.12 shows the corresponding SEDs.

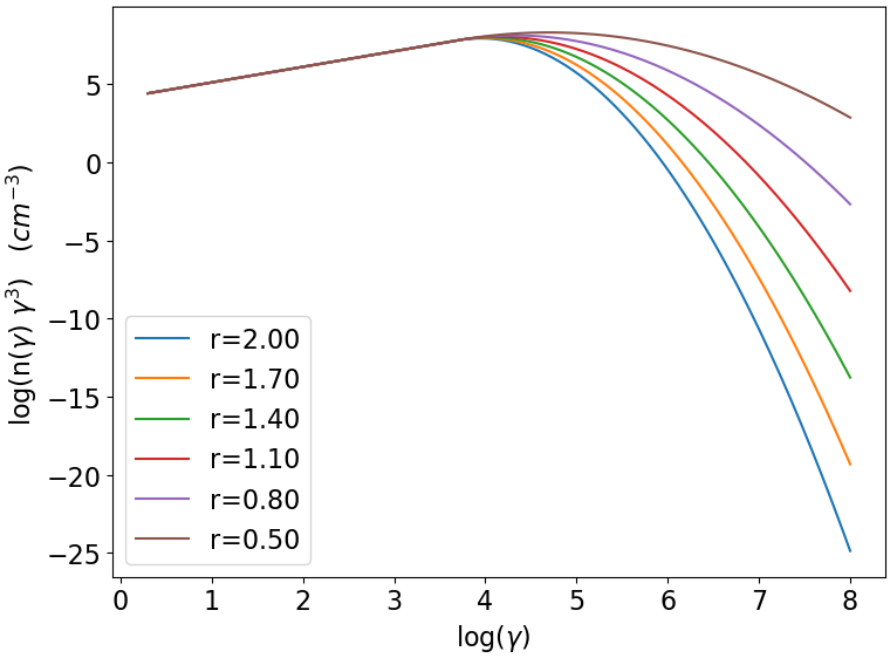


Figure 4.11: Electron energy distributions for different values of the curvature parameter r (0.5 to 2.0). The quantity $n(\gamma)\gamma^3$ is plotted to highlight the shape of the distribution. The curvature acts above the reference Lorentz factor: as r decreases, the high-energy part of the distribution broadens.

Several important observations emerge:

- **Synchrotron component:** Lower r (broader electron distribution) populates the high-energy tail with more electrons and produces a higher and broader synchrotron peak.
- **SSC component:** The SSC component changes more than the synchrotron one. This occurs because the SSC emission depends on the electron distribution twice: through the synchrotron seed photons and through the scattering electrons.
- **Peak position:** Both peak frequencies move to higher values as r decreases, since the electrons responsible for the peaks have higher energies.

This has a practical implication for fitting: the inverse Compton component also carries information on the shape of the electron distribution, so that well-sampled γ-ray data (such as from *Fermi*-LAT) can help constrain r when the synchrotron component is poorly sampled.

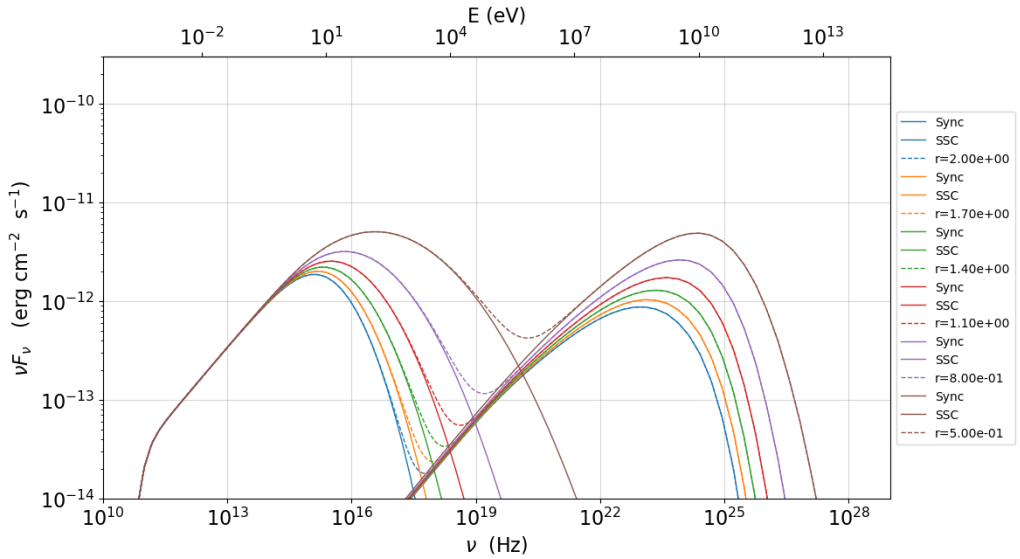


Figure 4.12: SED variation with the curvature parameter r (0.5 to 2.0). The left peak is synchrotron emission; the right peak is SSC. As r decreases (broader distribution), both peaks become higher and move to higher frequencies, and the SSC component changes more than the synchrotron one.

4.3.7 Summary of Parameter Degeneracies

Table 4.3 summarises the key parameter effects and degeneracies identified in this study.

Table 4.3: Summary of parameter effects and degeneracies

Parameter	Primary Effect	Secondary Effect	Degeneracy
$\gamma_{\rm min}$	Low-frequency cutoff	None strong	With N_0
s	Synchrotron slope	SSC slope	None strong
γ_0	Peak frequency	KN suppression	With B, δ
r	Peak width (Sync)	Peak width (SSC)	None strong

These degeneracies inform our fitting strategy for MG3 J163554+3629. The $\gamma_{\rm min}$ and s parameters are primarily constrained by the radio-to-IR spectrum. The peak Lorentz factor γ_0 determines the blazar subclass and influences the KN suppression of the SSC component. The curvature parameter r can be constrained by both the synchrotron and the inverse Compton components.

4.3.8 Implications for MG3 J163554+3629

The parameter study provides several key insights for modelling the real source:

1. The **radio-to-IR slope** would constrain s and $\gamma_{\rm min}$; in MG3 J163554+3629, however, the radio emission is dominated by extended regions of the jet, and these parameters are fixed in the fits.

2. The **synchrotron peak frequency** determines γ_0 and whether the source is LSP, ISP, or HSP. In MG3 J163554+3629 the peak falls in the infrared, where only upper limits are available, while the optical/UV emission is dominated by the accretion disk; the source is classified as an LSP FSRQ (Sánchez Zaballa et al., 2025).

3. If $\gamma_0 > 10^5$, KN suppression may affect the γ-ray spectrum, potentially favouring an external Compton interpretation.

4. The **curvature of the γ-ray spectrum** can help constrain r (Massaro et al., 2006; Tramacere et al., 2011).

In the following sections, we apply these insights to fit the actual observations of MG3 J163554+3629 using the broken power-law EED described in Section 3.2.2.

4.4 Broadband SED and Model Set-up

In this section we describe how the broadband SED of MG3 J163554+3629 was assembled for the fit and how the starting model was built. All the modelling was carried out with `JetSeT` version 1.3.0 (Tramacere, 2020), using the broken power-law EED of Section 3.2.2, external Compton emission from the accretion disk, the BLR and the dusty torus, and the `MultiBB` description of the accretion disk. The same set-up, the same data and the same treatment of the external photon fields are used for every fit presented in the rest of this chapter.

4.4.1 The SED Dataset

The SED contains 46 points, from radio frequencies to the highest-energy *Fermi*-LAT bins. Of these, 40 are detections and 6 are upper limits: the WISE W4 and W3 bands (1.318×10^{13} and 2.417×10^{13} Hz), the lowest-energy *Fermi*-LAT bin (1.709×10^{22} Hz) and the three highest-energy *Fermi*-LAT bins (4.188×10^{24}, 1.324×10^{25} and 7.646×10^{25} Hz). The upper limits were flagged by matching their frequencies with a relative tolerance of 10^{-3}, which avoids any confusion between bins with similar mantissas (for example 4.188×10^{22} and 4.188×10^{24} Hz).

Since the data were not collected simultaneously, a fractional systematic uncertainty of 15% was added in quadrature to all points between 10^6 and 10^{29} Hz. The data were not rebinned, so that every measurement, and in particular every *NuSTAR* flux point, enters the likelihood individually. Eleven points fall in the *NuSTAR* band (3–79 keV).

The fit range was set to 10^{10}–10^{29} Hz. The eight radio points below 10^{10} Hz are shown in all figures but are not fitted: the low-frequency radio emission of a powerful FSRQ is dominated by extended, self-absorbed regions of the jet that a single compact emission

zone cannot reproduce. As a consequence, the synchrotron self-absorption frequency and $\gamma_{\rm min}$ are not constrained by the fit, and $\gamma_{\rm min}$ is kept fixed (Section 4.5.1). The fit range contains 38 points, 32 detections and 6 upper limits. Upper limits are included in the likelihood through the `use_UL` option of the `JetSeT` minimiser. The resulting dataset is shown in Figure 4.13.

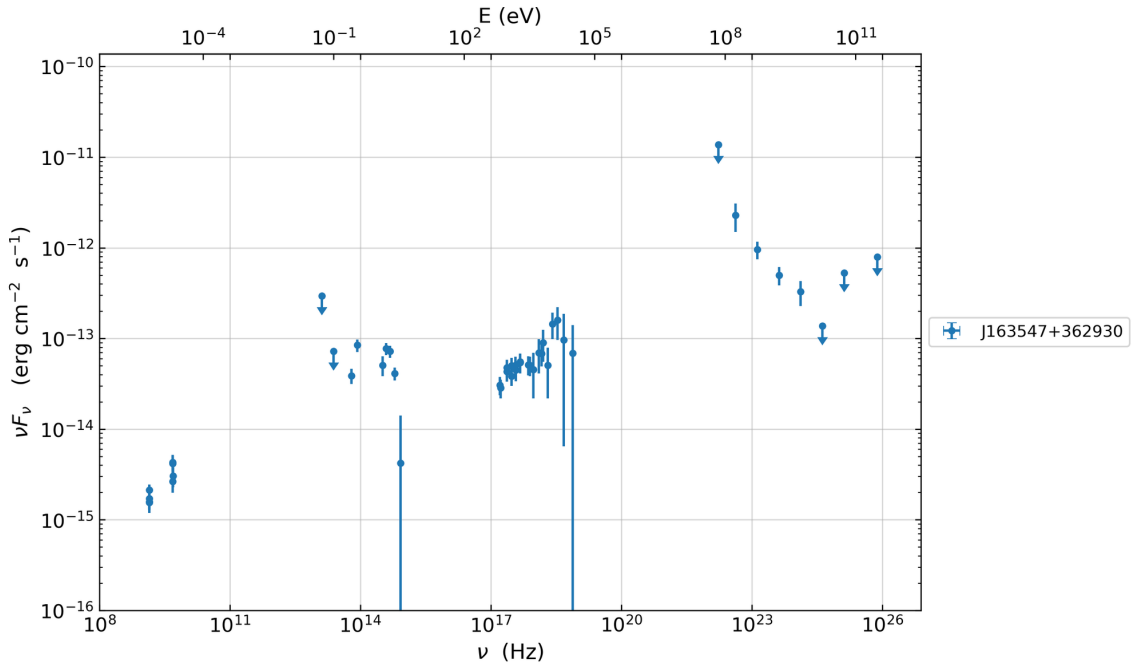


Figure 4.13: Broadband SED of MG3 J163554+3629 used in this work, after adding the 15% systematic uncertainty. Downward arrows mark the six upper limits. Points below 10^{10} Hz are not included in the fit.

4.4.2 Phenomenological Constraints

Before any physical model is fitted, the shape of the SED is characterised with the `SEDShape` tool of `JetSeT`. The low-energy hump is fitted between 10^{9} and 10^{16} Hz with a log-parabola plus a template for the big blue bump of the accretion disk, and the high-energy hump between 10^{16} and 10^{26} Hz with a log-cubic function. The fits give

$$\log \nu_{\rm s}^{\rm peak} = 13.31 \pm 0.57, \qquad \log(\nu F_\nu)_{\rm s}^{\rm peak} = -13.33 \pm 0.07,$$
$$\log \nu_{\rm IC}^{\rm peak} = 22.03 \pm 0.11, \qquad \log(\nu F_\nu)_{\rm IC}^{\rm peak} = -11.77 \pm 0.10,$$

with frequencies in Hz and fluxes in erg cm^{-2} s^{-1}. The source is classified as a low-synchrotron-peaked object, and the ratio of the two peak fluxes gives a Compton dominance of $A_C \approx 36$. The large uncertainty on the synchrotron peak frequency reflects the fact that the peak falls in a region with only WISE upper limits, while the optical and UV data are dominated by the accretion disk (Figure 4.14).

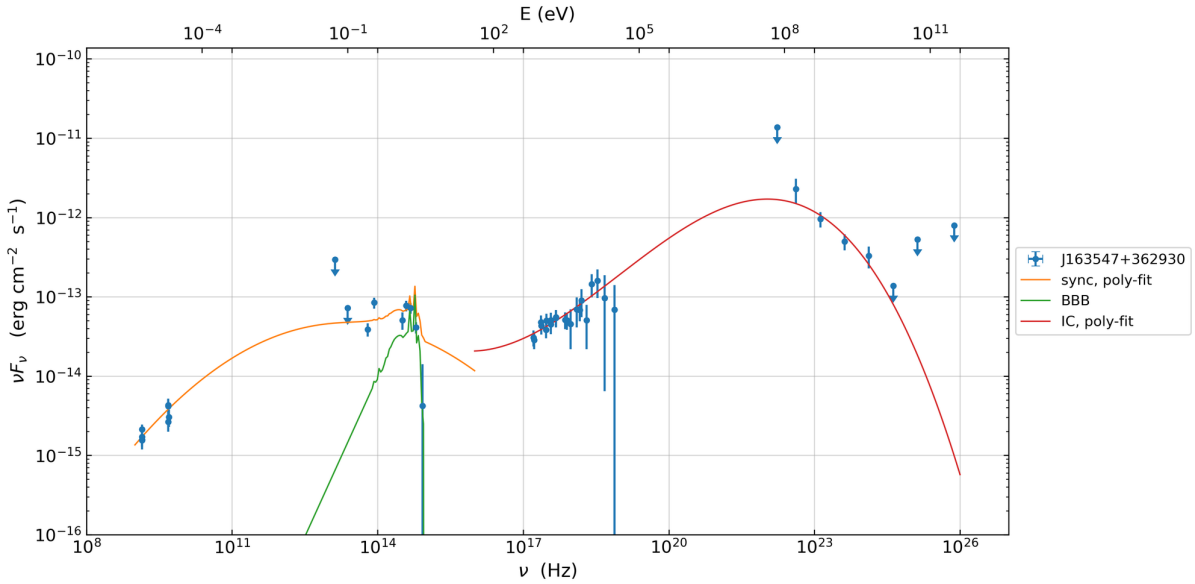


Figure 4.14: Phenomenological description of the SED with the `SEDShape` tool: log-parabola plus big blue bump template for the low-energy hump and log-cubic function for the high-energy hump.

These quantities were passed to the `ObsConstrain` module, which builds a first jet model from the observables. We adopted a viewing angle $\theta = 3°$, a bulk Lorentz factor $\Gamma = 18.5$, a magnetic field in the range 0.02–0.08 G, a variability timescale of 7 days and a broken power-law EED, with external Compton emission from the dusty torus, the BLR and the disk. In this module the radii of the BLR and of the torus are tied to the disk luminosity parameter L_D (the `L_Disk` parameter of `JetSeT`, see Section 4.4.4) through

$$R_\mathrm{BLR,in} = 3 \times 10^{17} \left(\frac{L_\mathrm{D}}{10^{46}\,\mathrm{erg\,s^{-1}}} \right)^{1/2} \mathrm{cm}, \qquad R_\mathrm{BLR,out} = 1.1\, R_\mathrm{BLR,in},$$
$$R_\mathrm{DT} = 2 \times 10^{19} \left(\frac{L_\mathrm{D}}{10^{46}\,\mathrm{erg\,s^{-1}}} \right)^{1/2} \mathrm{cm}, \tag{4.4}$$

which are the relations used by Sánchez Zaballa et al. (2025). These normalisations differ from the general scalings of Equation 3.36, in particular for the torus, whose radius is larger here by a factor of about 2.5. Because the radii follow L_D automatically, the geometry of the external photon fields changes during the fit whenever L_D changes. This point turns out to be central for the interpretation of the accretion-efficiency grid (Section 4.6.3).

4.4.3 The Template Model

The model produced by `ObsConstrain` was completed with the following starting values of the central-engine parameters: $L_\mathrm{D} = 1.38 \times 10^{46}$ erg s^{-1}, $M_\mathrm{BH} = 9.87 \times 10^{8}\, M_\odot$, a torus temperature $T_\mathrm{DT} = 882$ K and covering factors $\tau_\mathrm{BLR} = \tau_\mathrm{DT} = 0.1$. The accretion

efficiency was set to $\eta = 0.083$ ($= 1/12$, the Shakura–Sunyaev value), and its allowed range in `JetSeT` was widened to 0.04–0.35 so that the whole grid of Section 4.6 can be explored with the same model.

As in Sánchez Zaballa et al. (2025), the blob radius is not a free parameter but follows the dissipation radius R_H through a conical jet of half-opening angle $\theta_{\rm open} = 3°$,

$$R = R_H \tan\theta_{\rm open}, \tag{4.5}$$

so that a blob located further from the black hole is also larger. In `JetSeT` this is done by declaring R a dependent parameter of R_H and $\theta_{\rm open}$.

The template gives $R = 4.7 \times 10^{16}$ cm and $R_H = 8.9 \times 10^{17}$ cm, with $R_{\rm BLR,in} = 3.5 \times 10^{17}$ cm and $R_{\rm DT} = 2.4 \times 10^{19}$ cm. The jet model was finally multiplied by the Franceschini et al. (2008) EBL absorption template, whose redshift was linked to that of the jet,

$$F_{\rm model}(\nu) = F_{\rm jet}(\nu)\, e^{-\tau_{\rm EBL}(\nu,z)}, \tag{4.6}$$

and the composite model was saved and used as the starting point of every fit. We stress that the template is only a starting point: its parameters are not a fit to the data.

4.4.4 Disk Luminosity and Accretion Efficiency in `JetSeT`

The role of the accretion efficiency in the model requires some care, because the `JetSeT` parameter `L_Disk` does not coincide in general with the luminosity radiated by the disk. In the `MultiBB` model the accretion rate is obtained from the two parameters $L_{\rm D}$ and η as

$$\dot{M}_{\rm D} = \frac{L_{\rm D}}{\eta\, c^2}, \tag{4.7}$$

and the disk spectrum is a multi-colour blackbody for a black hole of mass $M_{\rm BH}$, with inner and outer radii $R_{\rm in}$ and $R_{\rm out}$ expressed in Schwarzschild radii. The temperature profile has the Shakura–Sunyaev shape, but `JetSeT` normalises it as

$$\sigma_{\rm SB} T^4(R) = \frac{R_{\rm in}\, L_{\rm D}}{8\pi\eta}\, R^{-3} \left(1 - \sqrt{R_{\rm in}/R}\right).$$

Integrating over the disk, $R_{\rm in}$ cancels, and the luminosity emitted by the model disk is

$$L_{\rm disc} = \frac{\dot{M}_{\rm D}\, c^2}{12} = \frac{L_{\rm D}}{12\,\eta} \tag{4.8}$$

for any inner radius. The factor 1/12 therefore follows from this normalisation, not from the radiative efficiency of the model disk: a Shakura–Sunyaev disk would radiate $\dot{M}c^2/12$ only for $R_{\rm in} = 3R_{\rm S}$, while here $R_{\rm in}$ is a free parameter. We verified Equation 4.8 with the disk luminosity computed by `JetSeT`: for $R_{\rm in} = 3$–$14\,R_{\rm S}$ and $\eta = 0.083$ and

0.3 the two agree to better than 1%, the small difference being due to the finite outer radius, and two models with the same ratio $L_{\rm D}/\eta$ produce identical disk spectra. The two quantities coincide only for $\eta = 0.083$ ($= 1/12$), which is why this value is used as the baseline. `JetSeT` converts this luminosity into a flux as $L_{\rm disc}/(4\pi d_L^2)$, so $L_{\rm disc}$ is an isotropic-equivalent luminosity. For a disk seen nearly face-on the intrinsic luminosity can be lower by up to a factor of about two ($2\cos\theta$ for a disk emitting as a flat surface); we follow the common practice of using the isotropic value.

Equation 4.8 has two consequences. First, the optical–UV data fix the disk spectrum, and therefore $L_{\rm disc}$ and $L_{\rm D}/\eta$, but not η itself: when η is changed, the fit simply rescales $L_{\rm D} \propto \eta$. Second, the luminosities of the BLR and of the torus are normalised to the radiated luminosity ($\tau_{\rm BLR} L_{\rm disc}$ and $\tau_{\rm DT} L_{\rm disc}$), as we checked by computing the comoving energy densities for different η at fixed $L_{\rm D}$, while their radii follow $L_{\rm D}$ through Equation 4.4. In our set-up, then, the accretion efficiency acts on the SED only through the size of the BLR and of the torus, $R \propto (12\,\eta\,L_{\rm disc})^{1/2}$, at fixed luminosity of the external fields. This is the key to understand the results of Section 4.6, and it means that the dependence of the SED on η is a property of this parameterisation. We checked that if the radii are scaled with the radiated luminosity, as the empirical radius–luminosity relations imply, models with the same $L_{\rm D}/\eta$ give SEDs identical to numerical precision for $\eta = 0.057$, 0.15 and 0.3: the accretion efficiency then has no effect on the SED, and every value of η reproduces the geometry of the baseline model.

Throughout this thesis we use $L_{\rm D}$ for the `JetSeT` parameter and $L_{\rm disc}$ for the radiated luminosity of Equation 4.8. The Eddington ratio is always computed with the radiated luminosity,

$$\lambda = \frac{L_{\rm disc}}{L_{\rm Edd}}, \qquad L_{\rm Edd} = 1.26 \times 10^{38} \left(\frac{M_{\rm BH}}{M_\odot}\right)\ {\rm erg\,s^{-1}}, \tag{4.9}$$

and the physical accretion rate for a radiative efficiency η is $\dot{M} = L_{\rm disc}/(\eta c^2)$. Note that $\dot{M}_{\rm D}$ of Equation 4.7 equals this physical accretion rate only for $\eta = 0.083$; for other values it is simply $12 L_{\rm disc}/c^2$, the accretion rate a Shakura–Sunyaev disk would need to produce the observed emission.

4.4.5 Transformation of the External Photon Fields

`JetSeT` can compute the external photon fields either directly in the blob rest frame or in the disk frame, transforming the emitted spectrum afterwards. The choice matters for the direct disk radiation: for a blob located at $R_H \sim 10^{18}$ cm, the EC(Disk) component changes by up to two orders of magnitude between the two options, while the EC(DT) and EC(BLR) components are less affected. We adopt the blob-frame transformation, which is the `JetSeT` default and the choice of Sánchez Zaballa et al. (2025), and we use it consistently for the Minuit fits, the MCMC sampling, the figures and the energetics. Using

the same frame everywhere is essential: a posterior sampled with one transformation and plotted or post-processed with the other would describe a different model.

4.5 The Baseline Model at $\eta = 0.083$

4.5.1 Free Parameters and Bounds

The baseline model has $\eta = 0.083$, for which $L_{\rm D} = L_{\rm disc}$. Eleven parameters are left free in the Minuit fit: the magnetic field B, the bulk Lorentz factor Γ, the electron density N, the break Lorentz factor $\gamma_{\rm break}$ and the high-energy index p_1 for the jet; $M_{\rm BH}$, $L_{\rm D}$, $R_{\rm in}$ and $R_{\rm out}$ for the disk; $T_{\rm DT}$ for the torus; and the dissipation radius R_H. The bounds, listed in Table 4.4, are deliberately wide and physically motivated: the same box is used later as the prior of the MCMC sampling, and no parameter should be constrained by the box rather than by the data. The torus temperature is limited to the dust sublimation temperature (~ 1500 K), and the lower bound on the inner disk radius, $2.07\,R_S$, corresponds to the innermost stable orbit for $\eta = 0.083$ ($a \simeq 0.52$).

Table 4.4: Free parameters of the Minuit fits and their bounds (also used as flat priors in the MCMC sampling). For the accretion-efficiency grid the upper bound on $L_{\rm D}$ is raised to 1.2×10^{47} erg s^{-1}, since $L_{\rm D}$ grows with η. The frozen parameters are listed at the bottom.

Parameter	Range	Notes
B [G]	0.01–0.20	
Γ	8–30	
N [cm^{-3}]	300–8000	
$\gamma_{\rm break}$	300–4000	
p_1	2.40–4.50	
$M_{\rm BH}$ [$M_\odot$]	3×10^8–8×10^9	
$L_{\rm D}$ [erg s^{-1}]	0.5–6 $\times 10^{46}$	grid: up to 1.2×10^{47}
$R_{\rm in}$ [R_S]	2.07–20	
$R_{\rm out}$ [R_S]	300–2×10^4	
$T_{\rm DT}$ [K]	300–1500	dust sublimation
R_H [cm]	5×10^{17}–8×10^{18}	
Frozen: $\theta = \theta_{\rm open} = 3°$, $z = 3.647763$, $\tau_{\rm BLR} = \tau_{\rm DT} = 0.1$, $\gamma_{\rm min} = 2.10$, $\gamma_{\rm max} = 1.853 \times 10^4$, $p = 1.59$, η, $N_{\rm H}/N_e$.		

The low-energy electron index was fixed at $p = 1.59$, the value of the reference model from which the fits were started. In the configuration adopted here the optically thin radio slope is not fitted, and the low-energy index is mainly constrained by the rising part of the inverse Compton components in the X-ray band; we have kept it fixed in order to limit the number of free parameters. The cutoff Lorentz factors $\gamma_{\rm min} = 2.10$ and

$\gamma_{\max} = 1.853 \times 10^4$ were also fixed. The ratio of cold protons to relativistic electrons, $N_{\rm H}/N_e$, does not change the SED and only enters the jet energetics (Section 4.9).

The fit was performed with Minuit (James & Roos, 1975) through the minimiser interface of `JetSeT`. The minimisation was repeated three times, each time starting from the previous minimum, to make sure that the solution was stable.

4.5.2 Minuit Best Fit

The fit converges to $\chi^2 = 26.34$ for 27 degrees of freedom (null-hypothesis probability 0.50); considering only the 32 detections, $\chi^2 = 26.25$ for 21 degrees of freedom. The three successive runs gave $\chi^2 = 26.51$, 26.38 and 26.34, so the minimum is stable. No parameter ends at a bound. The best-fit values are listed in Table 4.5 and the corresponding SED, with all spectral components, is shown in Figure 4.15.

Minuit flags the minimum as formally invalid (estimated distance to minimum above threshold, covariance matrix forced to be positive definite). This is common for SED models with strongly correlated parameters: the χ^2 surface is very flat along some directions, and the Hesse uncertainties become unreliable. For this reason we do not quote Minuit uncertainties, and all the uncertainties in this thesis come from the MCMC sampling of Section 4.7.

Table 4.5: Minuit best fit of the baseline model ($\eta = 0.083$). Derived quantities are computed from the best-fit parameters.

Parameter	Value	Component
B	5.98×10^{-2} G	blob
Γ	18.26	blob
δ (derived)	19.1	blob
N	1759 cm^{-3}	electrons
$\gamma_{\rm break}$	837	electrons
p (fixed), p_1	1.59, 3.30	electrons
R_H	1.20×10^{18} cm	blob position
R (derived)	6.3×10^{16} cm	blob size
$M_{\rm BH}$	$1.82 \times 10^9\, M_\odot$	disk
$L_{\rm D} = L_{\rm disc}$	1.90×10^{46} erg s^{-1}	disk
$R_{\rm in}$, $R_{\rm out}$	$8.2\, R_S$, $4.2 \times 10^3\, R_S$	disk
$T_{\rm DT}$	1036 K	torus
$R_{\rm BLR,in}$, $R_{\rm BLR,out}$	4.1, 4.5×10^{17} cm	derived
$R_{\rm DT}$	2.8×10^{19} cm	derived
$\lambda = L_{\rm disc}/L_{\rm Edd}$	0.083	derived
$\dot{M}$ ($\eta = 0.083$)	$4.0\, M_\odot$ yr^{-1}	derived
χ^2/dof	26.34/27	

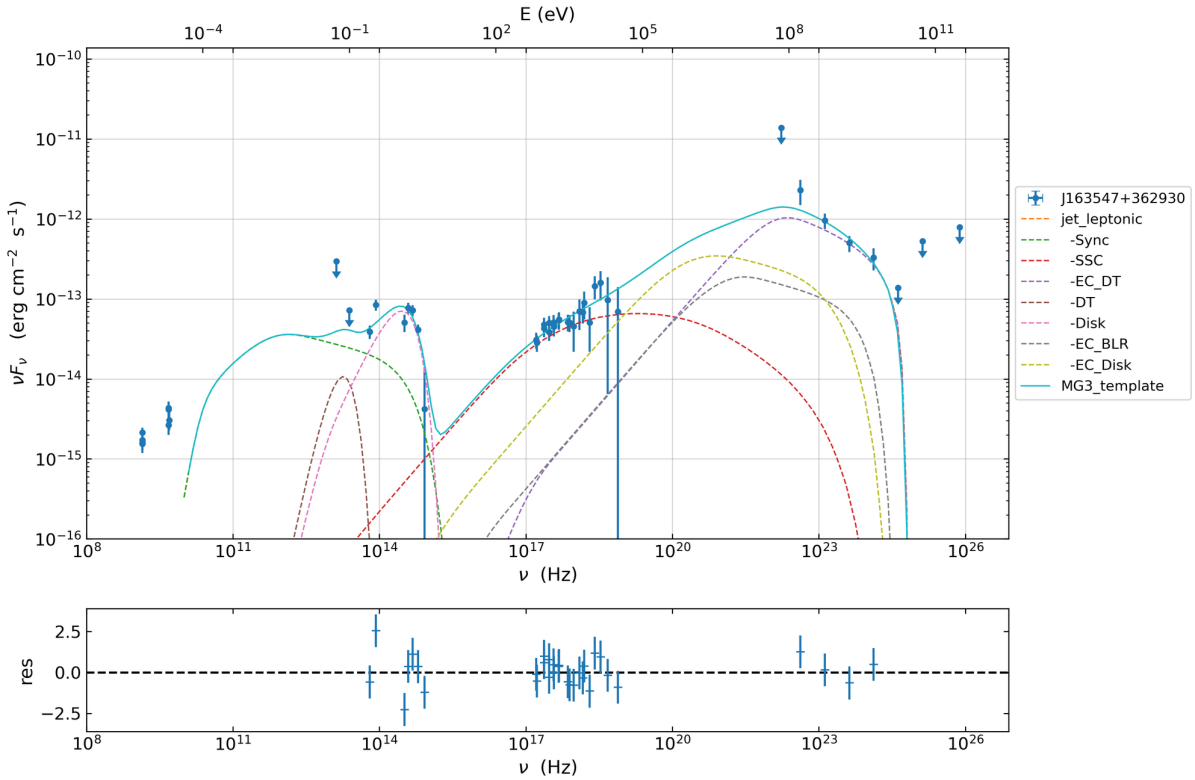


Figure 4.15: Minuit best fit of the baseline model ($\eta = 0.083$). The individual components are the synchrotron emission (Sync), synchrotron self-Compton (SSC), external Compton on the disk, BLR and torus photons (EC_Disk, EC_BLR, EC_DT), and the thermal emission of the disk and of the torus. The lower panel shows the residuals.

The physical picture is the following. The synchrotron component peaks in the far-infrared, around 10^{12}–$10^{12.5}$ Hz, below the WISE upper limits, and the optical–UV points are dominated by the accretion disk, with a small contribution of the torus in the near infrared. The X-ray band is produced by inverse Compton emission, and the γ-ray peak, at about 2×10^{22} Hz in the model, by external Compton scattering of torus photons. The EC(Disk) component peaks in the MeV band, where no data are available. Since $\Gamma = 18.3$ is very close to $1/\sin\theta = 19.1$, the Doppler factor is close to its maximum, $\delta \simeq 19.1$.

4.5.3 Geometry of the Emission Region

With $L_{\rm D} = 1.90 \times 10^{46}$ erg s^{-1}, the BLR extends between 4.1×10^{17} and 4.5×10^{17} cm and the torus is located at 2.8×10^{19} cm. The emission region sits at $R_H = 1.20 \times 10^{18}$ cm, that is at $R_H/R_{\rm BLR,out} = 2.6$ and $R_H/R_{\rm DT} = 0.043$: outside the BLR but well inside the torus. In this configuration the torus provides the dominant seed photons for the γ-ray emission. A location outside the BLR is also consistent with the absence of BLR absorption features in the *Fermi*-LAT spectrum discussed in Section 3.3.3. The blob radius $R = 6.3 \times 10^{16}$ cm corresponds to an observed light-crossing time $R(1+z)/(c\,\delta) \approx 6$ days, consistent with the 7-day variability timescale assumed in the template.

The black hole mass $M_{\rm BH} = 1.8 \times 10^9\, M_\odot$ and the disk luminosity $L_{\rm disc} = 1.9 \times 10^{46}$ erg s^{-1} give an Eddington ratio $\lambda = 0.083$. For $\eta = 0.083$ the accretion rate is $\dot{M} = 4.0\, M_\odot$ yr^{-1}.

4.5.4 The Hard X-ray Band

Figure 4.16 zooms on the X-ray band. The model reproduces the rising X-ray spectrum through the sum of two components. At 10 keV (observer frame) the SSC component contributes 64% of the flux and the EC(Disk) component 27%, with 5% from EC(BLR) and 4% from EC(DT). At 40 keV the SSC component has started to curve over, and the contributions become 44% (SSC), 42% (EC(Disk)), 7% (EC(BLR)) and 7% (EC(DT)). The hardness of the *NuSTAR* spectrum is therefore produced by an SSC component close to its peak together with the rising part of the external Compton components, and not by a single component.

This result refines the expectation of Section 3.7.2, based on the Thomson-limit relation for a single power law. Below the respective peaks, the slope of both the SSC and the EC components is set by the low-energy electron index p, not by p_1, and for $p = 1.59$ it corresponds to a photon index $(p+1)/2 \simeq 1.3$. The curvature of the SSC component close to its peak softens the total spectrum, and the local photon index of the model across the *NuSTAR* band is $\Gamma_X \simeq 1.6$. A simple power-law fit to the SED points in the same band gives $\Gamma_X \simeq 1.4$, while the XSPEC fit of the FPMB spectrum gives $\Gamma_X = 0.94 \pm 0.34$

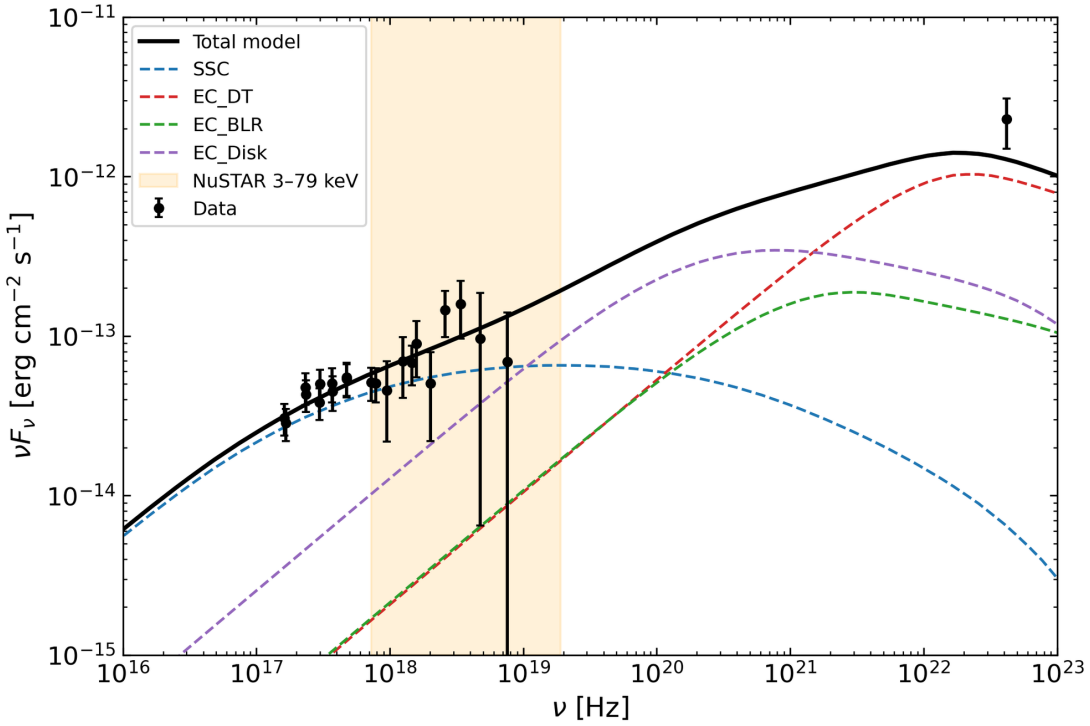


Figure 4.16: X-ray band of the baseline model. The shaded region marks the *NuSTAR* 3–79 keV band. Dashed lines show the SSC and the three external Compton components; the solid line is the total model.

(Section 4.1.2). The model is thus within $\sim 2\sigma$ of the XSPEC value and consistent with the SED points, whose uncertainties include the 15% systematic term. The index p_1, which controls the spectrum above the peaks, is constrained mainly by the *Fermi*-LAT data.

The main conclusion is that the *NuSTAR* band is not produced by a single mechanism: SSC is the largest single contribution, but the rising external Compton emission is needed to reproduce the hard spectrum. The relative weight of the two depends on the geometry of the external fields, as shown in the next section.

4.6 Exploring the Accretion Efficiency

4.6.1 Fitting Strategy

We repeated the fit for a grid of seven accretion efficiencies,

$$\eta \in \{0.057,\ 0.070,\ 0.083,\ 0.100,\ 0.150,\ 0.200,\ 0.300\}.$$

Through Equation 3.40 these correspond to prograde spins $a \simeq 0$, 0.31, 0.52, 0.67, 0.89, 0.96 and 0.996, from a Schwarzschild black hole to a nearly maximally rotating one. The value $\eta = 0.070$ was added to sample more finely the region between the Schwarzschild

and the Shakura–Sunyaev values.

Each value of η is fitted independently, starting from a fresh copy of the template, with the same eleven free parameters, the same frozen parameters and the same bounds as the baseline model (Table 4.4). The only exception is the upper bound on L_{D}, which is raised to 1.2×10^{47} erg s^{-1}: since the data fix L_{D}/η (Section 4.4.4), the value of L_{D} required at $\eta = 0.3$ is about 6.8×10^{46} erg s^{-1}, above the baseline bound. To avoid that Minuit wanders along the degenerate directions, every fit starts from the baseline solution rescaled to the new efficiency,

$$L_{\mathrm{D}}^{\mathrm{start}} = L_{\mathrm{D}}^{\mathrm{base}} \frac{\eta}{0.083}, \qquad R_H^{\mathrm{start}} = R_H^{\mathrm{base}} \left(\frac{\eta}{0.083}\right)^{1/2}, \tag{4.10}$$

which corresponds to the same disk spectrum and to the same position of the blob relative to the BLR and the torus. All the other parameters start from the baseline values. The minimisation was repeated twice for each η (`repeat=2`), and the fit was flagged if any parameter ended within 1% of a bound. The loop skips the values of η that have already been fitted, so that an interruption does not require repeating the completed fits.

4.6.2 Best-Fit Parameters Across the Grid

The results are summarised in Table 4.6. All seven fits converge, no best-fit value lies within 1% of a bound (although for $\eta = 0.057$ and 0.3 Minuit reports some parameters close to their limits according to its internal criterion), and the quality of the fit is the same for all values of η: χ^2 ranges from 25.9 to 26.7 for 27 degrees of freedom, a spread of only $\Delta\chi^2 = 0.76$. The SED does not prefer any accretion efficiency. The best-fit models are compared in Figure 4.17, and the individual fits are shown in Appendix A.1.

The total SEDs are almost indistinguishable wherever data are available. This is not surprising: all seven models are fitted to the same data, and the free parameters rearrange themselves to reproduce them. The ratio panel of Figure 4.17 shows that the differences are confined to two regions with little or no data: the far-infrared below the WISE limits, and especially the MeV band between *NuSTAR* and *Fermi*-LAT, constrained only by the lowest-energy *Fermi*-LAT upper limit, where the models with $\eta \geq 0.15$ exceed the baseline by a factor of 1.3–1.8 around 10^{22} Hz. The MeV band is precisely where the high-energy component peaks, and a measurement there would discriminate between the two geometries of the external fields, although not between the efficiencies as such (Section 4.4.4).

Table 4.6: Minuit best-fit parameters across the accretion-efficiency grid. The column $\eta = 0.083$ is the refit of the baseline with the grid settings. Derived quantities are computed from the best-fit $L_{\rm D}$ through Equations 4.4 and 4.8. No parameter is at a bound.

η	0.057	0.070	**0.083**	0.100	0.150	0.200	0.300
B [10^{-2} G]	7.13	6.32	**6.01**	5.89	6.47	8.77	7.01
Γ	17.7	18.0	**18.4**	18.4	19.5	20.5	21.0
N [cm^{-3}]	2094	1797	**1771**	1778	1688	1470	1137
$\gamma_{\rm break}$	751	830	**838**	870	850	805	854
p_1	3.24	3.32	**3.31**	3.30	3.35	3.56	3.49
$M_{\rm BH}$ [$10^9 M_\odot$]	1.74	1.78	**1.81**	1.81	1.70	1.12	1.96
$L_{\rm D}$ [10^{46} erg s^{-1}]	1.20	1.54	**1.87**	2.24	3.36	4.59	6.81
R_H [10^{18} cm]	1.06	1.18	**1.20**	1.20	1.18	1.16	1.42
$T_{\rm DT}$ [K]	1220	1230	**1110**	1110	984	1010	768
$R_{\rm in}$ [R_S]	8.1	8.1	**8.1**	8.1	8.5	14.2	7.6
$R_{\rm out}$ [$10^3 R_S$]	3.1	2.7	**3.0**	1.8	2.7	3.4	2.6
$L_{\rm D}/\eta$ [10^{47} erg s^{-1}]	2.10	2.20	**2.25**	2.24	2.24	2.29	2.27
$L_{\rm disc}$ [10^{46} erg s^{-1}]	1.75	1.83	**1.87**	1.87	1.87	1.91	1.89
$R_{\rm BLR,out}$ [10^{17} cm]	3.6	4.1	**4.5**	4.9	6.0	7.1	8.6
$R_{\rm DT}$ [10^{19} cm]	2.2	2.5	**2.7**	3.0	3.7	4.3	5.2
$R_H/R_{\rm BLR,out}$	2.95	2.88	**2.66**	2.44	1.96	1.64	1.65
$R_H/R_{\rm DT}$	0.049	0.048	**0.044**	0.040	0.032	0.027	0.027
χ^2 (27 dof)	26.1	25.9	**26.1**	26.1	26.7	26.4	26.1

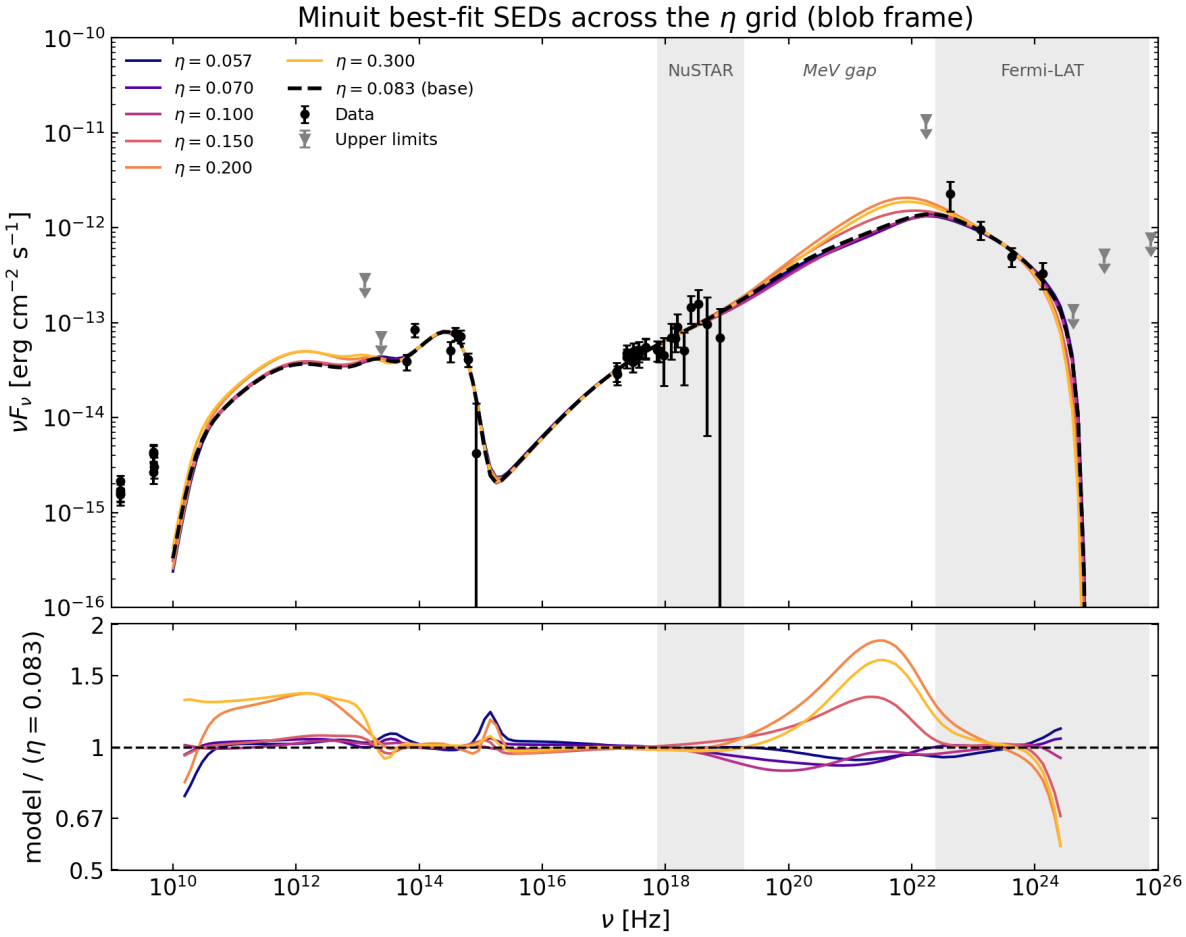


Figure 4.17: Minuit best-fit SEDs for the seven accretion efficiencies. The lower panel shows the ratio of each model to the $\eta = 0.083$ model, in the frequency range where the ratio is meaningful. Shaded bands mark the *NuSTAR* and *Fermi*-LAT energy ranges. The models agree within ∼10% where data are available and differ by up to ∼80% in the MeV band, between the two instruments.

4.6.3 What Changes with η: the Geometry of the External Fields

The differences between the fits are in the parameters, and they follow a clear logic once Section 4.4.4 is taken into account.

The disk. The ratio $L_{\rm D}/\eta$ is constant within $\pm5\%$ (2.10–2.29×10^{47} erg s^{-1}), and so is the radiated luminosity $L_{\rm disc} = (1.75$–$1.91) \times 10^{46}$ erg s^{-1}. The disk spectrum is the same in all models, as required by the optical–UV data. The black hole mass is also stable, $M_{\rm BH} \simeq (1.7$–$2.0) \times 10^9\, M_\odot$, with the exception of $\eta = 0.2$ discussed below.

The external fields. Since $L_{\rm D}$ grows proportionally to η, the radii of Equation 4.4 grow as $\eta^{1/2}$: from $\eta = 0.057$ to $\eta = 0.3$ the BLR and the torus become 2.4 times larger, while their luminosities ($\tau L_{\rm disc}$) do not change. The energy density of the torus radiation, $U_{\rm DT} \propto \tau_{\rm DT} L_{\rm disc}/R_{\rm DT}^2$, therefore decreases as $1/\eta$. The dissipation radius, on the other hand, does not follow the $\eta^{1/2}$ scaling of the starting values (Equation 4.10): Minuit brings it back to $R_H \simeq (1.1$–$1.4) \times 10^{18}$ cm for every η. As a result the blob, which sits at $2.9\, R_{\rm BLR,out}$ for $\eta = 0.057$, is only at $1.6\, R_{\rm BLR,out}$ for $\eta \geq 0.2$. In other words, increasing η moves the BLR towards the emission region and dilutes the torus radiation. The blob remains outside the BLR in all cases, so the absence of BLR absorption features in the *Fermi*-LAT spectrum (Section 3.3.3) is satisfied by every model and does not discriminate between the efficiencies.

The jet. The bulk Lorentz factor increases from 17.7 to 21.0 along the grid. At $\theta = 3^\circ$ the Doppler factor is almost insensitive to Γ in this range ($\delta \simeq 19.0$–19.1), so the synchrotron and SSC components are unaffected, while the external Compton emission scales with the comoving energy density $U'_{\rm ext} \propto \Gamma^2 U_{\rm ext}$ (Equation 3.16). The increase of Γ is the way the fit compensates the changing external fields. The electron density decreases from about 2100 to 1100 cm^{-3}, and the torus temperature decreases from about 1200 to 770 K, shifting the EC(DT) seed photons to lower energies. The magnetic field and the shape of the electron distribution ($\gamma_{\rm break} \simeq 750$–$870$, $p_1 \simeq 3.2$–3.6) show no systematic trend.

4.6.4 Spectral Decomposition Across the Grid

The change of geometry modifies which process produces the emission in each band. To quantify it, we computed the fraction of the jet emission (synchrotron, SSC and the three EC components) contributed by each component at four energies: 10 and 40 keV in the *NuSTAR* band, 10 MeV in the gap between the instruments, and 1 GeV in the *Fermi*-

LAT band. The results are given in Table 4.7; the full decomposition of each model is shown in Appendix A.1.

Table 4.7: Fraction (in %) of the jet emission contributed by the SSC and external Compton components at four observed energies, for the Minuit best fits. The synchrotron contribution is negligible at these energies.

	10 keV				40 keV			
η	SSC	Disk	BLR	DT	SSC	Disk	BLR	DT
0.057	67	25	3	5	48	39	5	8
0.070	70	22	4	5	51	36	6	8
0.083	67	24	5	4	48	38	8	7
0.100	74	16	6	4	56	28	10	7
0.150	65	22	11	3	44	34	18	4
0.200	60	20	19	2	37	30	31	3
0.300	64	17	17	2	42	27	28	3
	10 MeV				1 GeV			
η	SSC	Disk	BLR	DT	SSC	Disk	BLR	DT
0.057	4	31	12	54	0	8	7	86
0.070	3	27	16	54	0	5	7	88
0.083	3	28	22	47	0	6	12	82
0.100	4	21	34	41	0	4	21	75
0.150	2	19	57	21	0	5	50	45
0.200	1	10	80	10	0	2	77	20
0.300	1	9	77	13	0	2	74	24

Two results emerge. In the X-ray band the decomposition is stable: SSC is the largest single component at all efficiencies, with 60–74% of the emission at 10 keV and 37–56% at 40 keV, and the remaining part is external Compton emission that grows with energy. The identification of SSC plus rising EC as the origin of the *NuSTAR* emission is therefore independent of η.

In the γ-ray band, instead, the seed photon field changes with η. For $\eta \lesssim 0.1$ the torus dominates, with 75–88% of the emission at 1 GeV. For $\eta \gtrsim 0.15$ the BLR takes over, with up to 80% at 10 MeV and 77% at 1 GeV. The crossover occurs at $\eta \approx 0.15$, where the two contributions are equal at 1 GeV. This is the direct consequence of the geometry described above: as the BLR approaches the blob its photons become more important, while the torus radiation is diluted. Since BLR photons are more than an order of magnitude more energetic than torus photons, their scattering enters the Klein–Nishina regime at lower electron energies. In our models the EC(BLR) component therefore peaks at lower γ-ray energies than the EC(DT) component and fills the MeV band, which is why the high-η models exceed the baseline there, while a torus contribution remains important at the highest *Fermi*-LAT energies also at high η.Also, the origin of the γ-ray seed photons, torus or BLR, cannot be decided from the SED alone. With the radii scaled to the

radiated luminosity, every efficiency would give the torus-dominated geometry of the baseline model.

4.6.5 Remarks on Individual Fits

The fit at $\eta = 0.2$ converges to a larger inner disk radius, $R_{\rm in} = 14.2\,R_S$ instead of $\simeq 8\,R_S$, and to a lower black hole mass, $M_{\rm BH} = 1.1 \times 10^9\,M_\odot$. This is a known degeneracy of the disk model: at fixed luminosity the disk spectrum depends only on the inner radius in physical units, $R_{\rm in}R_S \propto R_{\rm in}M_{\rm BH}$ (Section 4.4.4), so that a larger inner radius in units of R_S is compensated by a smaller mass, $M_{\rm BH} \propto R_{\rm in}^{-1}$. Indeed, $1.8 \times 10^9\,M_\odot \times 8.1/14.2 \simeq 1.0 \times 10^9\,M_\odot$, and the product $M_{\rm BH}R_{\rm in}$ is constant within 10% across the whole grid. The quality of the fit is the same, and we consider this solution as a different point along the $M_{\rm BH}$–$R_{\rm in}$ degeneracy rather than a physical effect of η. It should be kept in mind when interpreting the black hole mass and the Eddington ratio at $\eta = 0.2$.

More generally, in `JetSeT` the inner radius of the disk is an independent parameter, measured in Schwarzschild radii, and it is not tied to the innermost stable circular orbit of a black hole with the spin corresponding to η. All fits except the one at $\eta = 0.2$ converge to $R_{\rm in} \simeq 8\,R_S$, larger than the ISCO even for a non-rotating black hole ($3\,R_S$). The disk spectrum therefore fixes the inner radius in physical units, $R_{\rm in} \simeq 4 \times 10^{15}$ cm, rather than the black hole mass, and all the values of $M_{\rm BH}$ and λ quoted in this thesis refer to $R_{\rm in} \simeq 8\,R_S$. With the inner radius at the ISCO, the mass would be $M_{\rm BH} \approx 5 \times 10^9\,M_\odot$ for $a = 0$ and up to $\approx 10^{10}\,M_\odot$ for $a = 0.8$, with $\lambda \approx 0.015$–0.03. Since only $\ln(M_{\rm BH}/M_{\rm seed})$ enters the growth of the black hole, the growth results of Section 4.10 would change by 10–20%. The accretion efficiency should therefore be read, within our modelling, as a parameter that sets the conversion between $L_{\rm D}$ and the accretion rate and, through Equation 4.4, the size of the BLR and of the torus, rather than as a self-consistent description of the black hole spin.

4.7 Bayesian Parameter Estimation with MCMC

The Minuit fits give the best-fit parameters but not reliable uncertainties, and they do not show how the parameters are correlated. We therefore sampled the posterior distribution of the parameters with the affine-invariant ensemble sampler `emcee` (Foreman-Mackey et al., 2013), through the `McmcSampler` class of `JetSeT`, starting from the Minuit solution at each value of η.

4.7.1 Sampler Set-up

Seven parameters are sampled at each η: the five jet parameters B, Γ, N, $\gamma_{\rm break}$ and p_1, and the two disk parameters $M_{\rm BH}$ and $L_{\rm D}$. These are the parameters that control the

SSC/EC balance and the disk emission. The dissipation radius is fixed at the Minuit value of each η, together with $T_{\rm DT}$, $R_{\rm in}$, $R_{\rm out}$ and the parameters that are frozen in the Minuit fits. Fixing R_H keeps the number of sampled parameters small; its effect is tested in Section 4.7.6 with an additional run in which R_H is also sampled.

The likelihood is the one used by Minuit, including the upper limits, and the priors are flat within the bounds of Table 4.4. In `JetSeT` the prior of the sampler is a flat box between the parameter bounds; by default the box is built from the Minuit uncertainties, which in our case are unreliable (Section 4.5) and would produce extremely narrow boxes. We therefore set the box explicitly to the physical bounds of the fit.

Each run uses 32 walkers, initialised within 5% of the Minuit values, and 800 steps, of which the first 300 are discarded as burn-in. This leaves $32 \times 500 = 16\,000$ samples per value of η, and each run takes about 16–17 minutes. The same external field transformation (blob frame) is used as in the Minuit fits. As a sanity check, each run verifies that the model inside the minimiser has the expected value of η and of R_H before sampling starts. The chains are saved immediately after the run, and all the following quantities (tables, corner plots, credible bands, energetics) are computed from the saved chains.

4.7.2 Convergence

Convergence was assessed with the integrated autocorrelation time τ of each parameter (Foreman-Mackey et al., 2013) and with the acceptance fraction. For all seven runs and all parameters, τ lies between 38 and 50 steps, so that the 500 post-burn-in steps correspond to 9.9–13.3 autocorrelation times, and the mean acceptance fraction is between 0.30 and 0.36, inside the range 0.2–0.5 usually recommended for `emcee`. The values for each run are given in Table A.1 of Appendix A.2. The chains are therefore long enough to provide stable estimates of the medians and of the 68% intervals; to estimate the extreme tails of the posteriors more accurately, longer chains would be needed.

4.7.3 Posterior Constraints Across the Grid

The posterior medians and 68% credible intervals (16th and 84th percentiles) are reported in Table 4.8. The posterior distribution for the baseline case is shown in Figure 4.18, and those for the other efficiencies in Appendix A.2. The Minuit values lie inside the 68% region for all parameters and all values of η.

For the baseline case the posterior gives $B = 0.059^{+0.025}_{-0.016}$ G, $\Gamma = 17.9^{+6.5}_{-4.2}$, $N = 1834^{+621}_{-503}$ cm^{-3}, $\gamma_{\rm break} = 935^{+298}_{-295}$, $p_1 = 3.38^{+0.29}_{-0.30}$, $M_{\rm BH} = (1.80^{+0.50}_{-0.44}) \times 10^9\,M_\odot$ and $L_{\rm disc} = (1.91^{+0.28}_{-0.35}) \times 10^{46}$ erg s^{-1}, corresponding to an Eddington ratio $\lambda = 0.083^{+0.012}_{-0.010}$. The relative uncertainties are of order 20–40% on B, Γ, N and $\gamma_{\rm break}$, about 10% on p_1, and 15–30% on the disk parameters. These are the uncertainties that we adopt throughout

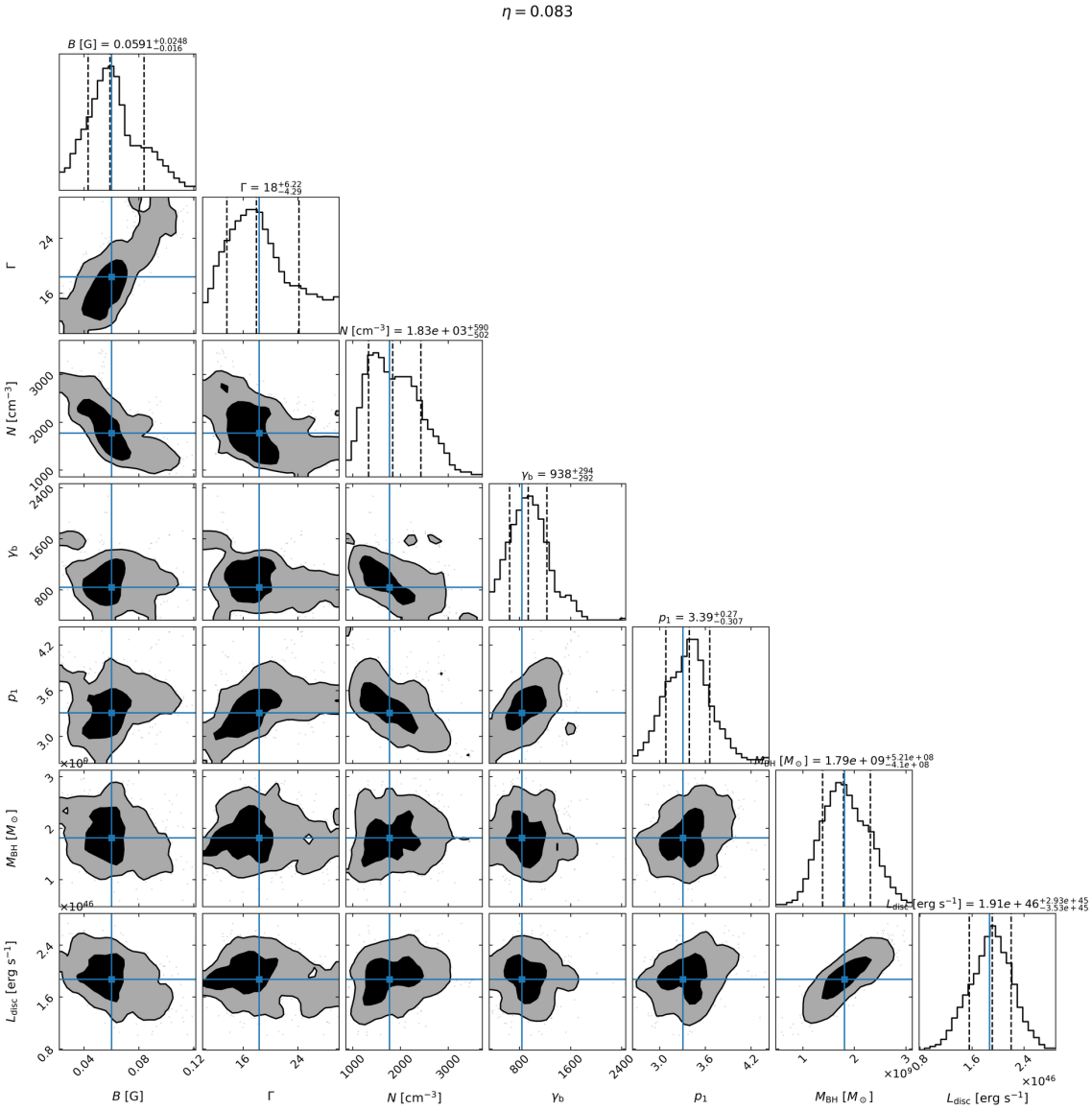


Figure 4.18: Posterior distribution of the seven sampled parameters for the baseline case $\eta = 0.083$. Contours enclose the 68% and 95% probability regions; dashed lines in the histograms mark the 16th, 50th and 84th percentiles. Blue lines show the Minuit best fit.

Table 4.8: MCMC posterior medians and 68% credible intervals for the seven sampled parameters, and derived quantities. $L_{\rm disc} = L_{\rm D}/(12\eta)$ is the radiated disk luminosity, $\lambda = L_{\rm disc}/L_{\rm Edd}$, and r_{NB} is the correlation coefficient between N and B. The last row gives the χ^2 of the posterior-median SED computed on the 32 detections in the fit range. A dagger marks the runs in which the upper tail of Γ is limited by the prior bound $\Gamma \leq 30$.

η	0.057	0.070	**0.083**	0.100	0.150	0.200	0.300
B [10^{-2} G]	$8.0^{+2.2}_{-1.9}$	$7.5^{+2.4}_{-2.1}$	$\mathbf{5.9^{+2.5}_{-1.6}}$	$7.5^{+1.9}_{-1.8}$	$7.3^{+2.2}_{-1.7}$	$9.6^{+2.6}_{-2.4}$	$7.6^{+1.8}_{-2.0}$
Γ	$20.5^{+5.8}_{-5.4}$	$21.5^{+6.2\dagger}_{-5.1}$	$\mathbf{17.9^{+6.5}_{-4.2}}$	$23.3^{+4.4\dagger}_{-5.2}$	$22.3^{+4.9\dagger}_{-5.1}$	$23.1^{+4.7\dagger}_{-6.2}$	$23.5^{+4.7\dagger}_{-4.8}$
N [cm^{-3}]	1872^{+430}_{-289}	1637^{+490}_{-338}	$\mathbf{1834^{+621}_{-503}}$	1586^{+453}_{-305}	1645^{+444}_{-376}	1454^{+457}_{-311}	1202^{+415}_{-200}
$\gamma_{\rm break}$	833^{+345}_{-204}	871^{+247}_{-237}	$\mathbf{935^{+298}_{-295}}$	898^{+258}_{-242}	851^{+330}_{-213}	837^{+293}_{-221}	796^{+291}_{-253}
p_1	$3.43^{+0.24}_{-0.21}$	$3.45^{+0.23}_{-0.20}$	$\mathbf{3.38^{+0.29}_{-0.30}}$	$3.45^{+0.28}_{-0.23}$	$3.43^{+0.31}_{-0.26}$	$3.65^{+0.30}_{-0.31}$	$3.51^{+0.27}_{-0.26}$
$M_{\rm BH}$ [$10^9 M_\odot$]	$1.85^{+0.42}_{-0.41}$	$1.84^{+0.44}_{-0.40}$	$\mathbf{1.80^{+0.50}_{-0.44}}$	$1.79^{+0.47}_{-0.39}$	$1.67^{+0.48}_{-0.37}$	$1.07^{+0.28}_{-0.24}$	$2.00^{+0.50}_{-0.52}$
$L_{\rm D}$ [10^{46} erg s^{-1}]	$1.26^{+0.18}_{-0.23}$	$1.56^{+0.27}_{-0.24}$	$\mathbf{1.91^{+0.28}_{-0.35}}$	$2.17^{+0.41}_{-0.36}$	$3.30^{+0.60}_{-0.66}$	$4.51^{+0.69}_{-0.65}$	$6.61^{+1.16}_{-1.05}$
$L_{\rm disc}$ [10^{46} erg s^{-1}]	1.84	1.85	**1.91**	1.81	1.83	1.88	1.84
λ	$0.078^{+0.013}_{-0.010}$	$0.081^{+0.011}_{-0.011}$	$\mathbf{0.083^{+0.012}_{-0.010}}$	$0.079^{+0.014}_{-0.009}$	$0.085^{+0.013}_{-0.011}$	$0.140^{+0.023}_{-0.021}$	$0.075^{+0.013}_{-0.011}$
$R_H/R_{\rm BLR,out}$	$2.88^{+0.31}_{-0.19}$	$2.87^{+0.25}_{-0.22}$	$\mathbf{2.63^{+0.28}_{-0.18}}$	$2.48^{+0.23}_{-0.20}$	$1.97^{+0.23}_{-0.16}$	$1.65^{+0.13}_{-0.11}$	$1.67^{+0.15}_{-0.13}$
r_{NB}	-0.54	-0.64	$\mathbf{-0.68}$	-0.59	-0.65	-0.56	-0.47
χ^2 (32 det.)	26.6	25.7	**26.8**	25.8	26.8	26.4	26.4

the thesis; they are considerably larger than the formal Minuit errors, as expected from the strong correlations discussed below.

The bulk Lorentz factor deserves a comment. Its posterior is broad and, for five of the seven efficiencies, more than 1% of the samples lie close to the upper bound $\Gamma = 30$. The upper tail of Γ is therefore partly limited by the prior, while the lower side is constrained by the data: the 16th percentile is $\Gamma \gtrsim 14$–19 at all efficiencies. This behaviour has a simple explanation. At $\theta = 3^\circ$ the Doppler factor reaches its maximum for $\Gamma = 1/\sin\theta \simeq 19$ and decreases for larger Γ, while the external Compton emission keeps growing with Γ^2; a larger Γ can then be compensated by the other parameters. We keep the bound at $\Gamma = 30$, a reasonable value for an FSRQ, and we quote the bulk Lorentz factor mainly through its lower limit.

4.7.4 Parameter Correlations

The corner plots show several correlations, the same at all efficiencies, which can be understood from the emission processes.

N and B. The electron density and the magnetic field are anticorrelated in all runs, with correlation coefficients between -0.47 and -0.68 (Table 4.8). The synchrotron peak of the source falls in the far-infrared, where only the WISE upper limits are available, and what the data constrain is the falling high-energy part of the synchrotron spectrum in the infrared, together with the SSC emission in the X-rays. Above the break, the synchrotron emissivity at a fixed frequency scales as $NB^{(p_1+1)/2}$, that is approximately as $NB^{2.2}$ for $p_1 \simeq 3.4$: a stronger field must be compensated by fewer electrons. The

anticorrelation is the signature of this degeneracy, and it is one of the main reasons why N and B are individually only constrained to 20–40%.

The dissipation radius plays an important role in this correlation. In a preliminary eight-parameter run at $\eta = 0.057$, in which R_H was sampled together with the other parameters (not used elsewhere in this thesis), both N and B are anticorrelated with R_H (correlation coefficients -0.59 and -0.43), because the emitting volume grows as R_H^3. When R_H varies, N and B decrease together, which introduces an apparent positive coupling and almost cancels the direct anticorrelation: the marginal N–B correlation drops to -0.18. The partial correlation, computed by inverting the correlation matrix and therefore holding all the other parameters fixed, is instead -0.77. The N–B degeneracy is thus always present, but it is clearly visible in the marginal distributions only when the size of the emitting region is fixed, as in the runs of Table 4.8.

B and Γ. The magnetic field and the bulk Lorentz factor are positively correlated (coefficient $+0.66$ in the run quoted above). In the X-ray band the ratio between the SSC and the EC emission depends on the ratio between the comoving synchrotron energy density, $U'_{\rm syn} \propto NB^2R$, and the comoving external energy density, $U'_{\rm ext} \propto \Gamma^2 U_{\rm ext}$. Keeping this ratio fixed requires B to increase with Γ.

$\gamma_{\rm break}$ and p_1. The two parameters are positively correlated ($+0.69$): a higher break can be compensated by a steeper high-energy slope in reproducing the falling part of the γ-ray spectrum.

$M_{\rm BH}$ and $L_{\rm D}$. The black hole mass and the disk luminosity are positively correlated ($+0.76$). At fixed η the luminosity sets the accretion rate, and the peak frequency of the disk, $\nu_p \propto \dot{M}^{1/4} M_{\rm BH}^{-1/2}$ (Equation 3.33), stays fixed only if $M_{\rm BH} \propto \dot{M}^{1/2}$. The optical–UV data sample the peak of the disk emission only partially, so this degeneracy is not fully broken.

4.7.5 Credible Bands and Residuals

To visualise the uncertainty on the model, we drew 300 random parameter sets from the posterior of each run, computed the corresponding SEDs and took, at each frequency, the 2.5th, 16th, 50th, 84th and 97.5th percentiles of νF_ν. The resulting 68% and 95% credible bands for the baseline case are shown in Figure 4.19, together with the normalised residuals $(\nu F_\nu - \nu F_{\nu,\rm model})/\sigma$ with respect to the posterior-median SED, where σ includes the 15% systematic uncertainty. The bands for the other efficiencies are shown in Appendix A.2.

The bands are narrow where the data are dense, across the optical–UV, the X-ray and the *Fermi*-LAT bands, and become wider where the model is not constrained: in

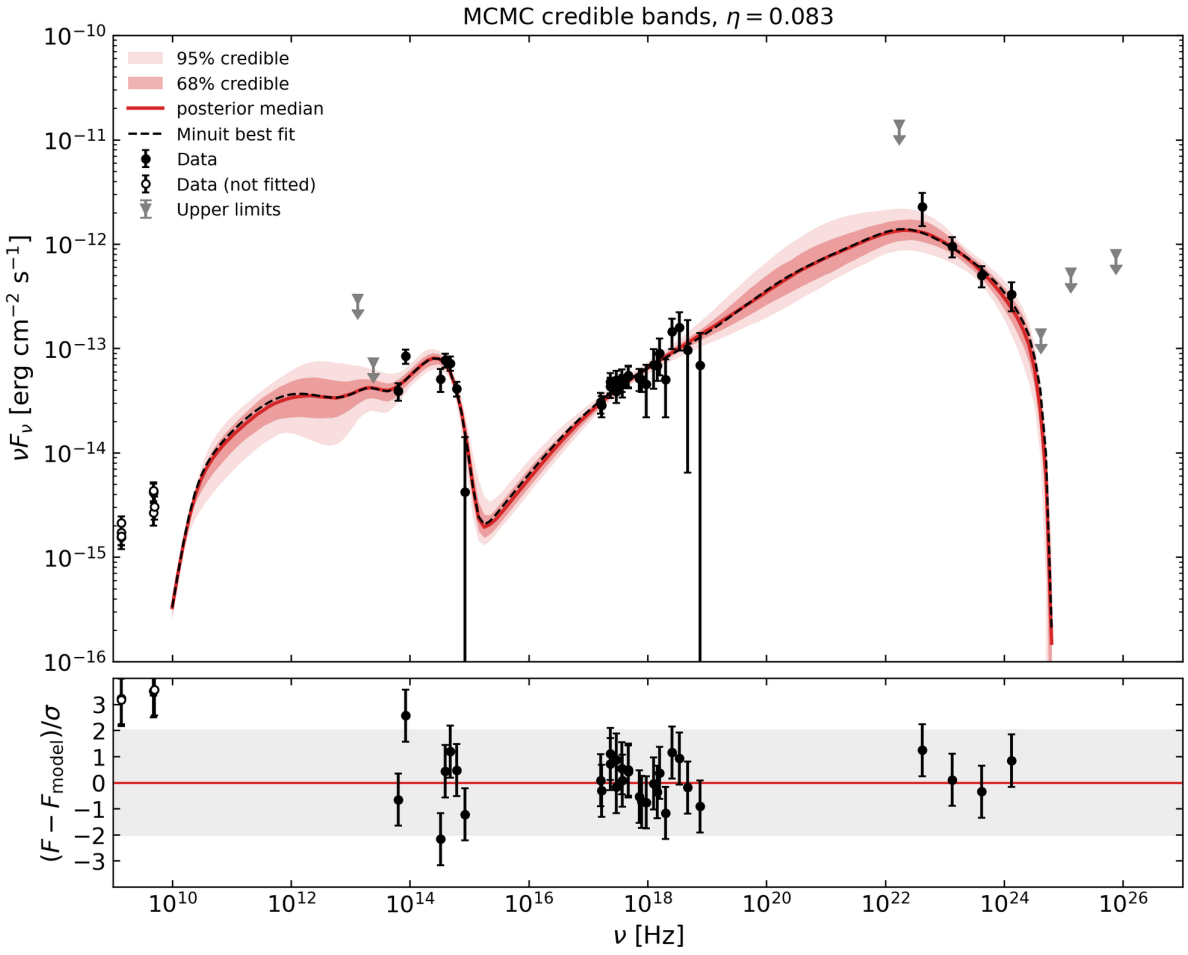


Figure 4.19: SED of the baseline case $\eta = 0.083$ with the 68% and 95% credible bands from 300 posterior draws, the posterior-median SED (red) and the Minuit best fit (dashed). Open circles are the radio points not included in the fit. The lower panel shows the normalised residuals of the detections with respect to the posterior-median SED; shaded bands mark $\pm 1\sigma$ and $\pm 2\sigma$.

the far-infrared around the synchrotron peak, in the MeV band and at the high-energy cutoff, where only upper limits are available. The residuals of the 32 fitted detections are distributed around zero, mostly within $\pm 2\sigma$, without systematic trends across the *NuSTAR* or *Fermi*-LAT bands. The χ^2 of the posterior-median SED on these points is 25.7–26.8 for all efficiencies (Table 4.8). The radio points below 10^{10} Hz lie well above the model, as expected, since they are not included in the fit and are produced by more extended regions of the jet.

The posterior-median SEDs of all efficiencies are compared in Figure 4.20. As for the Minuit fits, the models overlap in all the observed bands and separate in the MeV range, where the high-η models, whose γ-ray emission is dominated by EC(BLR), are brighter.

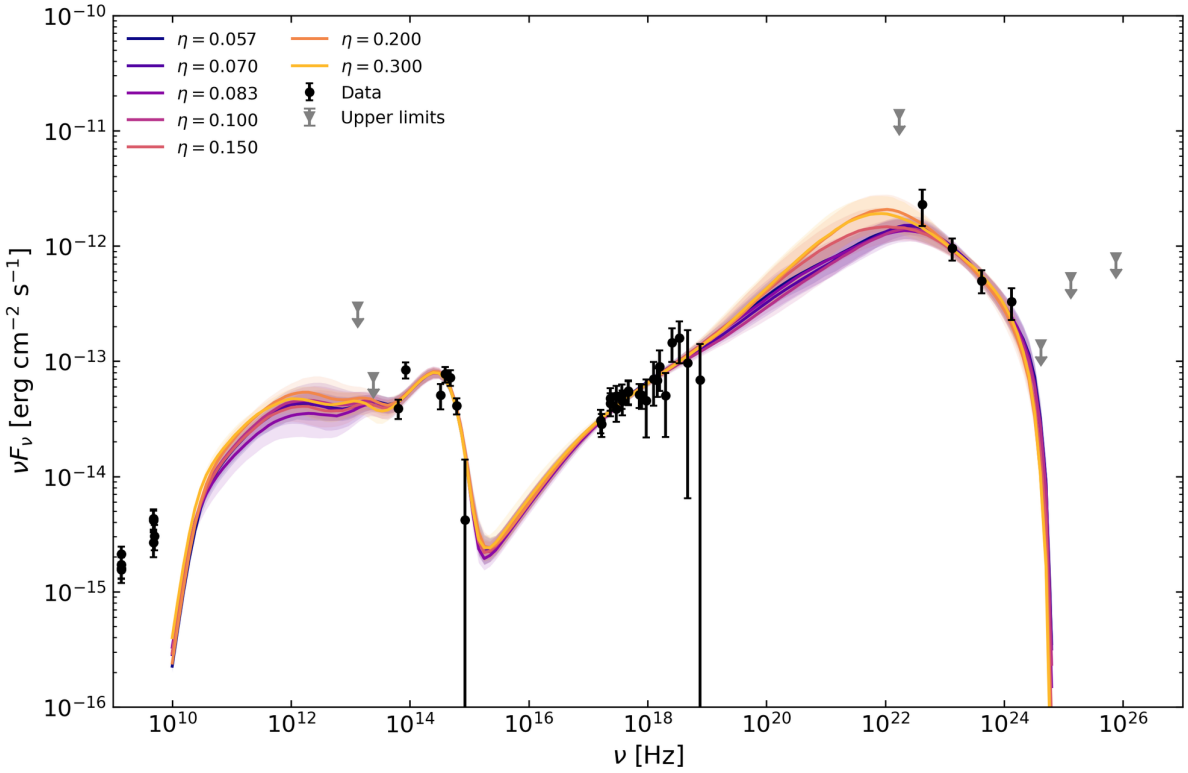


Figure 4.20: Posterior-median SEDs and 68% credible bands for the seven accretion efficiencies.

4.7.6 Robustness: Sampling the Dissipation Radius

To test the effect of fixing R_H, we ran an additional chain for the baseline model in which R_H is sampled together with the other seven parameters (eight parameters, same sampler settings, started from the baseline Minuit fit of Table 4.5). The posterior gives $R_H = (1.32^{+0.24}_{-0.25}) \times 10^{18}$ cm, consistent with the Minuit value of 1.20×10^{18} cm, and $R_H/R_{\rm BLR,out} = 2.9^{+0.6}_{-0.6}$. The other parameters agree with the seven-parameter run within a small fraction of their uncertainties (Table A.2 in Appendix A.3): for instance $B =$

$0.064^{+0.026}_{-0.020}$ G, $\Gamma = 18.2^{+6.0}_{-4.6}$ and $M_{\rm BH} = (1.85^{+0.46}_{-0.45}) \times 10^9\,M_\odot$. The main difference is a broader posterior on N, 1430^{+803}_{-456} cm^{-3}, which reflects the N–R_H degeneracy discussed above. This chain is less converged than the others (5.6–7.4 autocorrelation times), as expected for a parameter that is strongly correlated with N, but it shows that fixing R_H does not bias the results.

4.8 Trends with the Accretion Efficiency

Figure 4.21 shows how the main physical quantities change along the grid, with the MCMC uncertainties; the values of all the sampled parameters are given in Table 4.8. The quantities fall into three groups.

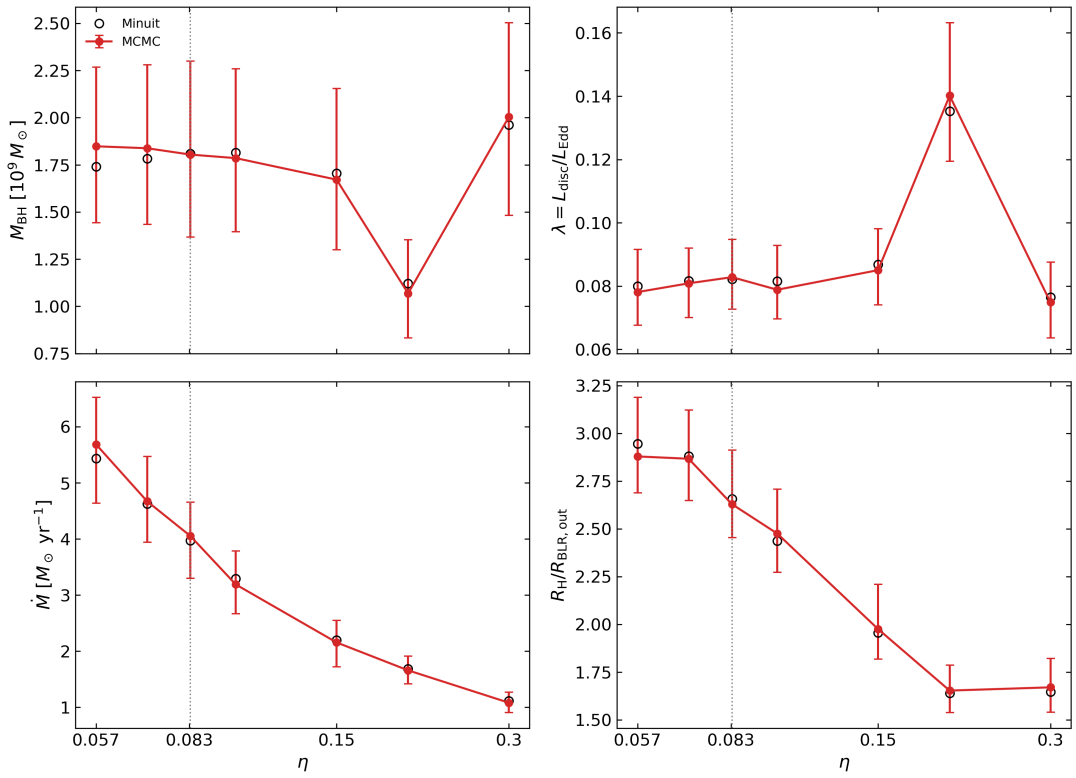


Figure 4.21: Black hole mass, Eddington ratio, accretion rate $\dot{M} = L_{\rm disc}/(\eta c^2)$ and position of the emission region relative to the BLR as a function of η: MCMC medians and 68% intervals (red) and Minuit best fits (open circles). The dotted line marks $\eta = 0.083$.

4.8.1 Quantities Fixed by the Data

The radiated disk luminosity, $L_{\rm disc} \simeq 1.8$–1.9×10^{46} erg s^{-1}, the black hole mass, $M_{\rm BH} \simeq 1.7$–$2.0 \times 10^9\,M_\odot$, and therefore the Eddington ratio, $\lambda \simeq 0.075$–0.085, are the same at all efficiencies within their uncertainties. The only exception is $\eta = 0.2$, where the $M_{\rm BH}$–$R_{\rm in}$ degeneracy of Section 4.6.5 gives $M_{\rm BH} = 1.1 \times 10^9\,M_\odot$ and $\lambda = 0.14$. The shape of the

electron distribution is also independent of η: $\gamma_{\rm break} \simeq 800$–$935$ and $p_1 \simeq 3.4$–3.65, with uncertainties larger than the differences. These are the quantities that can be quoted as properties of the source, independently of the assumed accretion efficiency.

The physical accretion rate, on the other hand, depends on η by definition, $\dot{M} = L_{\rm disc}/(\eta c^2)$: it decreases from $5.7\,M_\odot$ yr^{-1} at $\eta = 0.057$ to $4.0\,M_\odot$ yr^{-1} at $\eta = 0.083$ and $1.1\,M_\odot$ yr^{-1} at $\eta = 0.3$. Since the observed luminosity is fixed, a more efficient disk needs less mass to produce it. This is the quantity that enters the growth of the black hole in Section 4.10.

4.8.2 Geometry of the Emission Region

The clearest trend is geometric: the ratio $R_H/R_{\rm BLR,out}$ decreases from $2.88^{+0.31}_{-0.19}$ at $\eta = 0.057$ to $1.65^{+0.13}_{-0.11}$ at $\eta = 0.2$ and $1.67^{+0.15}_{-0.13}$ at $\eta = 0.3$, a change well beyond the uncertainties. As explained in Section 4.6.3, it is caused by the growth of the BLR radius with $L_{\rm D} \propto \eta$, while the emission region stays at $R_H \simeq 1.2 \times 10^{18}$ cm. The blob remains outside the BLR and well inside the torus for every η, but its distance from the BLR changes by almost a factor of two. This trend is partly a consequence of the radius–luminosity relations of Equation 4.4, which are calibrated on the luminosity rather than on η, and it should be interpreted within this prescription.

4.8.3 Jet Parameters

The jet parameters (Table 4.8) show no significant trend. The electron density shows a mild decrease, from 1872^{+430}_{-289} to 1202^{+415}_{-200} cm^{-3} between the two ends of the grid. The Minuit values follow the same trend, but each point has an uncertainty of 20–35%, and the difference between the two ends of the grid corresponds to only about 1.3σ: the trend is suggestive, but not significant. The magnetic field scatters between 0.06 and 0.10 G without a systematic trend. The bulk Lorentz factor of the Minuit fits increases from 17.7 to 21.0, but the MCMC posteriors are broad and partly limited by the prior, and the trend is not significant: what can be said is that $\Gamma \gtrsim 14$–19 at all efficiencies.

4.8.4 Summary of the Trends

The data fix the radiated luminosity of the disk, the black hole mass, the Eddington ratio and the electron distribution, and they do not depend on the assumed accretion efficiency. What changes with η is the geometry of the external photon fields relative to the emission region, and with it the origin of the γ-ray seed photons (torus for $\eta \lesssim 0.1$, BLR for $\eta \gtrsim 0.15$). These changes come from tying the radii to $L_{\rm D}$; with radii scaled to the radiated luminosity they would disappear (Section 4.4.4). The jet parameters adjust

to this change, mostly through Γ and N, but within uncertainties that are larger than the trends themselves.

4.9 Jet Energetics

4.9.1 Method

The powers carried by the jet were computed with the `energetic_report` method of `JetSeT`, which evaluates the components of Equation 3.20 for a single jet: the power in relativistic electrons P_e, in the magnetic field P_B, in cold protons P_p, and the radiated power $P_{\rm rad}$. The kinetic power is $P_{\rm kin} = P_e + P_p$ and the total power is $P_{\rm jet} = P_{\rm kin} + P_B + P_{\rm rad}$. Rather than evaluating the energetics only at the best fit, we propagated the posterior uncertainties: for each value of η, 200 parameter sets were drawn from the MCMC chain, the model was evaluated for each of them, and the energetics were computed. We quote the median and the 68% interval of the resulting distributions.

The proton power is not constrained by the SED, since cold protons do not radiate. It depends linearly on the ratio $N_{\rm H}/N_e$ between cold protons and relativistic electrons, which must be assumed. As a baseline we adopt one cold proton per relativistic electron, $N_{\rm H}/N_e = 1$ (Celotti & Ghisellini, 2008; Ghisellini et al., 2014), and we consider $N_{\rm H}/N_e = 0.1$ and 0.01 as alternatives, which correspond to jets containing a significant fraction of electron–positron pairs. Since P_p is linear in $N_{\rm H}/N_e$, it was computed once for $N_{\rm H}/N_e = 1$ and rescaled for the other values; we checked on one model that a direct evaluation gives exactly the same result.

The jet power is compared with two properties of the accretion flow: the radiated disk luminosity $L_{\rm disc}$ of Equation 4.8, and the accretion power $\dot{M}c^2 = L_{\rm disc}/\eta$, which is the total rest-mass energy flowing into the black hole per unit time for a radiative efficiency η.

4.9.2 Baseline Energetics

The results for the baseline case $\eta = 0.083$ are given in Table 4.9.

Several features stand out. The jet is strongly dominated by particles: the magnetic energy density is only about 2×10^{-3} of the energy density of the relativistic electrons, and the magnetic power is more than two orders of magnitude below the electron power. The jet is therefore far from equipartition in the emission region, as generally found for powerful FSRQs (Ghisellini et al., 2014). The radiated power is a very small fraction of the total, $P_{\rm rad}/P_{\rm jet} \approx 0.25\%$: almost all the energy is carried by the jet as kinetic energy, and only a small fraction is dissipated into radiation at the location of the emission region.

With one proton per electron, cold protons carry more than 97% of the jet power,

Table 4.9: Jet energetics for the baseline case $\eta = 0.083$ (one jet), computed from 200 posterior draws. Values are medians with 68% intervals; the disk luminosity refers to the same 200 draws and differs slightly from the value in Table 4.8, which uses the full chain.

Quantity	Symbol	Value
Magnetic field power	P_B	$1.5^{+3.9}_{-1.0} \times 10^{43}$ erg s^{-1}
Electron power	P_e	$7.5^{+4.0}_{-1.9} \times 10^{45}$ erg s^{-1}
Radiated power	$P_{\rm rad}$	$7.6^{+7.6}_{-2.8} \times 10^{44}$ erg s^{-1}
Cold proton power ($N_{\rm H}/N_e = 1$)	P_p	$3.0^{+2.4}_{-0.8} \times 10^{47}$ erg s^{-1}
Total jet power ($N_{\rm H}/N_e = 1$)	$P_{\rm jet}$	$3.1^{+2.4}_{-0.9} \times 10^{47}$ erg s^{-1}
Total jet power ($N_{\rm H}/N_e = 0.1$)	$P_{\rm jet}$	$3.8^{+2.9}_{-1.0} \times 10^{46}$ erg s^{-1}
Total jet power ($N_{\rm H}/N_e = 0.01$)	$P_{\rm jet}$	$1.2^{+0.7}_{-0.3} \times 10^{46}$ erg s^{-1}
Disk luminosity	$L_{\rm disc}$	$1.95^{+0.27}_{-0.35} \times 10^{46}$ erg s^{-1}
Accretion power	$\dot{M}c^2$	$2.33^{+0.33}_{-0.42} \times 10^{47}$ erg s^{-1}
Magnetisation	U'_B/U'_e	$2.2^{+3.0}_{-1.4} \times 10^{-3}$
Radiative efficiency of the jet	$P_{\rm rad}/P_{\rm jet}$	$2.5^{+0.9}_{-0.5} \times 10^{-3}$
Jet-to-disk ratio ($N_{\rm H}/N_e = 1$)	$P_{\rm jet}/L_{\rm disc}$	16^{+13}_{-5}
Jet-to-disk ratio ($N_{\rm H}/N_e = 0.1$)	$P_{\rm jet}/L_{\rm disc}$	2.0
Jet to accretion power ($N_{\rm H}/N_e = 1$)	$P_{\rm jet}/\dot{M}c^2$	$1.35^{+1.1}_{-0.4}$

and the total, $P_{\rm jet} \approx 3 \times 10^{47}$ erg s^{-1}, exceeds the disk luminosity by a factor of about 16. Even for $N_{\rm H}/N_e = 0.1$ the jet power is twice the disk luminosity, and only for $N_{\rm H}/N_e = 0.01$ does it become comparable to it. The result $P_{\rm jet} > L_{\rm disc}$ agrees with the finding of Ghisellini et al. (2014) for powerful *Fermi* blazars.

4.9.3 Energetics Across the Grid

The energetics for all efficiencies are summarised in Table 4.10.

The jet powers are similar at all efficiencies, within uncertainties of 30–60%: $P_{\rm jet} \approx$ (3–5.5) $\times 10^{47}$ erg s^{-1} for $N_{\rm H}/N_e = 1$. Since the radiated disk luminosity is also the same for all η, the ratio $P_{\rm jet}/L_{\rm disc}$ does not depend on the accretion efficiency: it is about 16–29 for $N_{\rm H}/N_e = 1$, 2–3.6 for $N_{\rm H}/N_e = 0.1$ and 0.6–1 for $N_{\rm H}/N_e = 0.01$.

4.9.4 The Energy Budget of the Jet

A more informative comparison is with the accretion power, $P_{\rm jet}/\dot{M}c^2 = \eta\, P_{\rm jet}/L_{\rm disc}$, which does depend on η because a more efficient disk requires a lower accretion rate. For one proton per electron, the power of a single jet exceeds the accretion power at every efficiency, from 1.2 times at $\eta = 0.057$ to almost 9 times at $\eta = 0.3$. A jet that carries more power than the total rest-mass energy accreted cannot be powered by accretion alone.

Table 4.10: Jet energetics across the accretion-efficiency grid (one jet, medians and 68% intervals from 200 posterior draws). $P_{\rm jet}$ is given for the three values of $N_{\rm H}/N_e$. The last two rows give the ratio between the jet power and the accretion power, $P_{\rm jet}/\dot{M}c^2 = \eta\, P_{\rm jet}/L_{\rm disc}$.

η	0.057	0.070	**0.083**	0.100	0.150	0.200	0.300
P_e [10^{45} erg s^{-1}]	$8.1^{+4.8}_{-3.0}$	$9.5^{+5.2}_{-2.6}$	$\mathbf{7.5^{+4.0}_{-1.9}}$	$11.0^{+3.6}_{-3.8}$	$9.9^{+4.4}_{-3.0}$	$9.6^{+3.3}_{-4.3}$	$11.4^{+5.1}_{-2.8}$
$P_{\rm rad}$ [10^{45} erg s^{-1}]	$1.1^{+1.3}_{-0.5}$	$1.2^{+1.2}_{-0.6}$	$\mathbf{0.8^{+0.8}_{-0.3}}$	$1.3^{+0.9}_{-0.6}$	$1.3^{+1.1}_{-0.6}$	$2.0^{+1.8}_{-1.1}$	$1.8^{+1.4}_{-0.8}$
U'_B/U'_e [10^{-3}]	$3.9^{+3.5}_{-1.9}$	$4.0^{+4.3}_{-2.4}$	$\mathbf{2.2^{+3.0}_{-1.4}}$	$3.7^{+3.0}_{-1.7}$	$3.2^{+4.0}_{-1.6}$	$8.2^{+7.6}_{-4.7}$	$5.2^{+3.2}_{-2.9}$
$P_{\rm jet}$, $N_{\rm H}/N_e = 1$ [10^{47}]	$3.6^{+2.5}_{-1.5}$	$4.3^{+2.4}_{-1.5}$	$\mathbf{3.1^{+2.4}_{-0.9}}$	$4.8^{+2.0}_{-1.9}$	$4.3^{+2.2}_{-1.3}$	$4.4^{+1.9}_{-2.1}$	$5.5^{+2.3}_{-1.6}$
$P_{\rm jet}$, $N_{\rm H}/N_e = 0.1$ [10^{46}]	$4.5^{+3.0}_{-1.8}$	$5.3^{+3.0}_{-1.8}$	$\mathbf{3.8^{+2.9}_{-1.0}}$	$6.0^{+2.1}_{-2.4}$	$5.3^{+2.8}_{-1.6}$	$5.5^{+2.3}_{-2.6}$	$6.7^{+2.8}_{-2.0}$
$P_{\rm jet}$, $N_{\rm H}/N_e = 0.01$ [10^{46}]	$1.3^{+0.8}_{-0.5}$	$1.5^{+0.9}_{-0.4}$	$\mathbf{1.2^{+0.7}_{-0.3}}$	$1.7^{+0.6}_{-0.7}$	$1.6^{+0.7}_{-0.5}$	$1.6^{+0.6}_{-0.8}$	$1.9^{+0.9}_{-0.5}$
$P_{\rm jet}/L_{\rm disc}$, $N_{\rm H}/N_e = 1$	22	24	**16**	25	24	23	29
$P_{\rm jet}/L_{\rm disc}$, $N_{\rm H}/N_e = 0.1$	2.7	2.9	**2.0**	3.1	3.0	2.9	3.6
$P_{\rm jet}/L_{\rm disc}$, $N_{\rm H}/N_e = 0.01$	0.76	0.83	**0.59**	0.90	0.87	0.86	1.00
$P_{\rm jet}/\dot{M}c^2$, $N_{\rm H}/N_e = 1$	1.2	1.7	**1.4**	2.5	3.6	4.6	8.7
$P_{\rm jet}/\dot{M}c^2$, $N_{\rm H}/N_e = 0.1$	0.15	0.21	**0.17**	0.31	0.45	0.57	1.1

Two possibilities remain. The first is that the jet is powered by the rotational energy of the black hole through the Blandford–Znajek mechanism (Blandford & Znajek, 1977): in magnetically arrested disks, numerical simulations show that jets can carry more than the accretion power if the black hole spins rapidly (Tchekhovskoy et al., 2011). The second is that the jet contains fewer cold protons than relativistic electrons, that is a significant fraction of pairs.

In general, energy conservation requires

$$\eta_{\rm tot}\,\dot{M}c^2 = L_{\rm disc} + 2P_{\rm jet} \quad\Longrightarrow\quad \eta_{\rm tot} = \eta\left(1 + \frac{2P_{\rm jet}}{L_{\rm disc}}\right), \tag{4.11}$$

where $\eta_{\rm tot}$ is the total energy released per unit accreted rest-mass energy and η is the fraction emitted as disk radiation. This is the energy budget $\eta_{\rm tot} = \eta_r + \eta_j$ of Jolley & Kuncic (2008) and Ghisellini et al. (2013), with $\eta_r = L_{\rm disc}/\dot{M}c^2 = \eta$ and $\eta_j = 2P_{\rm jet}/\dot{M}c^2$. $P_{\rm jet}$ is the power of one jet, as in Table 4.10; the factor 2 accounts for the counter-jet, which is powered by the same engine. The jet power can come from two sources.

Jets powered by accretion. If the jets take their energy from the accreting gas, $\eta_{\rm tot}$ cannot exceed the Novikov–Thorne efficiency of the black hole, at most $\simeq 0.32$ for $a \simeq 0.998$ (Thorne, 1974). A standard disk radiates all of this energy, so the jets can

be fed only if the disk radiates less than its Novikov–Thorne efficiency (Jolley & Kuncic, 2008; Ghisellini et al., 2013). For $N_\mathrm{H}/N_e = 0.1$, $2P_\mathrm{jet}/L_\mathrm{disc} \approx 4$–7, and Equation 4.11 requires $\eta \lesssim 0.04$–0.06 even for a maximally spinning black hole: the disk would radiate no more than that of a non-rotating black hole, and 80–90% of the energy released would go into the jets. For $N_\mathrm{H}/N_e = 0.01$ the condition is $\eta \lesssim 0.11$–0.15, again with a nearly maximal spin, and for $N_\mathrm{H}/N_e = 1$ it is $\eta \lesssim 0.01$.

Jets powered by the spin of the black hole. Alternatively, the disk is a standard one, radiating with $\eta = \eta_\mathrm{NT}(a)$, and the jets are powered by the rotation of the black hole through the Blandford–Znajek mechanism (Blandford & Znajek, 1977). In the magnetically arrested disks of Tchekhovskoy et al. (2011) the jet efficiency grows roughly as the square of the angular velocity of the horizon and reaches about $1.4\,\dot{M}c^2$ for $a = 0.99$. Scaling their result, the ratio between the power of the jets and the luminosity of a standard disk increases from about 1.6 at $a = 0.5$ to 3 at $a = 0.7$, and reaches a maximum of about 5.3 near $a = 0.99$. Requiring this ratio to equal the observed $2P_\mathrm{jet}/L_\mathrm{disc}$ gives a joint constraint on the proton content and the spin, summarised in Table 4.11. We take $N_\mathrm{H}/N_e = 0.1$, about five electron–positron pairs per proton, as the fiducial composition, and use the other values to bracket it.

Table 4.11: Joint constraint on the proton content of the jets and the spin of the black hole, for a standard disk and jets powered by the spin (both jets included), computed for the jet power of the baseline model ($\eta = 0.083$). The spread of the jet power across the grid reflects the parameterisation of the external fields (Section 4.4.4). The column $\langle\lambda\rangle$ gives the average Eddington ratio needed to grow a seed of $10^5\,M_\odot$ into $M_\mathrm{BH} = 1.8 \times 10^9\,M_\odot$ by $z = 3.65$.

N_H/N_e	$2P_\mathrm{jet}/L_\mathrm{disc}$	a	η	$\langle\lambda\rangle$	Comment
1	32	—	—	—	Excluded for a standard disk at any spin
0.1	4.0	0.82	0.13	≈ 1	Fiducial, median jet power, $\gamma_\mathrm{min} = 2.1$: Eddington-limited growth throughout
	2.9	0.69	0.10	0.47	Jet power at the lower end of its 68% interval
	2.2–2.8	0.59–0.67	0.09–0.10	0.33–0.44	Median jet power, $\gamma_\mathrm{min} = 5$–10
0.01	1.2	0.42	0.076	0.25	About 50 pairs per proton
0	0.9	0.36	0.072	0.23	No cold protons: minimum spin

For the fiducial composition, the baseline jet power, $2P_\mathrm{jet}/L_\mathrm{disc} \approx 4.0$, requires $a \approx$

0.82 and $\eta \approx 0.13$. Such a black hole could have grown from a heavy seed only by accreting at the Eddington limit for its whole life ($\langle\lambda\rangle \approx 1$). A jet power at the lower end of its 68% interval, about 30% below the median, gives $a \approx 0.7$, $\eta \approx 0.10$ and $\langle\lambda\rangle \approx 0.47$. The proton power also depends on the minimum Lorentz factor of the electrons, which was fixed at $\gamma_{\rm min} = 2.1$: for $p = 1.59$ the number of electrons, and with it the power in cold protons, scales as $\gamma_{\rm min}^{-0.59}$, and for $\gamma_{\rm min} = 5$–10 the median jet power is supplied by $a \approx 0.6$–0.67 with $\eta \approx 0.09$–0.10 and $\langle\lambda\rangle \approx 0.33$–0.44. The fiducial composition therefore points to a black hole that spins rapidly but not maximally, $a \approx 0.6$–0.7, with a disk radiating $\eta \approx 0.09$–0.10. If the disk is required to extend to the ISCO, as a standard disk would, the black hole mass becomes $(7.5\text{–}10)\times10^9\,M_\odot$ (Section 4.6.5) and the required $\langle\lambda\rangle$ increases by about 15%: the solution with $a \approx 0.8$ then needs super-Eddington accretion, while the range $a \approx 0.6$–0.7 remains sub-Eddington, with $\langle\lambda\rangle \approx 0.38$–0.54. These spins are somewhat below the equilibrium spin $a_{\rm eq} \approx 0.8$ found by Ricarte et al. (2023) for near-Eddington magnetically arrested disks.

One proton per electron requires jets 32 times more powerful than the disk, an order of magnitude more than a spinning black hole with a standard disk can provide. It is possible only if the spin supplies the jets and the disk radiates less than about 2–3% of the accreted rest-mass energy while accreting at a few tenths of the Eddington rate, a regime in which standard disks are radiatively efficient. The jets would then have to carry away almost all the binding energy of the disk, an extreme form of the scenario of Jolley & Kuncic (2008). This composition, usually adopted in studies of blazar energetics (Ghisellini et al., 2014), is therefore difficult to reconcile with a standard disk in MG3 J163554+3629, unless $\gamma_{\rm min}$ is much larger than assumed: $\gamma_{\rm min} \approx 50$ would reduce the proton power by a factor of about six and bring it close to the maximum that the spin can supply. Fewer cold protons than the fiducial value would allow a lower spin, $a \approx 0.4$ for $N_{\rm H}/N_e = 0.01$, and even jets without cold protons require $a \approx 0.35$. A low spin, $a \lesssim 0.3$, is therefore disfavoured for any composition: it would need jets weaker than estimated here, a jet efficiency at low spin about twice that of the scaling used, or a disk that gives part of its energy to the jets.

These numbers are order-of-magnitude estimates. They assume that the jet power estimated in the emission region equals the power extracted at the base of the jet, and that the jets are produced as in simulated magnetically arrested disks. A thin disk at $\lambda \approx 0.08$ probably carries less magnetic flux than the simulated ones and would produce weaker jets, so the spins in Table 4.11 are best read as lower limits, and the exclusion of $N_{\rm H}/N_e = 1$ as conservative.

Whatever the proton content, the jets of MG3 J163554+3629 carry a power at least comparable to the radiated disk luminosity. Unless the disk radiates far below its standard efficiency, part of this power has to be extracted from the rotation of the black hole. For the fiducial composition the spin required is $a \approx 0.6$–0.7, rapid but not maximal; lower

spins require fewer cold protons than one per ten electrons. This conclusion does not depend on the SED fits, which cannot constrain η. Its consequences for the growth of the black hole are discussed in the next section.

4.10 Black Hole Growth and Seed Mass

4.10.1 Method

We now use the black hole mass and the Eddington ratio of each posterior to reconstruct the possible growth history of the black hole. If a fraction η of the accreted rest-mass energy is radiated, the black hole grows at the rate $\dot{M}_{\rm BH} = (1-\eta)\dot{M}$, and with $L_{\rm disc} = \eta\dot{M}c^2 = \lambda L_{\rm Edd}$ one obtains (Shapiro, 2005)

$$\frac{dM_{\rm BH}}{dt} = \frac{1-\eta}{\eta}\,\lambda\,\frac{M_{\rm BH}}{t_{\rm Edd}}, \qquad t_{\rm Edd} = \frac{M_{\rm BH}c^2}{L_{\rm Edd}} = 0.450\ {\rm Gyr}, \tag{4.12}$$

where $t_{\rm Edd}$ is computed with the same Eddington luminosity used for λ (Equation 4.9, pure hydrogen). For constant η and λ the mass grows exponentially with e-folding time

$$\tau_{\rm Salp} = \frac{\eta}{1-\eta}\,\frac{t_{\rm Edd}}{\lambda}, \tag{4.13}$$

and in general

$$M_{\rm BH}(t_{\rm obs}) = M_{\rm seed}\,\exp\!\left[\frac{1-\eta}{\eta}\,\frac{1}{t_{\rm Edd}}\int_{t_{\rm seed}}^{t_{\rm obs}}\lambda(t)\,dt\right]. \tag{4.14}$$

which for constant λ reduces to Equation (16) of Shapiro (2005), with his radiative efficiency $\epsilon_M = \eta$ and Eddington fraction $\epsilon_L = \lambda$. Only the time integral of λ enters the final mass. Note also that the product $\lambda/t_{\rm Edd} = L_{\rm disc}/(M_{\rm BH}c^2)$ does not depend on the choice of the Eddington luminosity: for growth at the observed Eddington ratio the results are the same whether or not the helium content is included through μ_e (Equation 3.42). The choice matters only when $\lambda = 1$ is imposed, and the pure-hydrogen value adopted here is the more conservative one.

We place the seeds at $z = 30$ and compute cosmic times with the Planck 2018 cosmology (Planck Collaboration, 2020): the source is observed at $t_{\rm obs} = 1.715$ Gyr and $t_{\rm seed} = 0.099$ Gyr, leaving $\Delta t = 1.616$ Gyr for the growth. We compare the results with heavy seeds of 10^4–$10^6\,M_\odot$, formed by the direct collapse of gas (Volonteri, 2010), and adopt $10^5\,M_\odot$ as reference value. We do not consider light seeds, the 10–$10^3\,M_\odot$ remnants of Population III stars: their early growth is strongly limited by radiative feedback in their host minihaloes (Johnson & Bromm, 2007; Alvarez et al., 2009; Smith et al., 2018), and they would need to accrete close to the Eddington limit, almost without interruption, from their formation. For each η the calculation uses the posterior samples of $M_{\rm BH}$ and

λ, and the uncertainties quoted below are the 16th and 84th percentiles.

4.10.2 Growth at the Observed Eddington Ratio

The simplest assumption is that the black hole always accreted at its present rate, $\lambda = \lambda_{\rm obs} \simeq 0.08$. The resulting e-folding times are long, 350–930 Myr for $\eta \leq 0.15$ and 2.6 Gyr for $\eta = 0.3$, and the black hole could have grown only by a small factor: the seed required at $z = 30$ would be $M_{\rm seed} \simeq 2 \times 10^7\, M_\odot$ for $\eta = 0.057$ and $\simeq 10^9\, M_\odot$ for $\eta = 0.3$ (Table 4.12, Figure 4.22). These values are one to three orders of magnitude above the heaviest seeds predicted by any formation channel. Continuous accretion at the present Eddington ratio is therefore excluded: the present state of the source cannot be representative of its growth history, and the black hole must have accreted at a much higher rate in the past.

Table 4.12: Black hole growth from $z = 30$ to $z = 3.65$ for each accretion efficiency. $\tau_{\rm Salp}$ is the e-folding time for $\lambda = 1$ and for the observed Eddington ratio; $\mathcal{N}$ is the number of e-foldings available at $\lambda = 1$; the seed masses are those required at $z = 30$ for constant λ.

η	$\lambda_{\rm obs}$	$\tau_{\rm Salp}$ [Myr]		$\mathcal{N}$	$\log M_{\rm seed}/M_\odot$	
		$\lambda = 1$	$\lambda_{\rm obs}$	$(\lambda = 1)$	$\lambda = 1$	$\lambda_{\rm obs}$
0.057	0.078	27	348	59.5	-16.6	$7.3^{+0.3}_{-0.5}$
0.070	0.081	34	418	47.7	-11.5	$7.6^{+0.3}_{-0.3}$
0.083	0.083	41	494	39.5	-7.9	$7.8^{+0.3}_{-0.3}$
0.100	0.079	50	634	32.3	-4.8	$8.1^{+0.3}_{-0.2}$
0.150	0.085	79	933	20.4	0.4	$8.5^{+0.2}_{-0.2}$
0.200	0.140	112	802	14.4	2.8	$8.2^{+0.2}_{-0.3}$
0.300	0.075	193	2574	8.4	5.7	$9.0^{+0.2}_{-0.1}$

4.10.3 Eddington-Limited Growth

The opposite limit is growth at the Eddington rate during the whole time available. In this case the result depends strongly on η (Table 4.12). For $\eta \leq 0.1$ there are more than 30 e-foldings between $z = 30$ and $z = 3.65$, and a heavy seed could grow into the observed mass even with an average Eddington ratio of $\langle\lambda\rangle \simeq 0.2$–$0.3$, two to four times the present value. For $\eta = 0.3$, instead, a seed of $\sim 5 \times 10^5\, M_\odot$ is required even at the Eddington limit, and the required average Eddington ratio exceeds unity.

The condition that a seed of mass $M_{\rm seed}$ reaches $M_{\rm BH}$ at Eddington defines a critical efficiency,

$$\eta_{\rm crit} = \left[1 + \frac{t_{\rm Edd}}{\Delta t}\, \ln \frac{M_{\rm BH}}{M_{\rm seed}}\right]^{-1}. \tag{4.15}$$

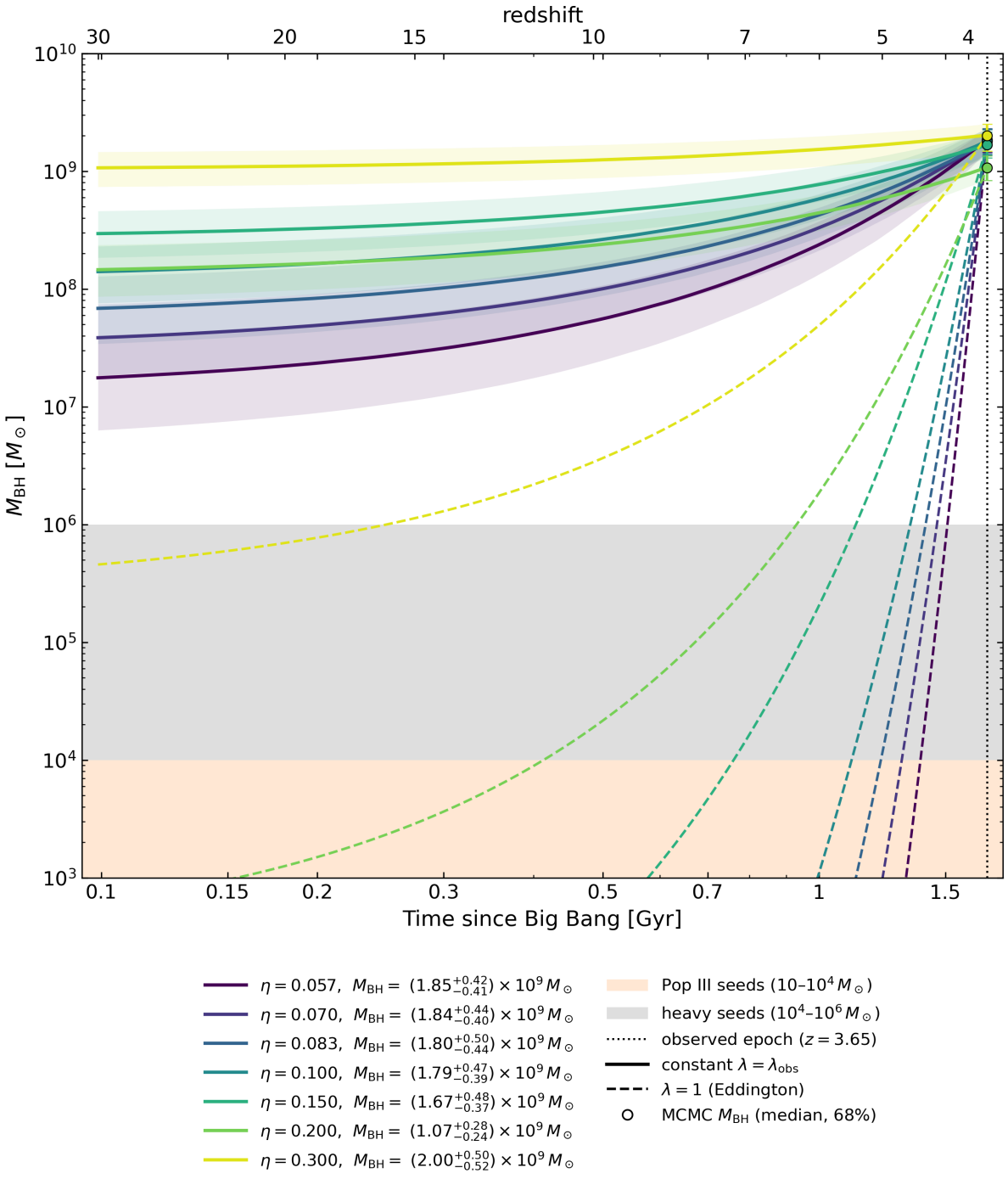


Figure 4.22: Black hole mass as a function of cosmic time for the seven accretion efficiencies, traced back from the observed mass at $z = 3.65$. Solid lines with 68% bands: constant Eddington ratio equal to the observed one. Dashed lines: Eddington-limited growth ($\lambda = 1$). Points with error bars at the observed epoch (dotted vertical line) show the MCMC black hole mass for each efficiency (median and 68% interval), with values given in the legend. The orange and grey shaded regions mark the mass ranges of Pop III remnant seeds (10–$10^4\, M_\odot$) and heavy (direct-collapse) seeds (10^4–$10^6\, M_\odot$), respectively.

For $M_{\rm BH} = 1.8 \times 10^9\,M_\odot$ we obtain $\eta_{\rm crit} \simeq 0.27$ for a seed of $10^5\,M_\odot$ and $\eta_{\rm crit} \simeq 0.32$ for a seed of $10^6\,M_\odot$, corresponding to spins $a \simeq 0.99$ and $a \simeq 0.998$. If the helium content is included in the Eddington luminosity, the limits become 0.30 and 0.35. A black hole radiating with an efficiency above these values cannot have grown from the corresponding seed by $z = 3.65$, even accreting at the Eddington limit for 1.6 Gyr.

These limits must be compared with the spin expected for the black hole. Accretion through a disk of fixed orientation spins the black hole up: a non-rotating black hole reaches the maximum spin after increasing its mass by a factor of about $\sqrt{6}$ (Bardeen, 1970), and photon capture limits the spin to $a \simeq 0.998$, with an efficiency of about 0.3 (Thorne, 1974). If this were the whole story, MG3 J163554+3629, which grew by at least three orders of magnitude, would be maximally spinning. Its mass could then be reproduced only from the most massive direct-collapse seeds, $\sim 10^6\,M_\odot$, accreting at the Eddington limit without interruption for 1.6 Gyr, which is very unlikely.

Spin-up by the disk is not the whole story, however. A jet powered by the black hole also exerts a torque on it and spins it down. Simulations of magnetically arrested disks give an equilibrium spin $a_{\rm eq} \approx 0.8$ near the Eddington limit (Ricarte et al., 2023), and $a_{\rm eq} \lesssim 0.1$ for geometrically thick disks, where accreting about 20% of its mass is enough to spin a black hole down from $a = 1$ to $a = 0.2$ (Lowell et al., 2024). Accretion in many randomly oriented episodes also keeps the spin low (King et al., 2008). For a source with jets as powerful as those of MG3 J163554+3629, a black hole kept at maximal spin is therefore unlikely, although this argument does not fix the spin to a particular value.

4.10.4 Growth with a Declining Eddington Ratio

Since only the time-averaged Eddington ratio matters, the observed mass can be reached if the black hole accreted at a higher rate at early times and then slowed down to its present rate. This behaviour is expected on physical grounds. Models of self-regulated black hole growth predict that, once triggered, accretion proceeds at about the Eddington rate until feedback removes the gas, after which the luminosity declines with time (Hopkins & Hernquist, 2009). Observationally, quasars at $z \gtrsim 6$ have a median Eddington ratio of about 0.7 (Mazzucchelli et al., 2023), and the characteristic Eddington ratio of the population decreases towards lower redshift and higher masses (Shankar et al., 2013).

As a simple description we assume an Eddington ratio that declines exponentially,

$$\lambda(t) = \min\left[1,\ \lambda_{\rm obs}\, e^{(t_{\rm obs}-t)/\tau_{\rm dec}}\right], \tag{4.16}$$

so that λ reaches the observed value at $z = 3.65$ and is capped at the Eddington limit at early times. The decline timescale $\tau_{\rm dec}$ is fixed by requiring that the observed mass is reached at $z = 3.65$. For a heavy seed of $10^5\,M_\odot$ and $\eta \leq 0.1$ the solutions have $\tau_{\rm dec} \approx 0.7$–$1.2$ Gyr, with $\lambda \approx 0.3$–0.8 at $z = 30$ and $\lambda \approx 0.15$–0.25 at $z = 6$ (Figure 4.23).

For $\eta = 0.15$ and 0.2 the black hole must accrete at the Eddington limit until $z \approx 12$ and $z \approx 7.5$, respectively, while for $\eta = 0.3$ no solution exists.

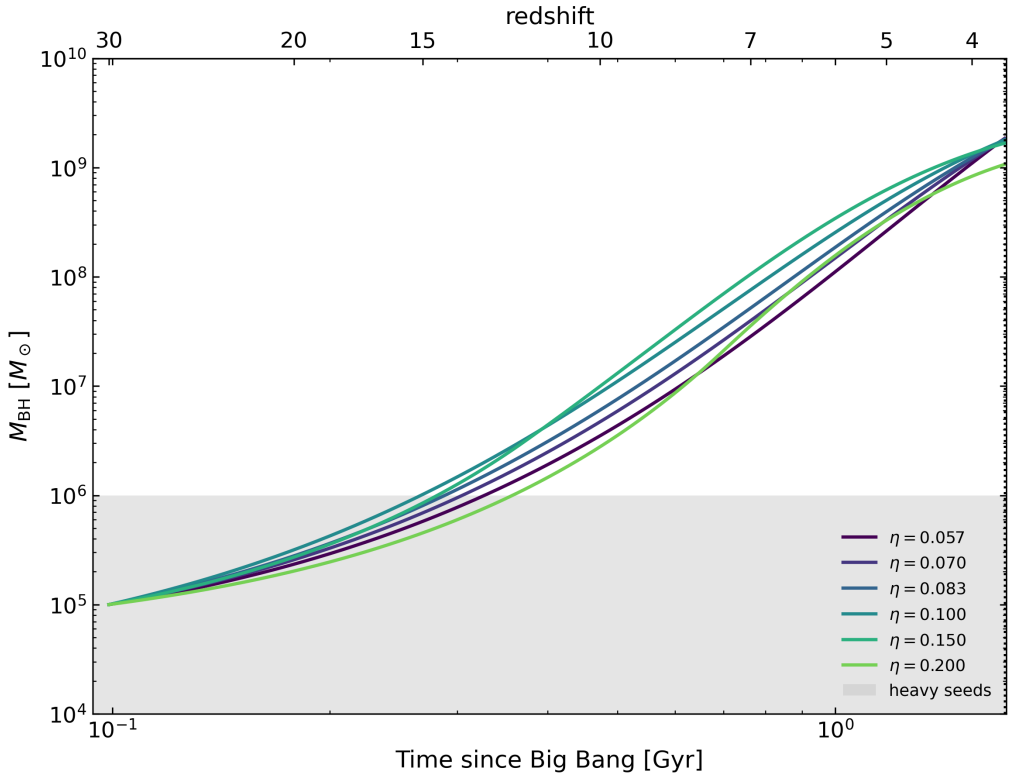


Figure 4.23: Black hole mass as a function of time for an Eddington ratio declining as in Equation 4.16, for a heavy seed of $10^5\,M_\odot$ formed at $z = 30$ and for each efficiency for which a solution exists. All tracks start from the same seed by construction; the efficiency determines the decline timescale τ_{dec} needed to reach the observed mass. The shaded region marks the mass range of heavy seeds.

This history is not a fit: with a single epoch of observation the shape of $\lambda(t)$ cannot be measured, and any history with the same average Eddington ratio gives the same final mass, for example Eddington-limited episodes with a duty cycle of 15–30% for $\eta \lesssim 0.1$. What the calculation shows is that the black hole must have accreted at a higher rate in the past: for $\eta \lesssim 0.1$ an average Eddington ratio two to four times the present one is sufficient, while for higher efficiencies the black hole must have spent most of its early life at the Eddington limit. On the smooth tracks of Figure 4.23, λ has already dropped to 0.15–0.25 at $z = 6$, below the median of 0.7 found for quasars at $z \gtrsim 6$ (Mazzucchelli et al., 2023). Flux-limited samples favour the most rapidly accreting objects, but the comparison also suggests that the decline need not have been smooth, and that a later phase of near-Eddington accretion is equally possible.

These solutions neglect the jets. For the fiducial composition, with spin-powered jets, the required average Eddington ratio rises to about 0.33–0.47, between the values needed without jets for $\eta = 0.1$ and $\eta = 0.15$. The black hole must then start at or near the

Eddington limit at $z \approx 30$ and remain above half of it until $z \approx 7$–10, declining on a timescale $\tau_{\rm dec} \approx 0.55$–$0.7$ Gyr (Section 4.10.5).

4.10.5 The Effect of the Jets on Growth

The constraints derived so far ignore the energy carried by the jets. Whether it comes from accretion or from the rotation of the black hole, this energy is not added to the black hole, which grows at $(1 - \eta_{\rm tot})\dot{M}$ with $\eta_{\rm tot}$ given by Equation 4.11. The radiation pressure that limits the accretion is instead set by $L_{\rm disc}$ only (Jolley & Kuncic, 2008), so that at a given Eddington ratio

$$\tau_{\rm Salp} = \frac{\eta}{1 - \eta_{\rm tot}} \frac{t_{\rm Edd}}{\lambda}. \tag{4.17}$$

At a fixed radiative efficiency the jets always slow the growth. For the baseline model, $\eta = 0.083$ and $M_{\rm BH} = 1.8 \times 10^9\,M_\odot$, the average Eddington ratio needed to grow a seed of $10^5\,M_\odot$ rises from 0.25 without jets to 0.28 for $N_{\rm H}/N_e = 0.01$ and 0.39 for $N_{\rm H}/N_e = 0.1$. The second value is only an illustration, since a black hole with $\eta = 0.083$ cannot supply jets of this power (Table 4.11).

A fair comparison between spins must include the jets in every case. Table 4.13 gives the average Eddington ratio needed to grow a seed of $10^5\,M_\odot$ for the two scenarios of Section 4.9.4: a maximally spinning black hole whose jets are powered by accretion, with a disk radiating $\eta = 0.32/(1 + 2P_{\rm jet}/L_{\rm disc})$ as proposed by Ghisellini et al. (2013), and a black hole with a standard disk and jets powered by its spin.

Table 4.13: Average Eddington ratio needed to grow a seed of $10^5\,M_\odot$ at $z = 30$ into the observed mass at $z = 3.65$ ($M_{\rm BH} = 1.8 \times 10^9\,M_\odot$), with both jets included.

Scenario	$N_{\rm H}/N_e$	a	η	$\langle\lambda\rangle$
Maximal spin, accretion-powered jets	0.1	0.998	0.04–0.06	0.16–0.26
	0.01	0.998	0.11–0.15	0.43–0.59
Standard disk, spin-powered jets	0.1	0.6–0.7	0.09–0.10	0.33–0.47
	0.01	≈ 0.4	≈ 0.076	≈ 0.25
No jets (reference)	—	0–0.67	0.057–0.1	0.16–0.30

Which scenario grows faster depends on the proton content: with ten percent cold protons the maximally spinning black hole grows faster, with one percent the moderately spinning one. The growth of the black hole therefore does not determine its spin. What all the viable scenarios have in common is a radiative efficiency well below the maximal-spin value of $\simeq 0.3$: $\eta \lesssim 0.1$ if the average Eddington ratio was about 0.3, and $\eta \lesssim 0.15$ if it was 0.5–0.6. The arguments against a maximal spin come from elsewhere: the spin-down

by the jets discussed in Section 4.10.3, and the extreme disk required by the maximal-spin scenario, which would radiate no more than that of a non-rotating black hole while 80–90% of the energy released goes into the jets.

4.10.6 Summary of the Growth Constraints

The growth analysis leads to three conclusions. First, the present Eddington ratio of MG3 J163554+3629, $\lambda \simeq 0.08$, cannot have been maintained over the history of the black hole: an earlier phase of faster accretion is required. Second, a black hole radiating with the maximal-spin efficiency, $\eta \approx 0.3$, could have produced the observed mass only from the most massive direct-collapse seeds accreting at the Eddington limit without interruption, which is very unlikely. The radiative efficiency must be well below this value, $\eta \lesssim 0.1$ for an average Eddington ratio of about 0.3. Third, once the jets are included in every scenario, the growth does not determine the spin: a maximally spinning black hole with accretion-powered jets and a moderately spinning one with spin-powered jets reach the observed mass with comparable Eddington ratios, and which one grows faster depends on the proton content of the jets. Episodes of super-Eddington accretion (Alexander & Natarajan, 2014; Madau et al., 2014) are another way to reach the observed mass.

Chapter 5

Results and Conclusions

This chapter collects the results of Chapter 4 into a physical picture of MG3 J163554+3629 and discusses their implications. Section 5.1 describes the jet, the electron population and the central engine. Section 5.2 returns to the main question of this thesis, the separation between synchrotron self-Compton and external Compton emission, and to the role of the *NuSTAR* data. Section 5.3 discusses what the analysis can and cannot say about the accretion efficiency and the spin of the black hole. Section 5.4 addresses the coupling between accretion and jet, Section 5.5 places the source in the population of high-redshift blazars, and Section 5.6 discusses the growth of the black hole. The systematic uncertainties are reviewed in Section 5.7, the conclusions are summarised in Section 5.8, and Section 5.9 outlines possible future work.

5.1 The Physical Picture of MG3 J163554+3629

5.1.1 The Emission Region

The broadband SED of MG3 J163554+3629, from 10^{10} Hz to the highest *Fermi*-LAT energies, is well described by a one-zone leptonic model with a broken power-law electron distribution, external Compton scattering of disk, BLR and torus photons, and EBL absorption. For the baseline efficiency $\eta = 0.083$ the fit gives $\chi^2 = 26.3$ for 27 degrees of freedom. The emission region is located at $R_H \simeq 1.2{\times}10^{18}$ cm from the black hole, outside the BLR ($R_{\rm BLR,out} \simeq 4.5{\times}10^{17}$ cm) and well inside the dusty torus ($R_{\rm DT} \simeq 2.8{\times}10^{19}$ cm), and it has a radius $R \simeq 6 \times 10^{16}$ cm. The magnetic field is $B = 0.059^{+0.025}_{-0.016}$ G and the bulk Lorentz factor $\Gamma = 17.9^{+6.5}_{-4.2}$; at the viewing angle of 3° this corresponds to a Doppler factor close to its maximum, $\delta \simeq 19$.

The synchrotron component peaks in the far-infrared, around 10^{12} Hz, where only upper limits are available, and the optical and ultraviolet emission is dominated by the accretion disk. The high-energy component peaks at about 10^{22} Hz in the observer frame, and the Compton dominance is about 36, as expected for a powerful low-synchrotron-

peaked FSRQ.

5.1.2 The Emitting Electrons

The electron distribution breaks at $\gamma_{\rm break} \simeq 800$–$950$ and steepens above the break to $p_1 \simeq 3.4$–3.65, with uncertainties of about 30% and 10% respectively. Both values are the same for all accretion efficiencies. The difference $p_1 - p \simeq 1.8$ is larger than the change of one unit expected for a pure cooling break; the break therefore encodes both radiative cooling and a change in the acceleration efficiency, as already noted in Section 3.2.5. The electron density is $N \simeq 1200$–1900 cm^{-3}, with a mild decrease at high η that is not statistically significant. N and B are anticorrelated at all efficiencies, as expected from the dependence of the synchrotron emissivity above the break on $NB^{(p_1+1)/2}$.

5.1.3 The Central Engine

The accretion disk radiates $L_{\rm disc} \simeq (1.8$–$1.9) \times 10^{46}$ erg s^{-1}, and the black hole mass is $M_{\rm BH} = (1.8^{+0.5}_{-0.4}) \times 10^9\, M_\odot$ for the baseline case and between 1.7 and $2.0 \times 10^9\, M_\odot$ for all efficiencies except $\eta = 0.2$, where a degeneracy with the inner radius of the disk gives $1.1 \times 10^9\, M_\odot$. The Eddington ratio is $\lambda \simeq 0.08$. These values refer to an inner disk radius of $\simeq 8\, R_S$: the disk spectrum fixes the inner radius in physical units, and with the inner radius at the ISCO the mass would be $(5$–$10) \times 10^9\, M_\odot$ and $\lambda \approx 0.015$–0.03 (Section 4.6.5). They do not depend on the assumed accretion efficiency, because the disk spectrum is the same in all fits; only the physical accretion rate depends on η, from $5.7\, M_\odot$ yr^{-1} for $\eta = 0.057$ to $1.1\, M_\odot$ yr^{-1} for $\eta = 0.3$. The black hole mass places MG3 J163554+3629 among the most massive black holes known in blazars at $z > 3$.

5.1.4 EBL Absorption

At $z = 3.65$ the extragalactic background light absorbs the γ-ray emission above a few tens of GeV. In the SED of MG3 J163554+3629 this region is constrained only by the upper limits of the three highest-energy *Fermi*-LAT bins, and the cutoff of the model is determined by the combination of the EBL absorption, the maximum electron energy and the Klein–Nishina suppression of the external Compton emission. The EBL model therefore affects only the highest-energy part of the model and has a negligible influence on the parameters of the jet and of the central engine.

5.2 Disentangling SSC and External Compton Emission

The main goal of this thesis was to understand the respective roles of SSC and external Compton emission in the high-energy component of MG3 J163554+3629, using for the first time the hard X-ray observation of *NuSTAR*. The results can be summarised as follows.

In the *NuSTAR* band the emission is a mixture of the two processes. SSC is the largest single component, providing 60–74% of the jet emission at 10 keV and 37–56% at 40 keV, and its share decreases with energy because the SSC component peaks around 10^{19} Hz. The rest is external Compton emission, mainly from disk photons at low efficiency and from disk and BLR photons at high efficiency, which produces the hardening of the spectrum towards 40 keV. This decomposition is the same for all accretion efficiencies, and it is therefore a robust property of the model.

The simple expectation of Section 3.7, that a hard photon index would identify a single rising EC component, is refined by the full model. Below the respective peaks, the slopes of both the SSC and the external Compton components are set by the low-energy electron index p, while the high-energy index p_1 is constrained by the *Fermi*-LAT spectrum. The model reproduces the *NuSTAR* points with a local photon index $\Gamma_X \simeq 1.6$, consistent with the SED points and within about 2σ of the XSPEC value $\Gamma_X = 0.94 \pm 0.34$.

In the γ-ray band the emission is entirely due to external Compton scattering. In the baseline geometry the seed photons come from the torus, and this geometry is reproduced by every efficiency once the radii are scaled with the radiated luminosity (Section 4.4.4). In our grid, where the radii follow $L_{\rm D}$, the BLR moves closer to the emission region as η increases and dominates for $\eta \gtrsim 0.15$. The two configurations fit the data equally well and differ mainly in the MeV band, where the models with BLR dominance are brighter by up to $\sim$80%. The degeneracy between SSC and EC emission is thus broken in the X-ray band by *NuSTAR*, while the position of the emission region relative to the BLR, and with it the origin of the γ-ray seed photons, remains uncertain. Observations in the MeV band, between *NuSTAR* and *Fermi*-LAT, where the high-energy component of the source peaks, would test it, but they would not measure η.

5.3 The Role of the Accretion Efficiency

5.3.1 What the SED Can and Cannot Constrain

The seven accretion efficiencies explored, corresponding to spins from $a = 0$ to $a \simeq 0.996$, give fits of the same quality, with $\Delta\chi^2 = 0.76$ across the grid. The reason is clear from the structure of the disk model: the optical–UV data fix the disk spectrum, which in

the `MultiBB` model depends on $L_{\rm D}$ and η only through the accretion rate $L_{\rm D}/\eta$, and the inner radius of the disk is a free parameter that is not tied to the spin. In our modelling, η acts on the SED only through the radii of the BLR and of the torus, which scale with $L_{\rm D} \propto \eta$.

The physical content of the η grid is therefore geometric. Increasing η at fixed radiated luminosity makes the BLR and the torus larger; since the emission region remains at $R_H \simeq 1.2 \times 10^{18}$ cm, it moves from 2.9 to 1.6 times the outer radius of the BLR, and the dominant γ-ray seed photons change from the torus to the BLR. The bulk Lorentz factor and the electron density adjust to the new geometry, while the other parameters are unchanged. This result holds within the parameterisation adopted, in which the radii follow $L_{\rm D}$ rather than the radiated luminosity. With the radii scaled to the radiated luminosity, as the empirical radius–luminosity relations imply, the SED is identical for every η, and the torus-dominated geometry of the baseline model applies to all efficiencies.

5.3.2 Constraints on the Spin from Energetics and Growth

Since all efficiencies give fits of similar quality, the growth of the black hole and the energetics of the jets give indirect constraints.

The growth requires a low radiative efficiency. A disk radiating with the maximal-spin efficiency, $\eta \approx 0.3$, could have grown the black hole only from the heaviest seeds accreting at the Eddington limit without interruption, while for $\eta \lesssim 0.1$ a heavy seed needs an average Eddington ratio of only about 0.3. This holds with and without jets, but it does not fix the spin, since a maximally spinning black hole whose jets take most of the accretion energy also radiates with a low efficiency.

The jet energetics constrain the spin together with the proton content (Table 4.11). With a standard disk, one proton per electron would need jets an order of magnitude more powerful than a spinning black hole can provide. For the fiducial composition of one cold proton per ten electrons, the median jet power would require $a \approx 0.8$ and growth at the Eddington limit for the whole history of the black hole. A jet power at the lower end of its range, or a minimum electron Lorentz factor $\gamma_{\rm min} \gtrsim 5$, gives $a \approx 0.6$–0.7. Fewer cold protons would allow $a \approx 0.4$, and even jets without cold protons need $a \approx 0.35$. The alternative, a maximally spinning black hole whose disk radiates no more than that of a non-rotating one, requires 80–90% of the accretion energy to go into the jets.

Two arguments favour the solutions with a standard disk: the extreme disk required by the alternative, and the spin-down produced by powerful jets, which makes a black hole kept at $a \simeq 0.998$ unlikely (Ricarte et al., 2023; Lowell et al., 2024). The picture that emerges for the fiducial composition is a black hole that spins rapidly but not maximally, $a \approx 0.6$–0.7, with a standard disk radiating $\eta \approx 0.09$–0.10. Its growth from a heavy seed requires an average Eddington ratio of 0.33–0.47, that is, a start at or near the Eddington

limit at $z \approx 30$ and accretion above half of it until $z \approx 7$–10. A low spin, $a \lesssim 0.3$, would require jets weaker than estimated or a disk that gives part of its energy to the jets. This is a consistent picture, not a measurement: it rests on the scaling of the jet efficiency of magnetically arrested disks and on the assumption that the jet power in the emission region equals the power extracted at the base of the jet.

5.4 Jet–Disk Coupling and the Jet Power Budget

The jet of MG3 J163554+3629 is strongly particle-dominated, with a magnetic energy density of only ~ 0.2–0.8% of the electron energy density in the emission region, and radiatively inefficient, with $P_{\rm rad}/P_{\rm jet} \approx 0.3\%$. The total power depends on the unknown proton content: for one cold proton per relativistic electron $P_{\rm jet} \approx (3\text{–}5.5) \times 10^{47}$ erg s^{-1}, about 16–29 times the disk luminosity and 1.4–3 times the Eddington luminosity of the black hole; for $N_{\rm H}/N_e = 0.1$ the power is ten times lower, $(4\text{–}7) \times 10^{46}$ erg s^{-1}, still two to four times the disk luminosity.

The ratio $P_{\rm jet}/L_{\rm disc}$ is independent of η, since both quantities are fixed by the data. The comparison that depends on the efficiency is with the accretion power $\dot{M}c^2 = L_{\rm disc}/\eta$. For one proton per electron the jet power exceeds the accretion power at all efficiencies, which cannot be explained by accretion alone and points to the extraction of rotational energy from the black hole (Blandford & Znajek, 1977; Tchekhovskoy et al., 2011), or to a jet rich in pairs. Counting both jets, one proton per electron cannot be supplied by a black hole with a standard disk at any spin, and for $N_{\rm H}/N_e = 0.1$ accretion-powered jets require a disk that radiates only $\eta \approx 0.04$–0.06 at maximal spin (Section 4.9.4).

These results agree with the general finding that the jets of powerful blazars carry more power than their disks radiate (Ghisellini et al., 2014). That work assumes one proton per electron; for MG3 J163554+3629 this composition would require an extreme disk, and jets with fewer cold protons are favoured.

5.5 MG3 J163554+3629 in the High-Redshift Blazar Population

5.5.1 Consistency with the Blazar Sequence

MG3 J163554+3629 shows the typical properties of the most luminous FSRQs: a synchrotron peak in the far-infrared, a strongly Compton-dominated SED, a γ-ray emission produced by external Compton scattering, a black hole of about $2 \times 10^9\, M_\odot$ and a jet more powerful than the accretion disk. These properties place it in the high-luminosity end of the blazar sequence (Fossati et al., 1998; Ghisellini et al., 1998), together with the

other *Fermi* blazars at $z \gtrsim 3$. The physical parameters, a dissipation region outside the BLR, a magnetic field of a few 10^{-2} G, a bulk Lorentz factor of about 20, are within the ranges found for this population (Ghisellini et al., 2010a; Sbarrato et al., 2014).

5.5.2 The Hidden Population of Jetted Black Holes

Since the jet is observed at an angle $\theta \lesssim 1/\Gamma$, MG3 J163554+3629 is one of the few sources of its kind whose jet points towards us. For randomly oriented jets emitting within a cone of half-opening angle $1/\Gamma$, the probability of observing a source as a blazar is about $1/(2\Gamma^2)$, and each blazar therefore implies about $2\Gamma^2$ similar jetted sources with misaligned jets (Ghisellini et al., 2010b). With the lower limit $\Gamma \gtrsim 14$–19 from our posteriors, MG3 J163554+3629 implies at least several hundred, and for $\Gamma \approx 20$ about 800, black holes with $M_{\rm BH} \sim 2 \times 10^9\, M_\odot$ hosting powerful jets at similar redshifts. This argument, applied to complete samples of high-redshift blazars, suggests that massive jetted black holes were already common at $z \gtrsim 3$, and that their number density peaks at earlier times than that of radio-quiet quasars of similar mass (Ghisellini et al., 2010b, 2013). From a single source the estimate is only indicative, but it shows why individual high-redshift blazars are important probes of the early growth of supermassive black holes.

5.5.3 Hard X-rays as a Diagnostic

The analysis of MG3 J163554+3629 shows that hard X-ray data are essential to separate the SSC and EC contributions in high-redshift FSRQs. At $z = 3.65$ the *NuSTAR* band corresponds to rest-frame energies of 14–370 keV, where the SSC component reaches its peak and the external Compton components start to rise. The same diagnostic can be applied to other high-redshift blazars, and it becomes more powerful when combined with γ-ray data, which constrain the high-energy electron slope, and with optical–UV data, which fix the disk luminosity and therefore the external photon fields.

5.6 Implications for Early Supermassive Black Hole Growth

The mass of MG3 J163554+3629, $\sim 2 \times 10^9\, M_\odot$ about 1.7 Gyr after the Big Bang, sets a strong constraint on its growth history. The main results of Section 4.10 are the following.

The present state is not representative. At its present Eddington ratio, $\lambda \simeq 0.08$, the black hole would have grown only by a small factor since $z = 30$: the required seed would be of 10^7–$10^9\, M_\odot$, far above any proposed formation channel. This seed-mass

tension shows that the present state of the source is not representative of its growth history: the black hole must have accreted at a much higher rate in the past.

An early phase of rapid growth. For $\eta \lesssim 0.1$, an Eddington ratio that was higher at early times and declined to its present value on a timescale of 0.7–1.2 Gyr reproduces both the mass and the present Eddington ratio, starting from a heavy seed. Such histories are in line with models of self-regulated growth (Hopkins & Hernquist, 2009), and quasars at $z \gtrsim 6$, which accrete close to the Eddington limit (Mazzucchelli et al., 2023), may represent the earlier phase of objects like MG3 J163554+3629. The smooth exponential decline is only one possibility: it gives $\lambda \approx 0.15$–0.25 at $z = 6$, below the median of 0.7 of those quasars, and a later phase of near-Eddington growth, or growth in episodes with a duty cycle of 15–30%, reaches the same mass. These numbers neglect the jets: for the fiducial composition, with spin-powered jets, the required average Eddington ratio is 0.33–0.47, and the black hole must start at or near the Eddington limit at $z \approx 30$ and remain above half of it until $z \approx 7$–10, declining on a timescale $\tau_{\rm dec} \approx 0.55$–$0.7$ Gyr.

A limit on the radiative efficiency. If all the accretion energy is radiated, Eddington-limited growth from a heavy seed requires $\eta \lesssim 0.27$ ($\eta \lesssim 0.32$ for a seed of $10^6\,M_\odot$). A black hole spun up by accretion to $a \approx 0.998$, with $\eta \approx 0.3$, could have reached the observed mass only from the heaviest seeds and with uninterrupted Eddington-limited accretion. For $\eta \lesssim 0.1$ growth from a heavy seed requires an average Eddington ratio of only about 0.3.

The role of the jets. At a fixed radiative efficiency the jets slow the growth, since the energy they carry is not added to the black hole. They can speed up the growth of a maximally spinning black hole only if they take the energy that its disk would otherwise radiate (Ghisellini et al., 2013). Once the jets are included in every scenario, a maximally spinning black hole of this kind and a moderately spinning one with spin-powered jets need comparable average Eddington ratios, and which one grows faster depends on the proton content. The growth therefore constrains the radiative efficiency, but not the spin.

In summary, Eddington-limited growth of a black hole radiating with the maximal-spin efficiency is not sufficient to explain the mass of MG3 J163554+3629. The mass is explained by growth from a heavy seed with a radiative efficiency well below 0.3, $\eta \lesssim 0.1$ for a past Eddington ratio a few times higher than today. The spin is constrained by the jet energetics instead (Section 5.3), which favour a rapid but not maximal spin, $a \approx 0.6$–0.7, for the fiducial composition. Episodes of super-Eddington accretion (Alexander & Natarajan, 2014; Madau et al., 2014) and mergers are possible alternatives.

5.7 Systematic Uncertainties and Caveats

The results of this thesis depend on several assumptions, which we summarise here.

One-zone model. All the non-thermal emission in the fitted range is attributed to a single homogeneous region. The radio emission below 10^{10} Hz is not reproduced and was excluded from the fit; a multi-zone model would be needed to describe it, and it could modify the infrared part of the SED.

Geometry of the external fields. The radii of the BLR and of the torus are tied to $L_{\rm D}$ through fixed relations (Equation 4.4), with covering factors fixed at 0.1. Because $L_{\rm D} \propto \eta$ at fixed radiated luminosity, the trends with η, and in particular the transition from torus to BLR dominance, depend on these relations. If the radii are scaled with the radiated luminosity, the transition from torus to BLR dominance disappears. Different normalisations or covering factors would instead change the position of the emission region relative to the BLR and the torus.

External field transformation. The external photon fields were computed in the blob frame. The EC(Disk) component is strongly affected by this choice, which changes its contribution in the X-ray band.

Fixed parameters. The viewing angle, the jet opening angle, the cutoff Lorentz factors and the low-energy index p were fixed in all fits, and in the MCMC sampling the dissipation radius, the torus temperature and the disk radii were also fixed at the Minuit values. A dedicated run showed that sampling R_H does not change the results, but the uncertainties on the other parameters might be somewhat larger if all of them were sampled.

Prior on Γ. The upper tail of the posterior of Γ is limited by the prior $\Gamma \leq 30$ for most efficiencies. The lower limit on Γ is robust, but the upper values depend on the prior.

Spin inference. The constraints on the spin assume that the jet power estimated in the emission region equals the power extracted at the base of the jet, and that the jets are produced as in simulated magnetically arrested disks, with an efficiency scaled from Tchekhovskoy et al. (2011). This scaling is an order-of-magnitude estimate. Thin disks probably produce weaker jets, in which case the inferred spins are lower limits. The power in cold protons scales with the number of electrons, and therefore as $\gamma_{\rm min}^{-0.59}$ for the fixed $\gamma_{\rm min} = 2.1$: a larger $\gamma_{\rm min}$ would lower the jet power and the inferred spin.

Disk model. The `MultiBB` disk radiates with the Shakura–Sunyaev efficiency and its inner radius is not tied to the ISCO. The disk spectrum fixes the inner radius in physical units, so the black hole mass is degenerate with the inner radius, $M_{\rm BH} \propto R_{\rm in}^{-1}$ (Section 4.6.5). The values quoted refer to $R_{\rm in} \simeq 8\, R_S$; with the inner radius at the ISCO the mass would be $\approx (5\text{–}10) \times 10^9\, M_\odot$ and $\lambda \approx 0.015\text{–}0.03$, which would change the growth results by 10–20%. An independent mass estimate would remove this uncertainty.

A relativistic disk model would be needed to connect the disk emission directly to the spin.

Non-simultaneity. The SED combines observations taken at different times. The 15% systematic uncertainty partially accounts for variability, but a simultaneous SED would give tighter constraints.

Hard X-ray data. The *NuSTAR* spectrum is based on a 32.8 ks exposure and on the FPMB module only, and the photon index has a large uncertainty. The model photon index, $\simeq 1.6$, is within about 2σ of the measured value.

Proton content. The jet power depends linearly on the cold proton fraction, which the SED cannot constrain. We have bracketed it with $N_{\rm H}/N_e = 1$, 0.1 and 0.01.

Growth model. The growth calculations assume a constant efficiency, heavy seeds formed at $z = 30$, continuous accretion and growth by gas accretion only. Light seeds were not considered, since radiative feedback limits their early growth. Mergers and a different seed formation redshift would change the numbers, but not the qualitative conclusions.

EBL model. Only the Franceschini et al. (2008) model was used. Since the EBL affects only the region constrained by upper limits, its influence on the results is small.

5.8 Conclusions

The principal conclusions of this thesis are the following.

1. **The broadband SED of MG3 J163554+3629 is well described by a one-zone leptonic model.** With a broken power-law electron distribution, external Compton emission from disk, BLR and torus photons and EBL absorption, the model reproduces the data from 10^{10} Hz to the *Fermi*-LAT band with $\chi^2 \simeq 26$ for 27 degrees of freedom, for every accretion efficiency explored.

2. **The *NuSTAR* band is produced by a combination of SSC and external Compton emission.** SSC provides 60–74% of the jet emission at 10 keV and 37–56% at 40 keV, and the rising external Compton components produce the hardening of the spectrum. This decomposition is independent of the accretion efficiency. The slopes of both components in this band are set by the low-energy electron index, while the high-energy index is constrained by the *Fermi*-LAT data.

3. **The disk luminosity is well constrained and independent of η.** The disk radiates $L_{\rm disc} \simeq 1.9 \times 10^{46}$ erg s^{-1}. For an inner disk radius of $8\,R_S$ the black hole mass is $M_{\rm BH} = (1.8^{+0.5}_{-0.4}) \times 10^9\,M_\odot$ and the Eddington ratio is $\lambda \simeq 0.08$. The disk spectrum fixes the inner radius in physical units rather than the mass: with the inner radius at the ISCO, $M_{\rm BH} \approx (5\text{–}10) \times 10^9\,M_\odot$ and $\lambda \approx 0.015$–0.03.

4. **The γ-ray seed photons come from the torus in the baseline geometry.** With the radii scaled to the radiated luminosity this holds at every η. In our grid, where the radii follow $L_{\rm D}$, the BLR dominates for $\eta \gtrsim 0.15$; this transition is a property of the parameterisation, not of the efficiency. The two geometries differ by up to ∼80% in the MeV band, where future observations could locate the emission region relative to the BLR.

5. **The jet is particle-dominated, radiatively inefficient and more powerful than the disk.** $U'_B/U'_e \approx 2\text{–}8 \times 10^{-3}$, $P_{\rm rad}/P_{\rm jet} \approx 0.3\%$, and $P_{\rm jet}/L_{\rm disc} \approx 16$–$29$ for one proton per electron (2–3.6 for $N_{\rm H}/N_e = 0.1$), independently of η. For one proton per electron the jet power exceeds the accretion power. Counting both jets, this composition cannot be supplied by a black hole with a standard disk at any spin, and for $N_{\rm H}/N_e = 0.1$ accretion-powered jets require a disk radiating no more than that of a non-rotating black hole. Unless the disk is strongly suppressed, part of the jet power is extracted from the rotation of the black hole.

6. **The present Eddington ratio cannot have been maintained during the growth of the black hole.** Growth at $\lambda \simeq 0.08$ would require seeds of 10^7–$10^9\,M_\odot$. For $\eta \lesssim 0.1$, an earlier phase of faster accretion, followed by a decline, reproduces

the observed mass from a heavy seed. Without jets an average Eddington ratio of about 0.3 is sufficient; for the fiducial jets it is 0.33–0.47, with a start at or near the Eddington limit at $z \approx 30$ and accretion above half of it until $z \approx 7$–10.

7. **The growth requires a low radiative efficiency, and the jet energetics favour a rapid but not maximal spin.** A disk radiating with the maximal-spin efficiency ($\eta \approx 0.3$) could have grown the black hole only from the heaviest seeds, accreting at the Eddington limit without interruption, while for $\eta \lesssim 0.1$ a heavy seed needs an average Eddington ratio of only about 0.3. The growth alone does not fix the spin. Combined with the jet power, for the fiducial composition of one cold proton per ten electrons, a standard disk requires $a \approx 0.6$–0.7, with $\eta \approx 0.09$–0.10, for a jet power slightly below its median or $\gamma_{\rm min} \gtrsim 5$. The median jet power would require $a \approx 0.8$ and growth at the Eddington limit for the whole history of the black hole. A lower spin would require fewer cold protons, and a maximal spin a strongly suppressed disk, which is extreme and is also disfavoured by the spin-down expected from powerful jets.

8. **Jet launching.** The launching of the jet is controlled by the particle distribution (Fabian, 2012). Any change in the distribution affects the synchrotron self-Compton component (Volonteri et al., 2008), which dominates the hard X-ray band measured by *NuSTAR*. These data constrain an electron density about twice that found by Sánchez Zaballa et al. (2025). This suggests that part of the total efficiency goes into launching the jet.

5.9 Future Perspectives

The analysis presented here can be extended in several directions.

MeV observations. The models that describe the data equally well differ mainly in the MeV band, where the high-energy component of MG3 J163554+3629 peaks. Observations in this band, with future MeV missions, would directly test the origin of the γ-ray seed photons and, through it, the position of the emission region relative to the BLR and the torus.

Deeper hard X-ray observations. A new *NuSTAR* observation of MG3 J163554+3629 will soon be available. Its deeper exposure, ideally with both focal-plane modules, will reduce the uncertainty on the hard X-ray photon index and allow the low-energy electron index to be fitted rather than fixed.

A relativistic disk model. Replacing the `MultiBB` disk with a model in which the inner radius is tied to the ISCO and the spectrum depends on the spin would allow the optical–UV data to constrain the spin directly, at least in principle.

Independent black hole mass. A near-infrared spectrum covering the broad Mg II or C IV lines would give a virial estimate of the black hole mass, independent of the disk fit, and would help to break the degeneracy between the mass and the inner radius of the disk.

Jet composition. The spin inferred from the energetics depends on the proton content of the jets. Independent constraints on the pair content would turn the joint constraint of Table 4.11 into a constraint on the spin.

Simultaneous and multi-epoch SEDs. Simultaneous observations would remove the uncertainty due to variability, and a multi-epoch analysis could test whether the emission region changes position along the jet.

Samples of high-redshift blazars. Applying the same analysis, including the energetics and the growth constraints, to a sample of high-redshift blazars with hard X-ray data would turn the constraints obtained here for a single source into constraints on the population of the first massive jetted black holes.

Appendix A

Results for the Individual Accretion Efficiencies

This appendix collects the results of the fits and of the MCMC sampling for each value of the accretion efficiency, which are summarised in the main text. All models use the same data, the same external field transformation and the same bounds (Table 4.4).

A.1 Minuit Fits

Figures A.1–A.7 show the Minuit best fit for each value of η, with the individual spectral components and the residuals. The parameters are listed in Table 4.6. Figure A.8 compares the decomposition of the high-energy emission for all efficiencies: as η increases, the EC(BLR) component grows at the expense of the EC(DT) component, while the SSC component, which dominates the *NuSTAR* band, changes little.

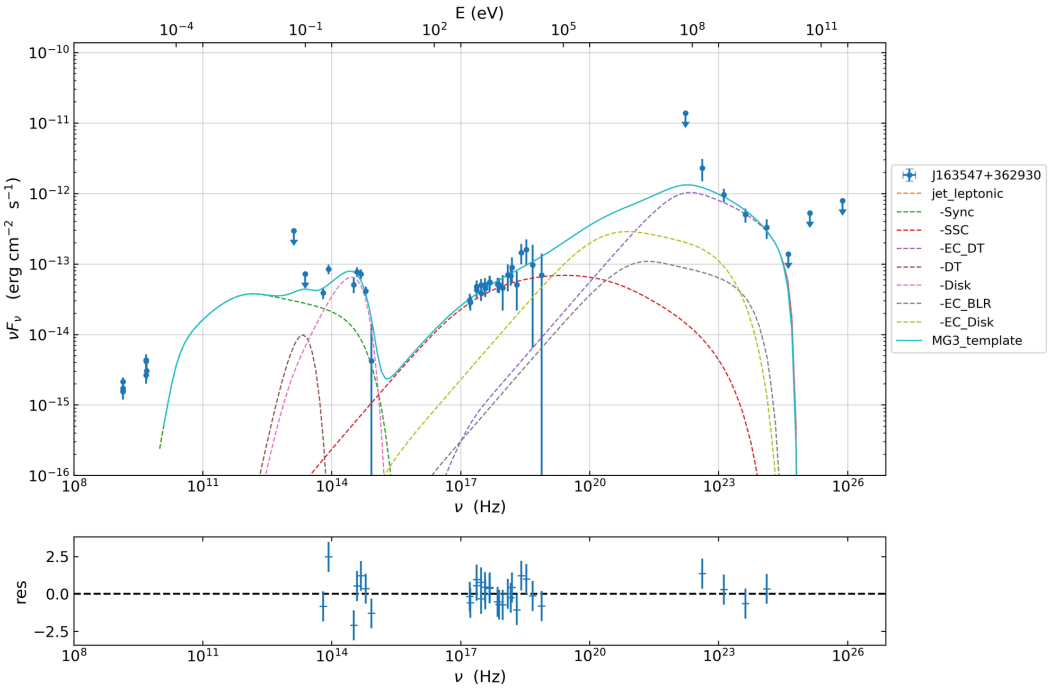


Figure A.1: Minuit best fit for $\eta = 0.057$, with the individual components and the residuals.

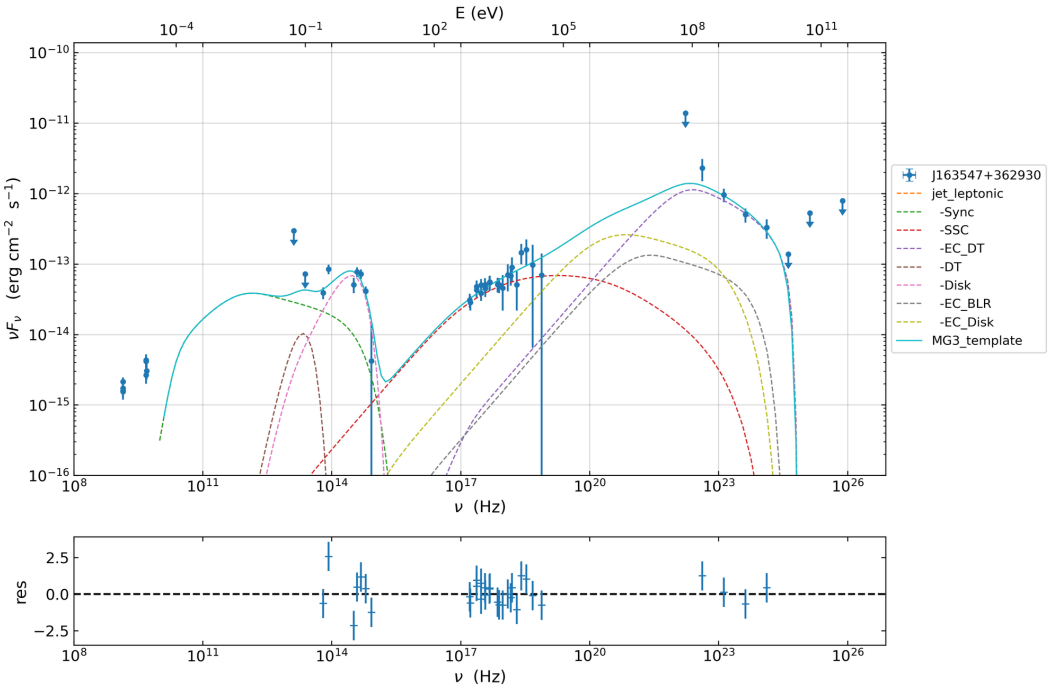


Figure A.2: Minuit best fit for $\eta = 0.070$, with the individual components and the residuals.

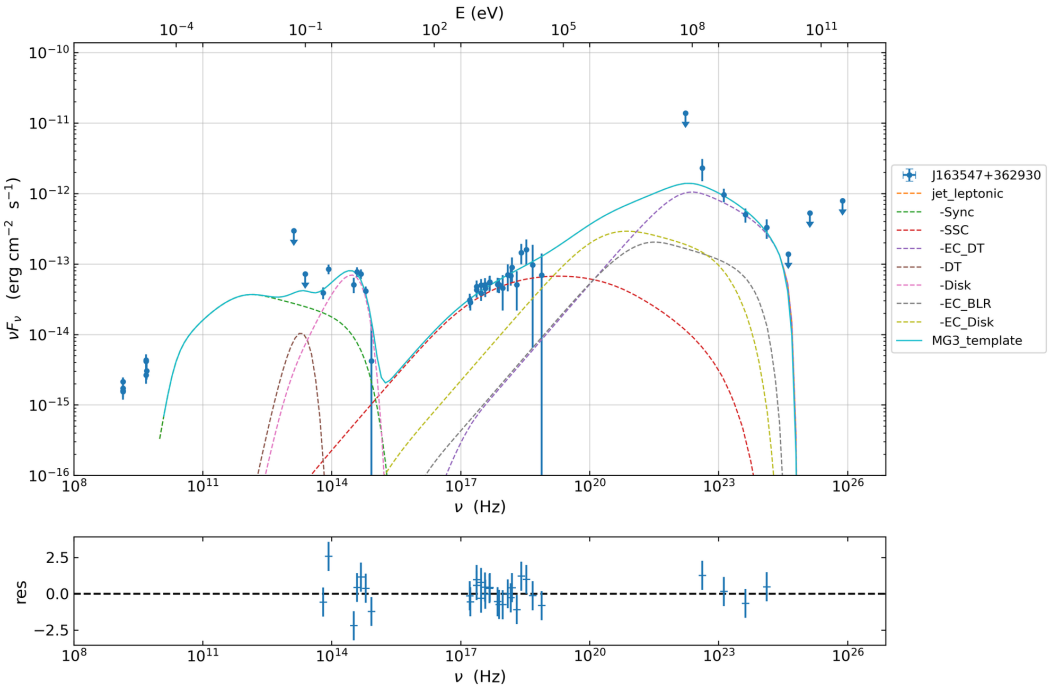


Figure A.3: Minuit best fit for $\eta = 0.083$, with the individual components and the residuals.

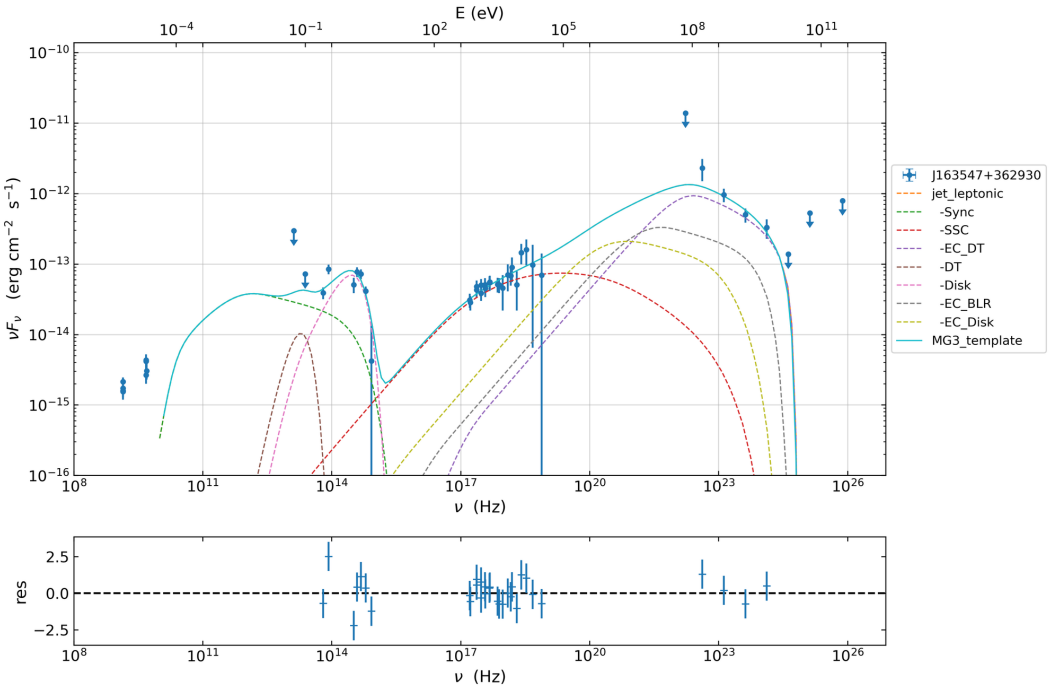


Figure A.4: Minuit best fit for $\eta = 0.100$, with the individual components and the residuals.

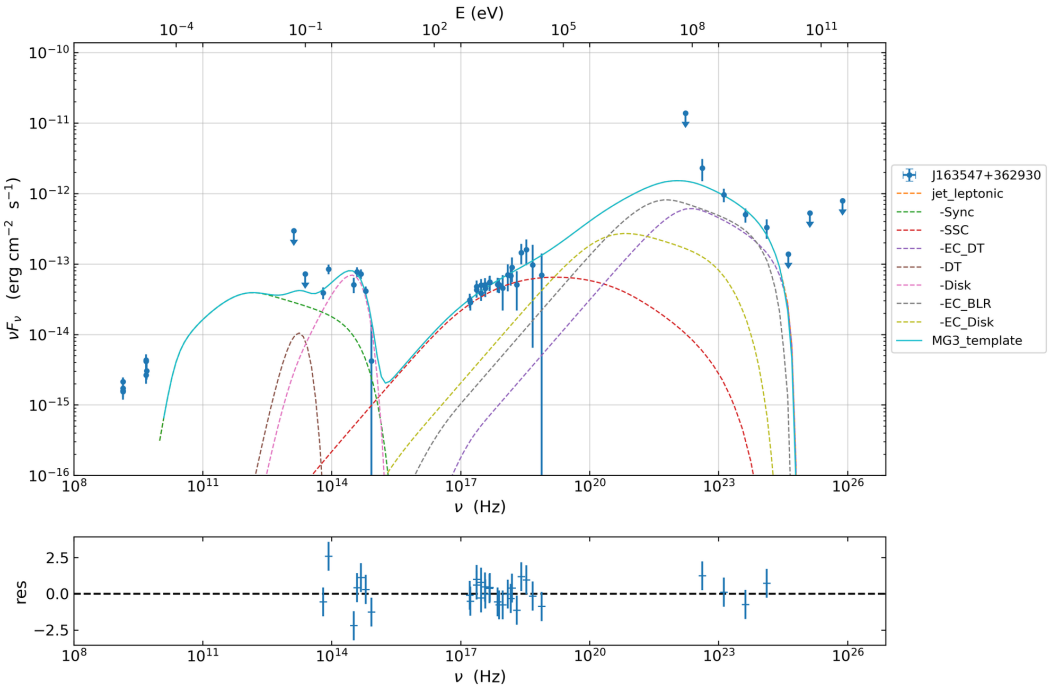


Figure A.5: Minuit best fit for $\eta = 0.150$, with the individual components and the residuals.

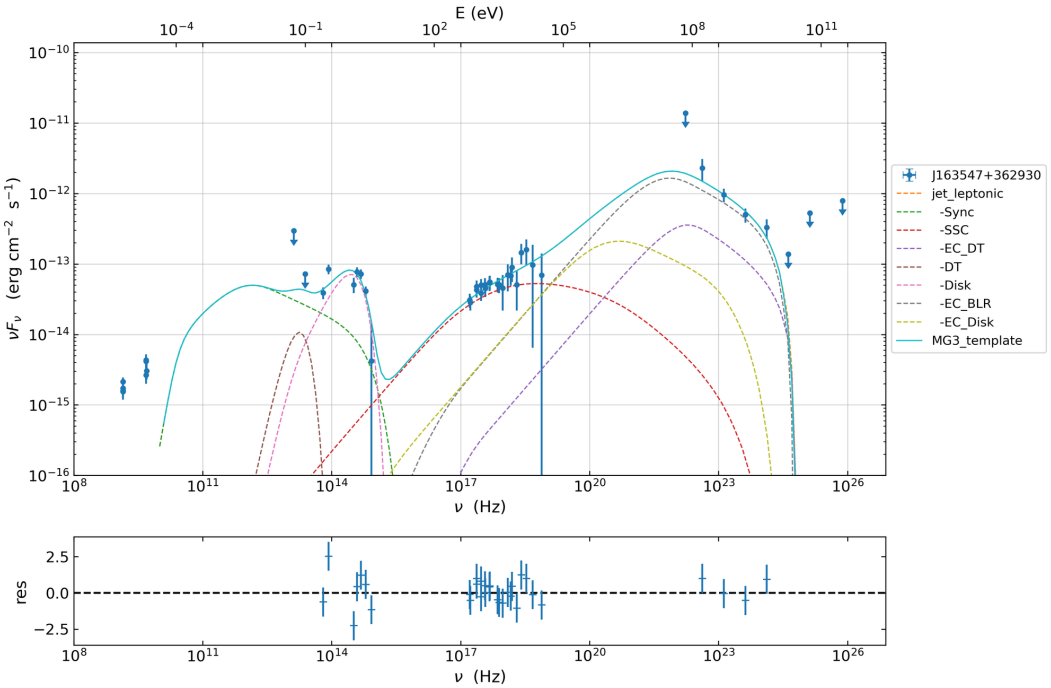


Figure A.6: Minuit best fit for $\eta = 0.200$, with the individual components and the residuals.

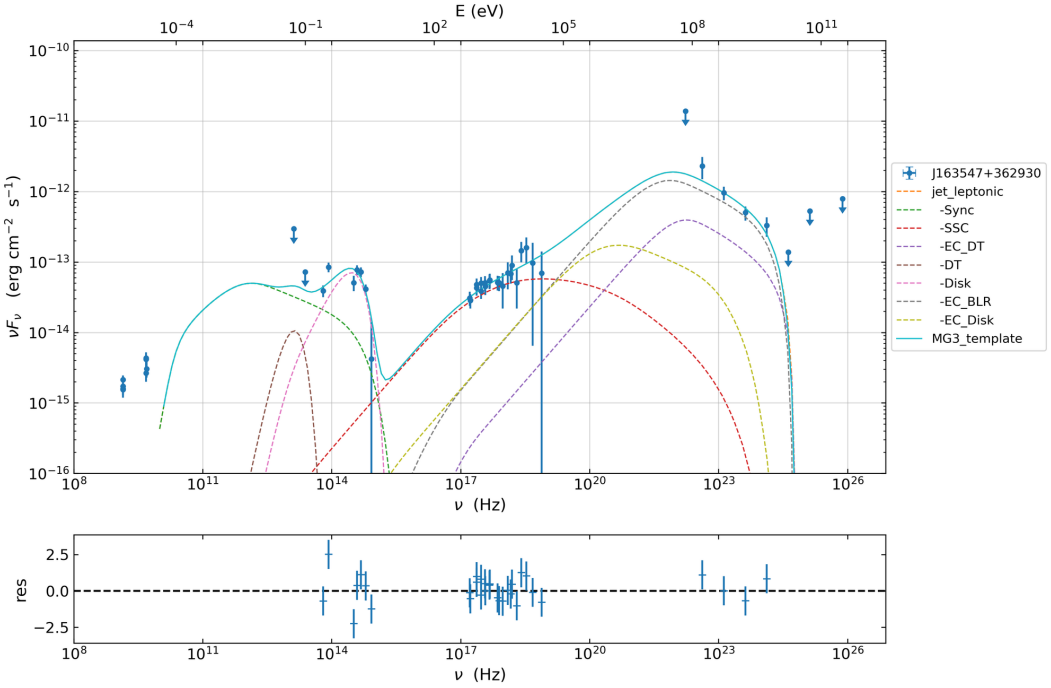


Figure A.7: Minuit best fit for $\eta = 0.300$, with the individual components and the residuals.

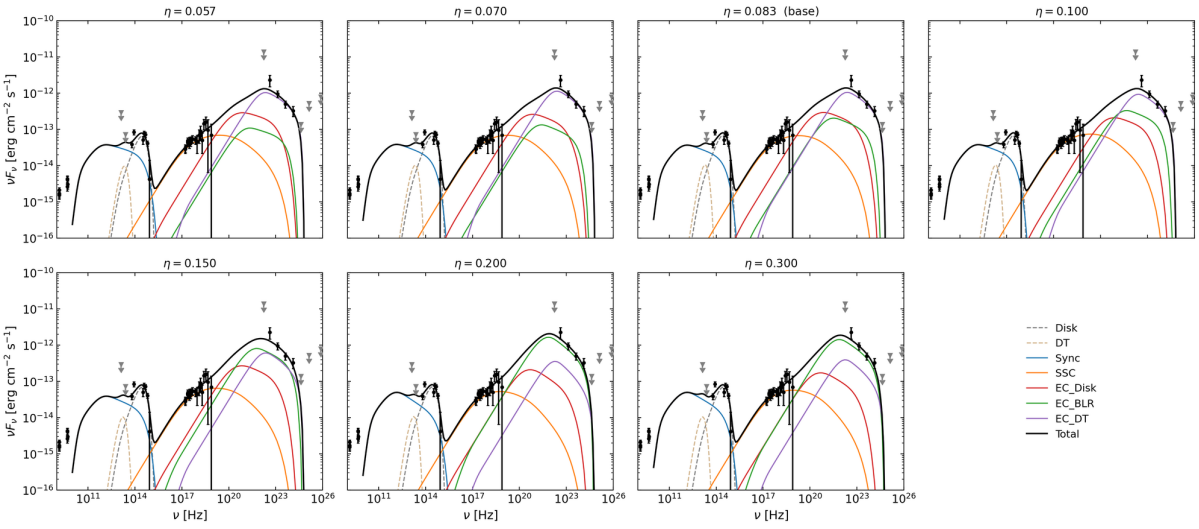


Figure A.8: Spectral decomposition of the Minuit best fits for the seven accretion efficiencies: synchrotron, SSC and external Compton components (solid), thermal emission of the disk and of the torus (dashed), and total model.

A.2 MCMC Sampling

Table A.1 gives the convergence diagnostics of the seven runs. The integrated autocorrelation times were computed with the `emcee` implementation on the post-burn-in part of the chains. The posterior distributions and the SEDs with the 68% and 95% credible bands and the residuals are shown in Figures A.9–A.15.

Table A.1: Convergence of the MCMC runs: integrated autocorrelation time τ (in steps) of each sampled parameter, minimum number of autocorrelation times covered by the 500 post-burn-in steps, mean acceptance fraction, and fixed value of the dissipation radius.

η	B	Γ	N	$\gamma_{\rm break}$	p_1	$M_{\rm BH}$	$L_{\rm D}$	$\min(500/\tau)$	acc.	R_H [10^{18} cm]
0.057	49.0	46.2	43.6	48.7	45.0	44.9	50.4	9.9	0.35	1.065
0.070	45.7	45.9	47.1	44.8	42.3	45.9	44.9	10.6	0.34	1.180
0.083	49.5	50.4	49.7	47.6	45.5	43.2	42.8	9.9	0.30	1.200
0.100	48.9	47.0	37.5	39.6	38.9	40.6	42.6	10.2	0.36	1.202
0.150	44.6	43.0	47.6	45.0	44.6	43.1	43.4	10.5	0.34	1.183
0.200	46.7	49.3	49.0	47.7	44.0	40.4	39.7	10.1	0.33	1.158
0.300	48.3	48.6	47.2	43.4	43.5	41.2	42.7	10.3	0.34	1.417

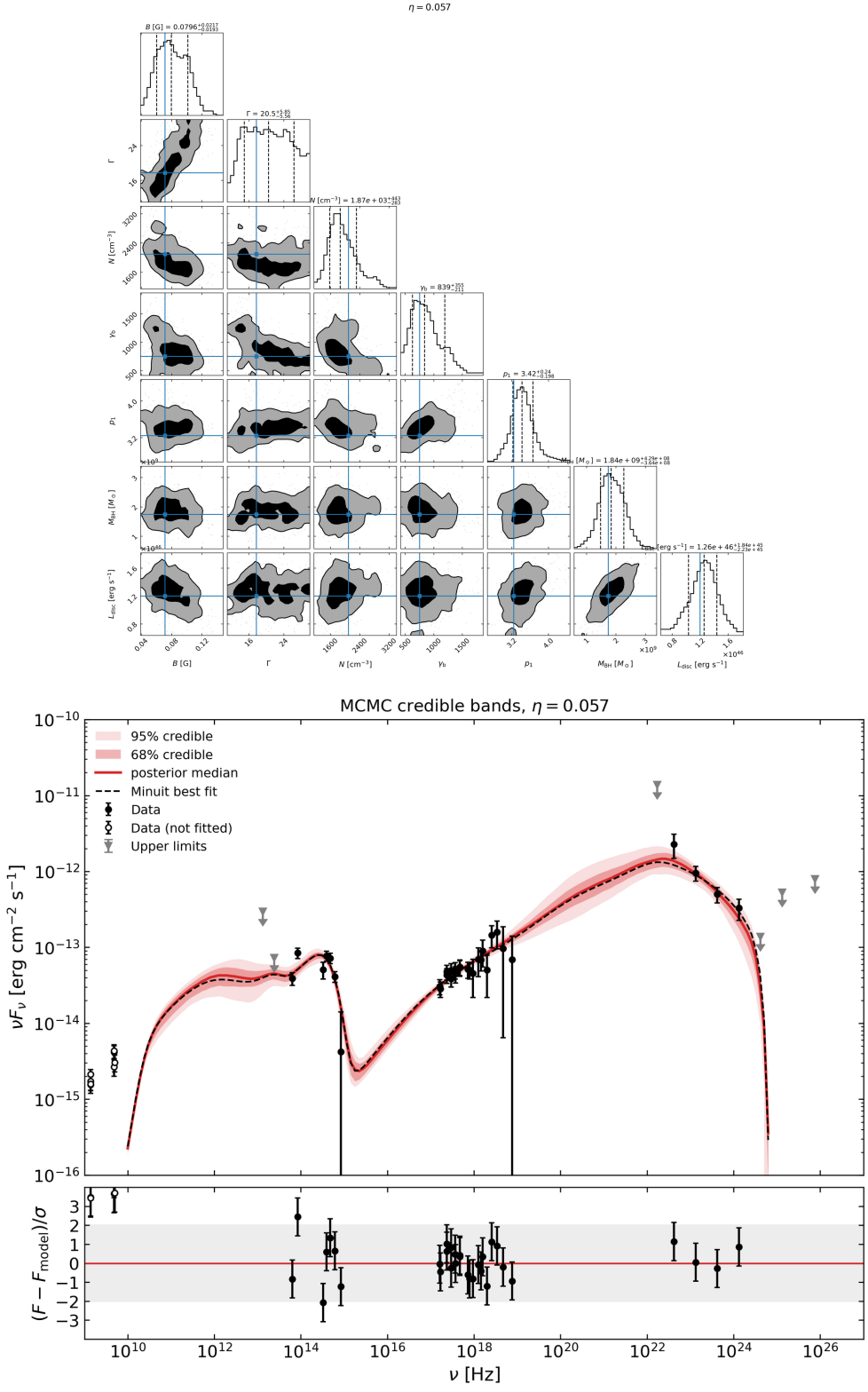


Figure A.9: MCMC results for $\eta = 0.057$. Top: posterior distribution of the seven sampled parameters, with 68% and 95% contours and the Minuit best fit (blue lines). Bottom: SED with the 68% and 95% credible bands, posterior-median SED, Minuit best fit and normalised residuals of the detections.

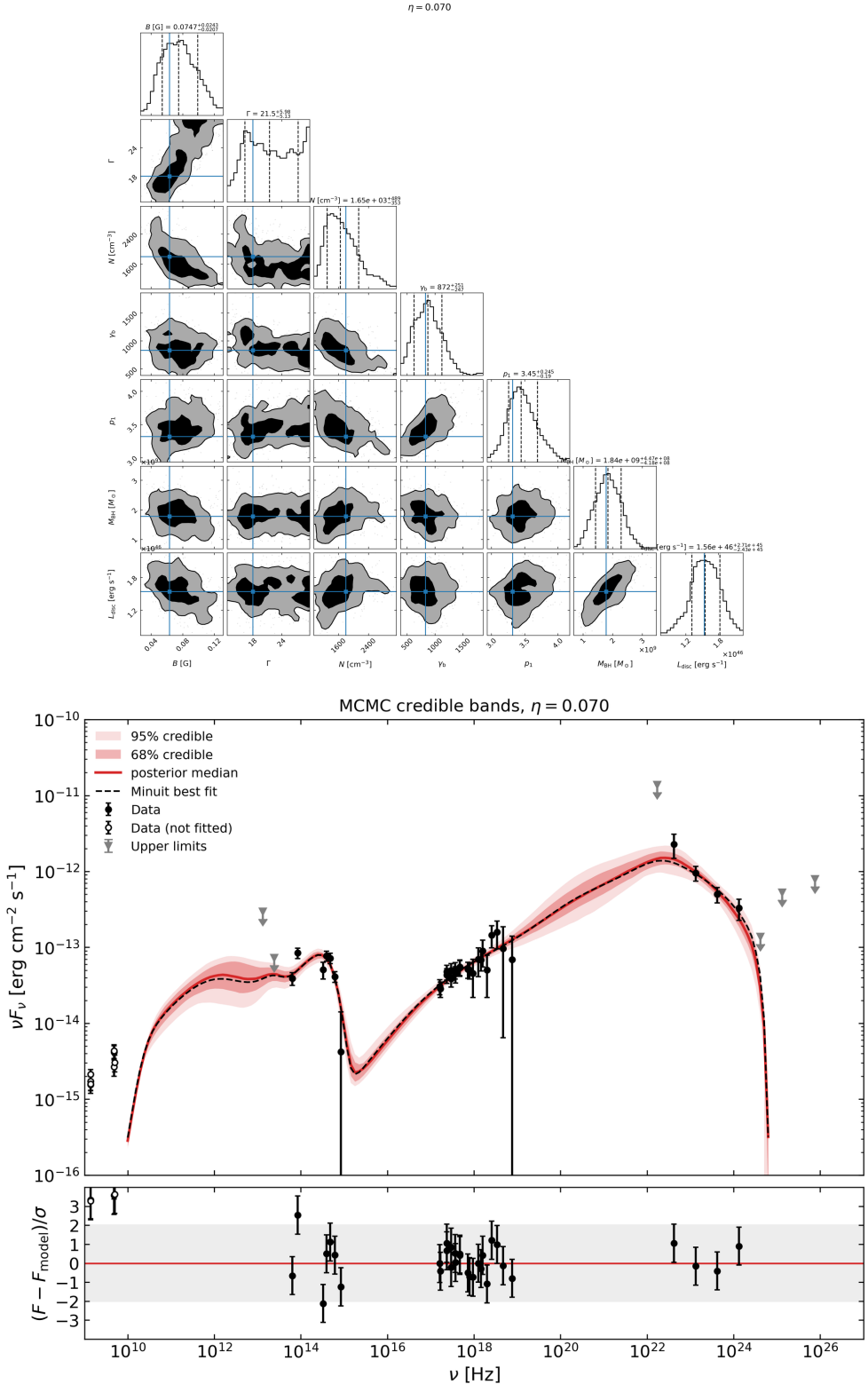


Figure A.10: MCMC results for $\eta = 0.070$. Top: posterior distribution of the seven sampled parameters, with 68% and 95% contours and the Minuit best fit (blue lines). Bottom: SED with the 68% and 95% credible bands, posterior-median SED, Minuit best fit and normalised residuals of the detections.

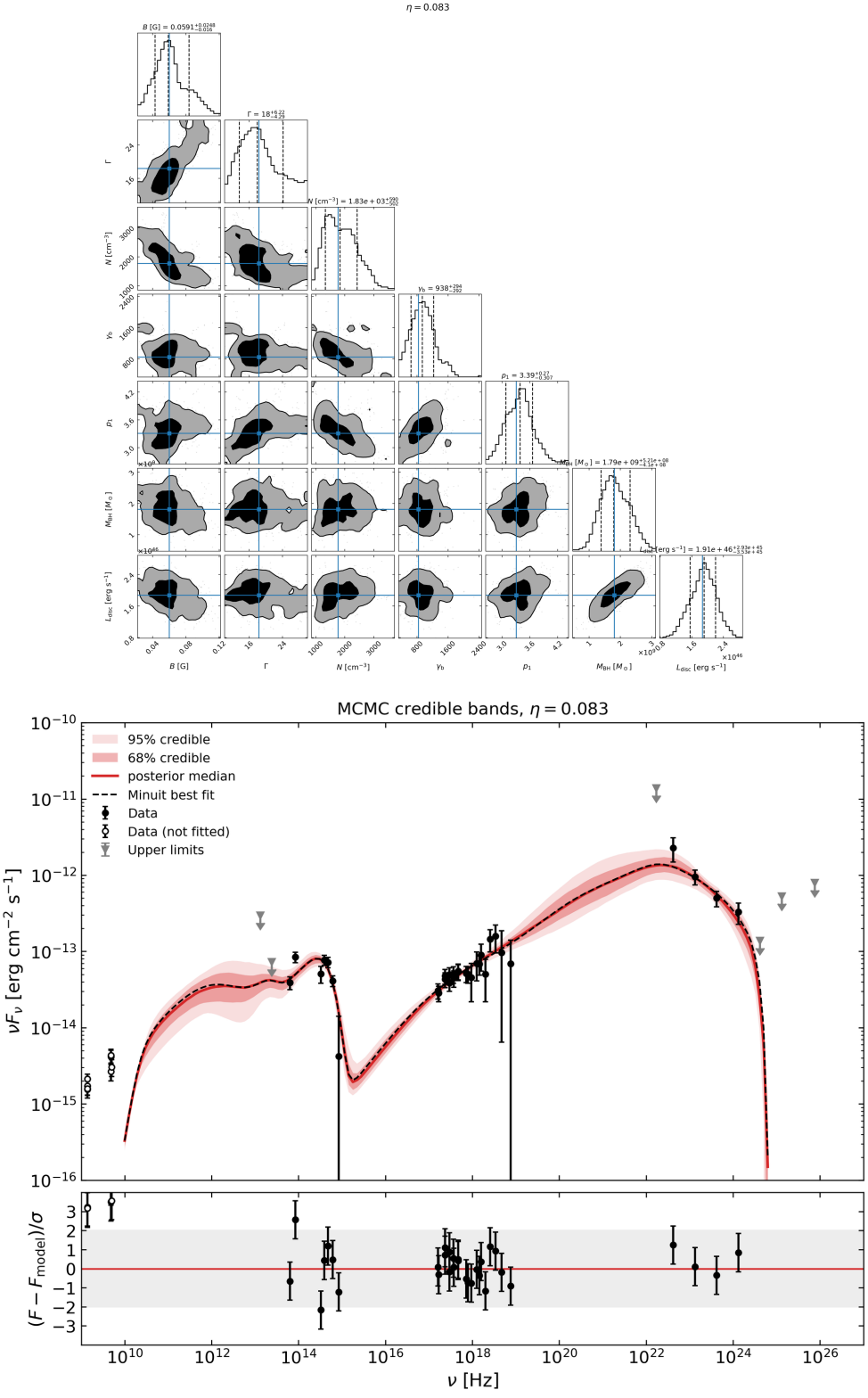


Figure A.11: MCMC results for $\eta = 0.083$. Top: posterior distribution of the seven sampled parameters, with 68% and 95% contours and the Minuit best fit (blue lines). Bottom: SED with the 68% and 95% credible bands, posterior-median SED, Minuit best fit and normalised residuals of the detections.

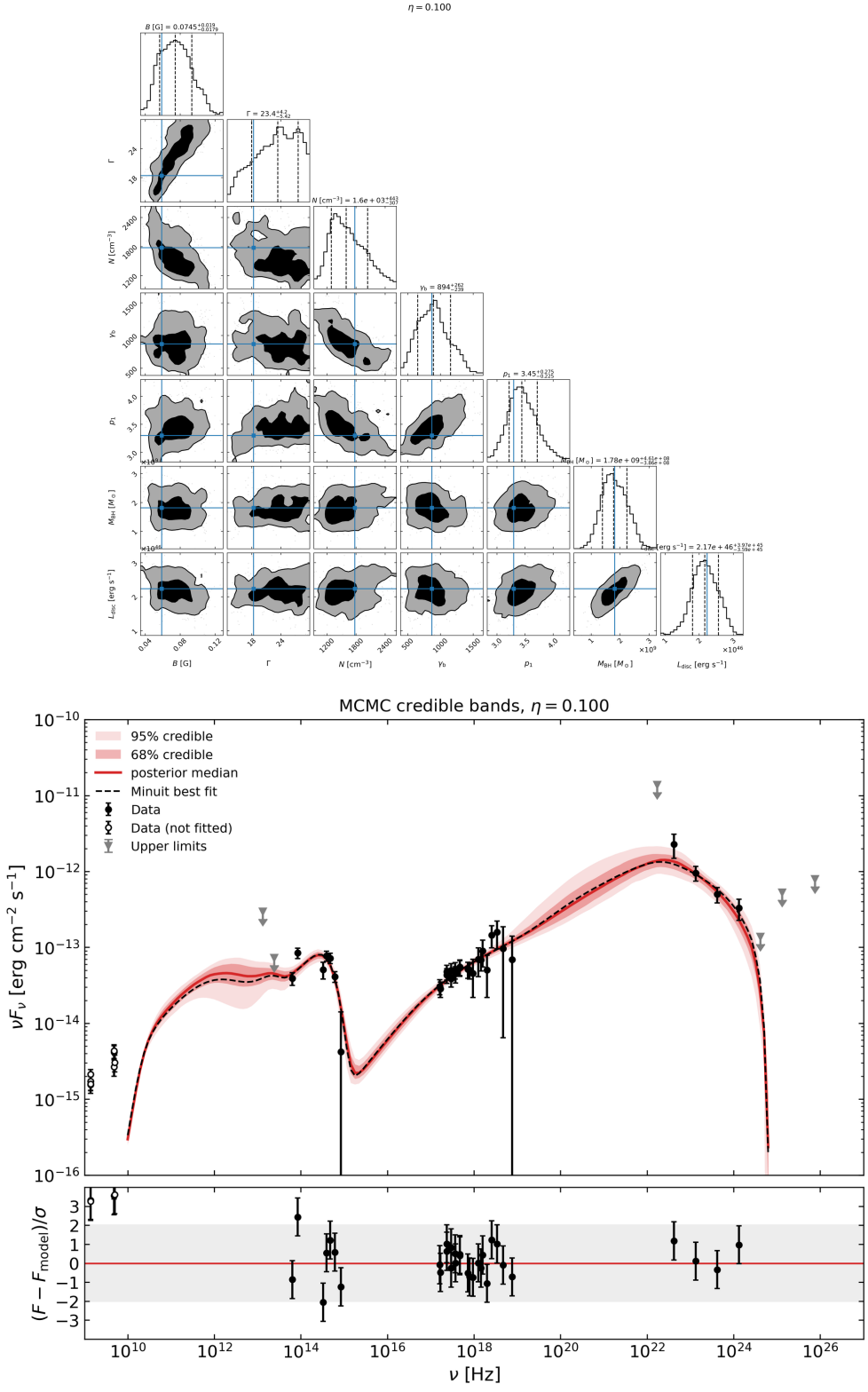


Figure A.12: MCMC results for $\eta = 0.100$. Top: posterior distribution of the seven sampled parameters, with 68% and 95% contours and the Minuit best fit (blue lines). Bottom: SED with the 68% and 95% credible bands, posterior-median SED, Minuit best fit and normalised residuals of the detections.

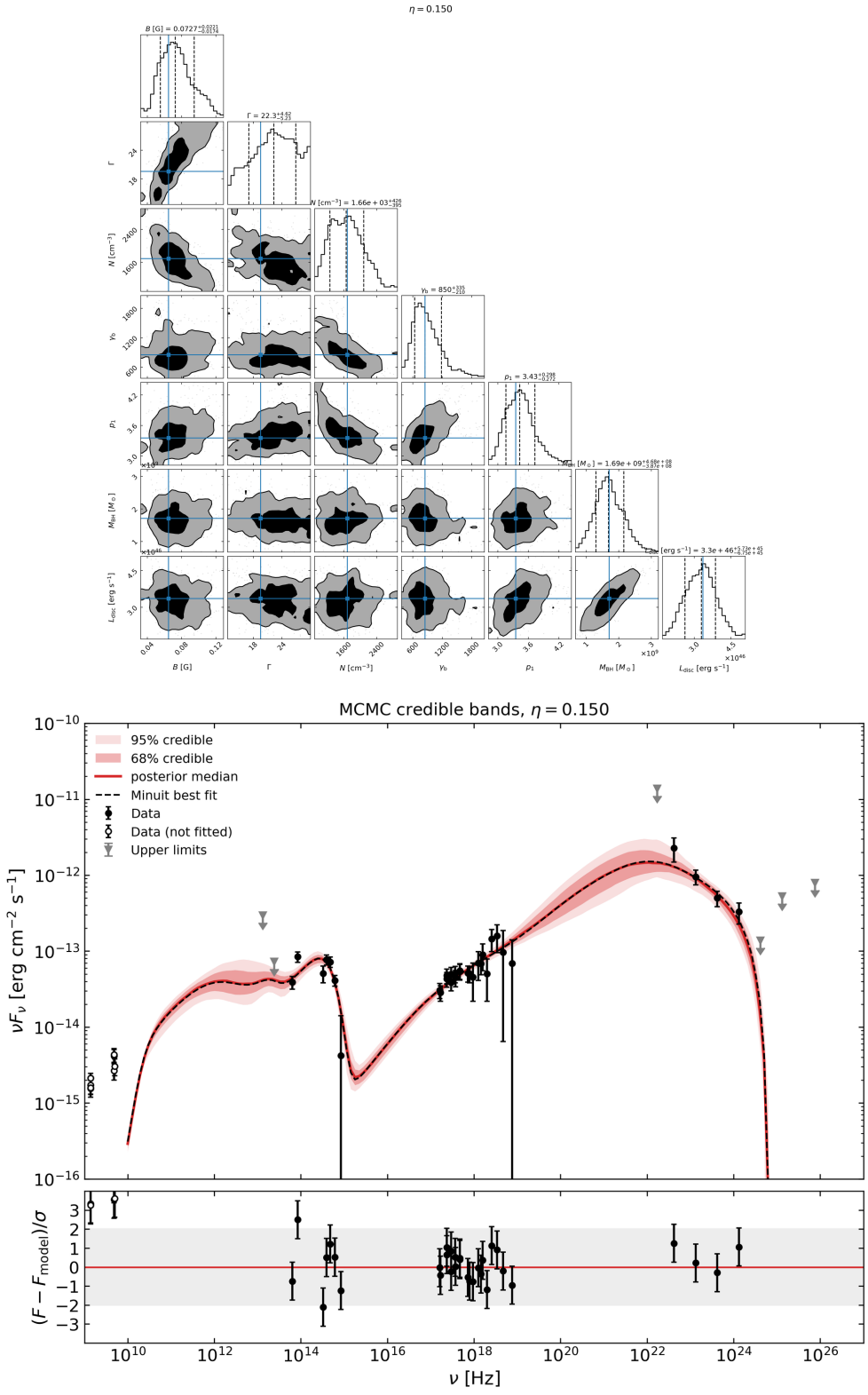


Figure A.13: MCMC results for $\eta = 0.150$. Top: posterior distribution of the seven sampled parameters, with 68% and 95% contours and the Minuit best fit (blue lines). Bottom: SED with the 68% and 95% credible bands, posterior-median SED, Minuit best fit and normalised residuals of the detections.

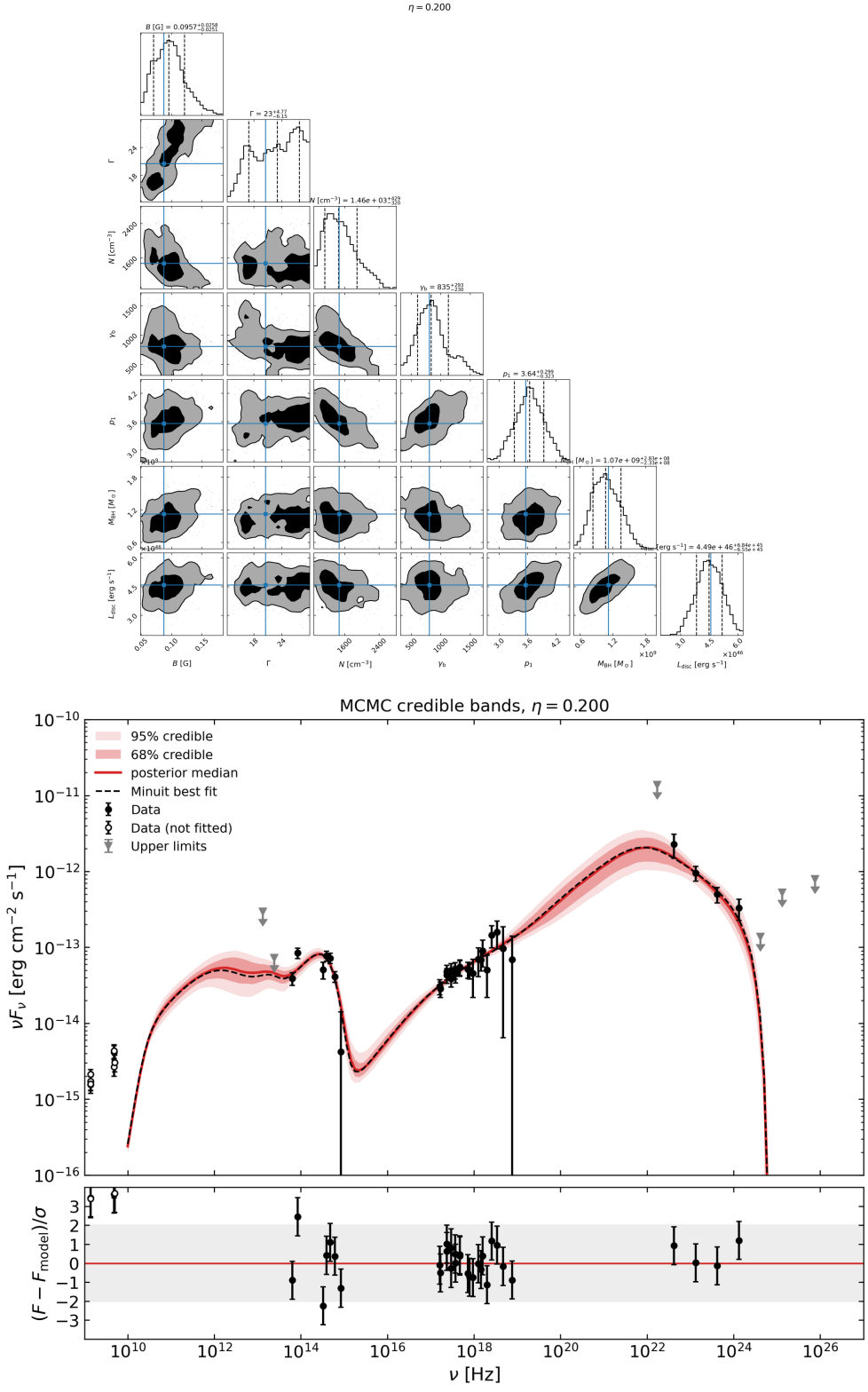


Figure A.14: MCMC results for $\eta = 0.200$. Top: posterior distribution of the seven sampled parameters, with 68% and 95% contours and the Minuit best fit (blue lines). Bottom: SED with the 68% and 95% credible bands, posterior-median SED, Minuit best fit and normalised residuals of the detections.

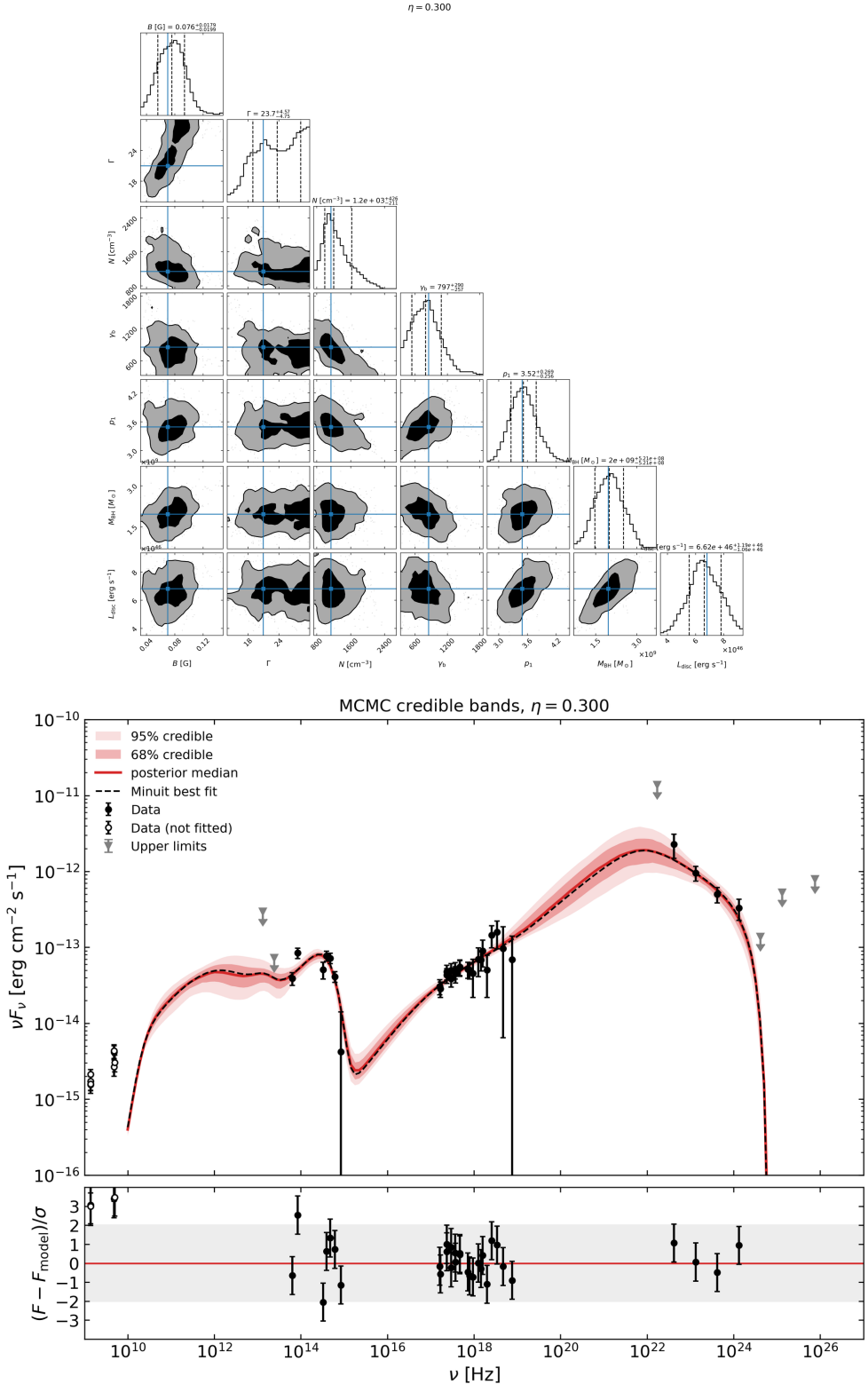


Figure A.15: MCMC results for $\eta = 0.300$. Top: posterior distribution of the seven sampled parameters, with 68% and 95% contours and the Minuit best fit (blue lines). Bottom: SED with the 68% and 95% credible bands, posterior-median SED, Minuit best fit and normalised residuals of the detections.

A.3 Baseline Model with Free Dissipation Radius

Table A.2 compares the posterior of the baseline model obtained with R_H fixed at the Minuit value (seven sampled parameters) with that obtained when R_H is also sampled (eight parameters). The posterior of the second run is shown in Figure A.16. Apart from the broader distribution of N, due to its strong anticorrelation with R_H, the two runs agree.

Table A.2: Posterior medians and 68% intervals for the baseline model ($\eta = 0.083$) with R_H fixed and with R_H sampled.

Parameter	R_H fixed	R_H sampled
B [10^{-2} G]	$5.9^{+2.5}_{-1.6}$	$6.4^{+2.6}_{-2.0}$
Γ	$17.9^{+6.5}_{-4.2}$	$18.2^{+6.0}_{-4.6}$
N [cm^{-3}]	1834^{+621}_{-503}	1430^{+803}_{-456}
$\gamma_{\rm break}$	935^{+298}_{-295}	985^{+292}_{-214}
p_1	$3.38^{+0.29}_{-0.30}$	$3.39^{+0.42}_{-0.22}$
$M_{\rm BH}$ [$10^9 M_\odot$]	$1.80^{+0.50}_{-0.44}$	$1.85^{+0.46}_{-0.45}$
$L_{\rm D}$ [10^{46} erg s^{-1}]	$1.91^{+0.28}_{-0.35}$	$1.87^{+0.32}_{-0.36}$
R_H [10^{18} cm]	1.20 (fixed)	$1.32^{+0.24}_{-0.25}$
λ	$0.083^{+0.012}_{-0.010}$	$0.081^{+0.012}_{-0.010}$
$R_H/R_{\rm BLR,out}$	$2.63^{+0.28}_{-0.18}$	$2.91^{+0.62}_{-0.55}$
Acceptance fraction	0.30	0.28
min($500/\tau$)	9.9	5.6

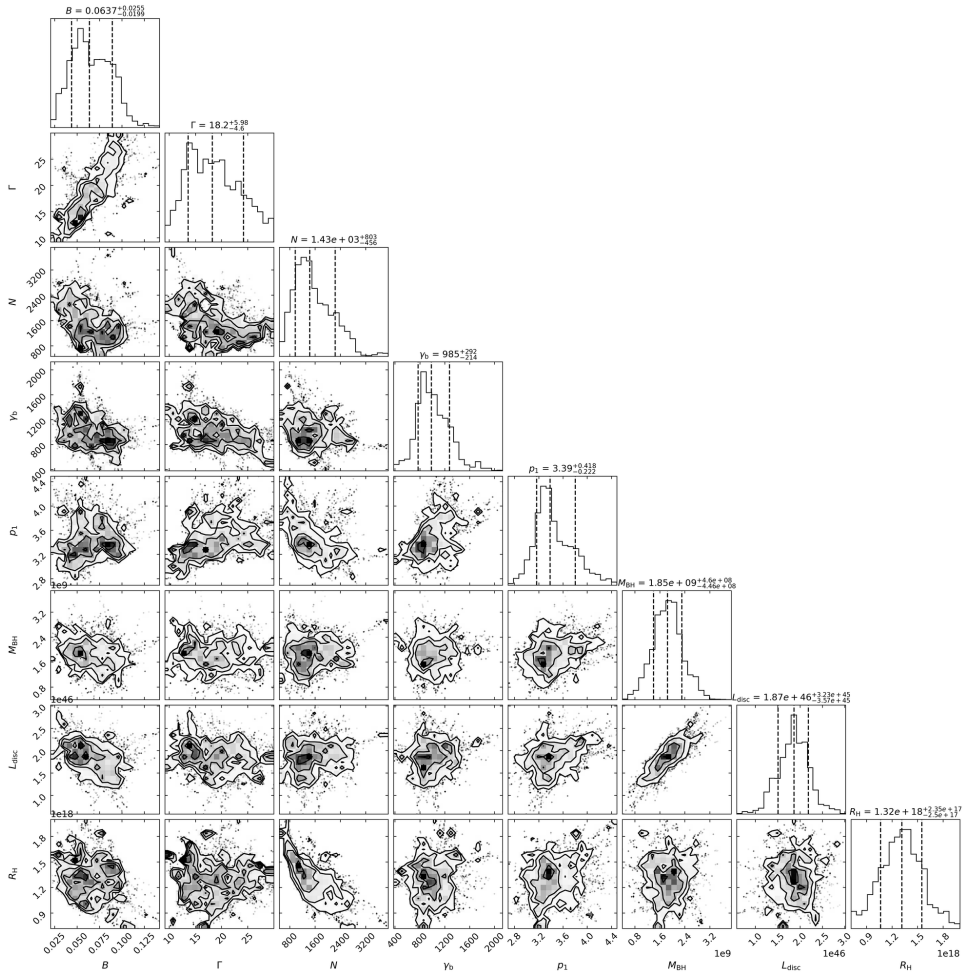


Figure A.16: Posterior distribution of the baseline model with the dissipation radius sampled together with the other seven parameters.

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